JESS Thermodynamic Database v8.9

Silicon

24-Jan-23

Reaction No. 1423							
H+1 + SiH2O4-2 = H+1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:13.2(0.05SD)	0	CMN	447
3	25	0	Inf. Dilution	lgK:13.1(3SF)	0	ENS	6405
Note (3): p. 161							
4	25	0	Inf. Dilution	lgK:13.06(4SF)	0	CRV	8740
5	25	0	Inf. Dilution	lgK:13.10(4SF)	0	CRV	9402
6	25	0	Inf. Dilution	lgK:13.45(0.07SD)	4	EOD	30201
7	25	0	UCT_Sea	lgK:13.10(0.01SD)	0	EDH	5390
8	25	0.1	UCT_Sea	lgK:12.72(0.01SD)	0	EDH	5390
9	25	0.1	Unknown	lgK:12.96(4SF)	0	CRV	552
10	25	0.2	UCT_Sea	lgK:12.65(0.01SD)	0	EDH	5390
11	25	0.3	UCT_Sea	lgK:12.61(0.01SD)	0	EDH	5390
12	25	0.4	UCT_Sea	lgK:12.58(0.01SD)	0	EDH	5390
13	25	0.5	NaCl	lgK:12.7(0.05SD)	0	MIS	430
14	25	0.5	NaClO4	lgK:12.6(0.05SD)	0	MIS	430
15	25	0.5	UCT_Sea	lgK:12.56(0.01SD)	0	EDH	5390
16	25	0.5	Unknown	lgK:12.71(4SF)	0	ENS	6405
Note (16): p. 161							
17	25	0.6	UCT_Sea	lgK:12.55(0.01SD)	0	EDH	5390
18	25	0.7	UCT_Sea	lgK:12.54(0.01SD)	0	EDH	5390
19	25	3	NaClO4	lgK:12.7(0.05SD)	0	MIS	430
20	37	0.15	Blood Plasma	lgK:11.4(0.05SD)	0	ENS	94
21	50	0	Inf. Dilution	lgK:12.6(0.05SD)	0	CMN	447
22	50	0	Inf. Dilution	lgK:12.95(0.25SD)	4	EOD	30201
23	75	0	Inf. Dilution	lgK:12.56(0.25SD)	4	EOD	30201
24	100	0	Inf. Dilution	lgK:12.28(0.25SD)	4	EOD	30201
25	0:50	0	Inf. Dilution	dH:-14.(1.SD)	0	EVH	12
26	20:50	0.5	Unknown	dH:-10.00(4SF)	0	CRV	552

doi: 10.5281/zenodo.7700024

27	25	0	Inf. Dilution	dH:-10.00(4SF)	0	CRV	9402
28	25:50	0.5	Unknown	dH:-11.(1.SD)	0	EVH	430
29	25	0	Inf. Dilution	Cp:0.0(1SF)	0	CRV	9402

Reaction No. 1424

SiH2O4-2 + 2<H+1> = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.8(3SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:23.0(0.1SD)	4	ENS	557
3	25	0	Inf. Dilution	lgK:23.0(3SF)	4	ENS	6496
4	25	0	Inf. Dilution	lgK:22.96(4SF)	4	CRV	9402
5	25	0	Inf. Dilution	lgK:23.140(0.090SD)	7	CRV	14923
6	25	0	Inf. Dilution	lgK:23.140(0.045SD)	5	CRV	27292
7	25	0	Inf. Dilution	lgK:22.90(4SF)	4	CRV	32224
8	25	0	CHEMVAL1	lgK:21.47(0.01SD)	0	ENS	5156
9	25	0	CHEMVAL2	lgK:21.47(0.01SD)	0	ENS	5156
10	25	0	CHEMVAL3	lgK:21.47(0.01SD)	0	ENS	5156
11	25	0	CHEMVAL4	lgK:21.50(0.01SD)	0	ENS	5156
12	25	0	UCT_Sea	lgK:22.96(0.01SD)	4	EDH	5390
13	25	0.05	Na(Cl)	lgK:22.74(0.06MD)	3	MHY	10238
14	25	0.1	Na(Cl)	lgK:22.55(0.03MD)	5	MHY	10238
15	25	0.1	UCT_Sea	lgK:22.36(0.01SD)	4	EDH	5390
16	25	0.2	Na(Cl)	lgK:22.39(0.02MD)	6	MHY	10238
17	25	0.2	UCT_Sea	lgK:22.23(0.01SD)	4	EDH	5390
18	25	0.3	UCT_Sea	lgK:22.14(0.01SD)	4	EDH	5390
19	25	0.4	Na(Cl)	lgK:22.25(0.02MD)	6	MHY	10238
20	25	0.4	UCT_Sea	lgK:22.06(0.01SD)	4	EDH	5390
21	25	0.5	UCT_Sea	lgK:22.02(0.01SD)	4	EDH	5390
22	25	0.6	Na(Cl)	lgK:22.07(0.01MD)	4	MHY	10238
23	25	0.6	NaCl	lgK:22.07(0.01SD)	4	MIS	438
24	25	0.6	NaCl	lgK:22.12(4SF)	5	CRV	30909
25	25	0.6	UCT_Sea	lgK:22.01(0.01SD)	4	EDH	5390
26	25	0.7	UCT_Sea	lgK:22.00(0.01SD)	4	EDH	5390
27	25	2	Na(Cl)	lgK:22.11(0.05MD)	6	MHY	10238
28	37	0.15	Blood Plasma	lgK:20.5(0.05SD)	0	ENS	94
29	25	0	Inf. Dilution	dG:-132.080(0.26SD)kJ	5	CRV	27292

30	25	0	Inf. Dilution	dH:-17.6(3SF)	4	ENS	6496
31	25	0	Inf. Dilution	dH:-15.00(4SF)	0	CRV	9402
32	25	0	Inf. Dilution	dH:-75.000(15.000SD)kJ	6	CRV	14923
33	25	0	Inf. Dilution	dH:-75.000(7.5SD)kJ	5	CRV	27292
34	25	0	Inf. Dilution	Cp:0.0(1SF)	3	CRV	9402

Reaction No. 1425

SiH2O4-2 + Ca+2 = Ca+2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:4.36(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:3.58(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:3.41(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:3.32(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:3.25(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:3.21(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:3.17(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:3.14(0.01SD)	4	EDH	5390
9	25	1	NaClO4	lgK:3.1(0.05SD)	5	MIS	455
10	37	0.15	Blood Plasma	lgK:3.0(0.05SD)	4	ENS	94

Reaction No. 1426

SiH2O4-2 + Ca+2 + H+1 = Ca+2_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.3(3SF)	1	ELC	
2	25	0	UCT_Sea	lgK:14.08(0.01SD)	0	EDH	5390
3	25	0.1	UCT_Sea	lgK:13.31(0.01SD)	0	EDH	5390
4	25	0.2	UCT_Sea	lgK:13.13(0.01SD)	0	EDH	5390
5	25	0.3	UCT_Sea	lgK:13.08(0.01SD)	0	EDH	5390
6	25	0.4	UCT_Sea	lgK:13.02(0.01SD)	0	EDH	5390
7	25	0.5	UCT_Sea	lgK:12.98(0.01SD)	0	EDH	5390
8	25	0.6	UCT_Sea	lgK:12.96(0.01SD)	0	EDH	5390
9	25	0.7	UCT_Sea	lgK:12.94(0.01SD)	0	EDH	5390
10	37	0.15	Blood Plasma	lgK:11.8(0.05SD)	0	ENS	94

Reaction No. 1427

2<SiH2O4-2> + Ca+2 + 2<H+1> = Ca+2_H+1(2)_SiH2O4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	UCT_Sea	lgK:30.02(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:28.65(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:28.39(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:28.24(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:28.13(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:28.06(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:28.02(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:27.99(0.01SD)	4	EDH	5390
9	37	0.15	Blood Plasma	lgK:25.7(0.05SD)	4	ENS	94

Reaction No. 1428

SiH2O4-2 + Fe+3 + H+1 = Fe+3_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:22.44(0.01SD)	4	EDH	5390
2	25	0.1	ClO4-1 salt	lgK:9.3(0.05SD)	0	MSP	5293
3	25	0.1	UCT_Sea	lgK:21.62(0.01SD)	4	EDH	5390
4	25	0.2	UCT_Sea	lgK:21.48(0.01SD)	4	EDH	5390
5	25	0.3	UCT_Sea	lgK:21.41(0.01SD)	4	EDH	5390
6	25	0.4	UCT_Sea	lgK:21.36(0.01SD)	4	EDH	5390
7	25	0.5	UCT_Sea	lgK:21.33(0.01SD)	4	EDH	5390
8	25	0.6	UCT_Sea	lgK:21.32(0.01SD)	4	EDH	5390
9	25	0.7	UCT_Sea	lgK:21.30(0.01SD)	4	EDH	5390
10	37	0.15	Blood Plasma	lgK:20.4(0.05SD)	3	ENS	94

Reaction No. 1429

SiH2O4-2 + Mg+2 = Mg+2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:5.32(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:4.58(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:4.43(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:4.35(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:4.30(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:4.26(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:4.23(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:4.21(0.01SD)	4	EDH	5390
9	25	1	NaClO4	lgK:4.2(0.05SD)	5	MIS	455

10	37	0.15	Blood Plasma	lgK:4.2(0.05SD)	4	ENS	94

Reaction No. 1430

SiH2O4-2 + Mg+2 + H+1 = Mg+2_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:14.20(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:13.48(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:13.35(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:13.28(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:13.24(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:13.21(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:13.19(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:13.18(0.01SD)	4	EDH	5390
9	37	0.15	Blood Plasma	lgK:12.0(0.05SD)	4	ENS	94

Reaction No. 1431

2<SiH2O4-2> + Mg+2 + 2<H+1> = Mg+2_H+1(2)_SiH2O4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:30.82(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:29.50(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:29.26(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:29.12(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:29.03(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:28.96(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:28.93(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:28.90(0.01SD)	4	EDH	5390
9	37	0.15	Blood Plasma	lgK:26.6(0.05SD)	4	ENS	94

Reaction No. 1432

SiH2O4-2 + Pu+4 + H+1 = Pu+4_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.1(3SF)	1	EES	
2	37	0.15	Blood Plasma	lgK:20.9(0.05SD)	2	ENS	94

Reaction No. 1433

SiH2O4-2 + Sr+2 = Sr+2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:3.1(2SF)	1	ESC	
2	37	0.15	Blood Plasma	lgK:2.9(0.05SD)	0	ENS	94

Reaction No. 2614

H+1_SiH2O4-2 + H+1 = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.8(0.05SD)	4	CMN	447
2	25	0	Inf. Dilution	lgK:9.77(0.05SD)	4	MGL	450
3	25	0	Inf. Dilution	lgK:9.86(3SF)	4	ENS	6405
4	25	0	Inf. Dilution	lgK:9.825(0.033SD)	4	CRV	6478
5	25	0	Inf. Dilution	lgK:9.83(3SF)	4	ENS	6496
6	25	0	Inf. Dilution	lgK:9.82(3SF)	4	CRV	8740
7	25	0	Inf. Dilution	lgK:9.810(0.020SD)	7	CRV	14923
8	25	0	Inf. Dilution	lgK:9.810(0.01SD)	5	CRV	27292
9	25	0	Inf. Dilution	lgK:9.68(0.14SD)	4	EOD	30201
10	25	0	Inf. Dilution	lgK:9.80(3SF)	4	CRV	32224
11	25	0	CHEMVAL1	lgK:9.77(0.01SD)	0	ENS	5156
12	25	0	CHEMVAL2	lgK:9.77(0.01SD)	0	ENS	5156
13	25	0	CHEMVAL3	lgK:9.77(0.01SD)	0	ENS	5156
14	25	0	CHEMVAL4	lgK:9.77(0.01SD)	4	ENS	5156
15	25	0	Na(Cl)	lgK:9.84(3SF)	6	MHY	10238
16	25	0	Unknown	lgK:9.929(4SF)	3	ENS	7373
17	25	0.05	Na(Cl)	lgK:9.654(0.005MD)	6	MHY	10238
18	25	0.1	Na(Cl)	lgK:9.587(0.008MD)	6	MHY	10238
19	25	0.1	Unknown	lgK:9.58(0.01SD)	6	CRV	552
20	25	0.2	Na(Cl)	lgK:9.547(0.005MD)	6	MHY	10238
21	25	0.4	Na(Cl)	lgK:9.495(0.011MD)	6	MHY	10238
22	25	0.5	NaClO4	lgK:9.5(0.05SD)	5	MIS	430
23	25	0.5	Unknown	lgK:9.61(3SF)	5	ENS	6405
Note (23): p. 161							
24	25	0.6	Na(Cl)	lgK:9.472(0.002MD)	6	MHY	10238
25	25	0.6	NaCl	lgK:9.5(0.05SD)	5	MIS	438
26	25	0.6	NaCl	lgK:9.473(4SF)	5	CRV	30909
27	25	1	NaClO4	lgK:9.5(0.05SD)	5	MIS	449
28	25	2	Na(Cl)	lgK:9.460(0.008MD)	4	MHY	10238
29	25	2	NaCl	lgK:9.46(3SF)	3	CRV	552

30	25	3	NaCl	lgK:9.46(3SF)	0	CRV	552
31	25	3	NaClO4	lgK:9.4(0.05SD)	2	MIS	430
32	25	4.5	NaCl	lgK:9.54(3SF)	0	CRV	552
33	35	0	Inf. Dilution	lgK:9.70(0.05SD)	4	MGL	450
34	37	0.15	Blood Plasma	lgK:9.6(2SF)	0	ENS	1086
35	50	0	Inf. Dilution	lgK:9.4(0.05SD)	4	CMN	447
36	50	0	Inf. Dilution	lgK:9.34(0.16SD)	4	EOD	30201
37	75	0	Inf. Dilution	lgK:9.10(0.17SD)	4	EOD	30201
38	90	0	Inf. Dilution	lgK:9.18(3SF)	4	CRV	8740
39	100	0	Inf. Dilution	lgK:8.94(0.16SD)	4	EOD	30201
40	100	0	Inf. Dilution/m	lgK:9.10(3SF)	5	MPT	4914
41	100	0.1	NaCl/m	lgK:8.80(3SF)	5	MPT	4914
42	100	1	NaCl/m	lgK:8.52(3SF)	4	MPT	4914
43	100	3	NaCl/m	lgK:8.53(3SF)	5	MPT	4914
44	100	5	NaCl/m	lgK:8.64(3SF)	5	MPT	4914
45	125	0	Inf. Dilution/m	lgK:8.98(3SF)	5	MPT	4914
46	125	0.1	NaCl/m	lgK:8.66(3SF)	5	MPT	4914
47	125	1	NaCl/m	lgK:8.35(3SF)	5	MPT	4914
48	125	3	NaCl/m	lgK:8.32(3SF)	5	MPT	4914
49	125	5	NaCl/m	lgK:8.40(3SF)	4	MPT	4914
50	130	0	Inf. Dilution/m	lgK:8.88(3SF)	5	MPT	4914
51	150	0	Inf. Dilution/m	lgK:8.90(3SF)	5	MPT	4914
52	150	0.1	NaCl/m	lgK:8.55(3SF)	5	MPT	4914
53	150	1	NaCl/m	lgK:8.20(3SF)	5	MPT	4914
54	150	3	NaCl/m	lgK:8.13(3SF)	5	MPT	4914
55	150	5	NaCl/m	lgK:8.18(3SF)	4	MPT	4914
56	175	0	Inf. Dilution/m	lgK:8.85(3SF)	5	MPT	4914
57	175	0.1	NaCl/m	lgK:8.48(3SF)	5	MPT	4914
58	175	1	NaCl/m	lgK:8.09(3SF)	5	MPT	4914
59	175	3	NaCl/m	lgK:7.97(3SF)	5	MPT	4914
60	175	5	NaCl/m	lgK:7.97(3SF)	4	MPT	4914
61	200	0	Inf. Dilution/m	lgK:8.85(3SF)	5	MPT	4914
62	200	0.1	NaCl/m	lgK:8.45(3SF)	5	MPT	4914
63	200	1	NaCl/m	lgK:8.00(3SF)	5	MPT	4914
64	200	3	NaCl/m	lgK:7.82(3SF)	5	MPT	4914
65	200	5	NaCl/m	lgK:7.77(3SF)	5	MPT	4914

66	225	0	Inf. Dilution/m	lgK:8.89(3SF)	5	MPT	4914
67	225	0.1	NaCl/m	lgK:8.44(3SF)	4	MPT	4914
68	225	1	NaCl/m	lgK:7.94(3SF)	5	MPT	4914
69	225	3	NaCl/m	lgK:7.68(3SF)	5	MPT	4914
70	225	5	NaCl/m	lgK:7.58(3SF)	5	MPT	4914
71	250	0	Inf. Dilution/m	lgK:8.96(3SF)	5	MPT	4914
72	250	0.1	NaCl/m	lgK:8.47(3SF)	3	MPT	4914
73	250	1	NaCl/m	lgK:7.88(3SF)	5	MPT	4914
74	250	3	NaCl/m	lgK:7.55(3SF)	5	MPT	4914
75	250	5	NaCl/m	lgK:7.39(3SF)	5	MPT	4914
76	275	0	Inf. Dilution/m	lgK:9.07(3SF)	5	MPT	4914
77	275	0.1	NaCl/m	lgK:8.51(3SF)	4	MPT	4914
78	275	1	NaCl/m	lgK:7.84(3SF)	5	MPT	4914
79	275	3	NaCl/m	lgK:7.41(3SF)	3	MPT	4914
80	275	5	NaCl/m	lgK:7.17(3SF)	3	MPT	4914
81	300	0	Inf. Dilution/m	lgK:9.22(3SF)	4	MPT	4914
82	300	0.1	NaCl/m	lgK:8.57(3SF)	4	MPT	4914
83	300	1	NaCl/m	lgK:7.78(3SF)	0	MPT	4914
84	300	3	NaCl/m	lgK:7.24(3SF)	0	MPT	4914
85	300	5	NaCl/m	lgK:6.92(3SF)	0	MPT	4914
86	25	0	Inf. Dilution	dG:-55.996(0.06SD)kJ	5	CRV	27292
87	20:50	0.5	Unknown	dH:-5.(1SF)	3	CRV	552
88	25	0	Inf. Dilution	dH:-6.12(0.3SD)	4	CRV	6478
89	25	0	Inf. Dilution	dH:-6.12(3SF)	4	ENS	6496
90	25	0	Inf. Dilution	dH:-25.600(2.000SD)kJ	6	CRV	14923
91	25	0	Inf. Dilution	dH:-25.600(1.0SD)kJ	5	CRV	27292
92	25	0	Unknown	dH:-8.936(4SF)	0	ENS	7373

Note (92): Low wgt for internal consistency

Reaction No. 2615

4<H+1(2)_SiH2O4-2> = H+1(4)_SiH2O4-2(4) + 4<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-36.7(3SF)	1	EOR	
2	25	3	NaClO4	lgK:-32.5(0.05SD)	3	MIS	430

Reaction No. 2616

2<H+1(2)_SiH2O4-2> = H+1(2)_SiH2O4-2(2) + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-18.0(0.06SD)	5	MGC	5521
2	25	0.6	NaCl	lgK:-18.00(4SF)	3	CRV	30909
Note (2): Corrected misprint in source							
3	25	3	NaClO4	lgK:-18.1(0.05SD)	5	MIS	430

Reaction No. 2617

4<H+1(2)_SiH2O4-2> = H+1(6)_SiH2O4-2(4) + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.6(3SF)	1	ELC	
2	25	0.5	NaClO4	lgK:-12.6(0.05SD)	0	MIS	430
3	50	0.5	NaClO4	lgK:-12.6(0.05SD)	3	MIS	430

Reaction No. 2618

2<H+1> + 4<H+1_SiH2O4-2> = H+1(6)_SiH2O4-2(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.7(0.05SD)	5	CMN	447
2	50	0	Inf. Dilution	lgK:24.5(0.05SD)	5	CMN	447

Reaction No. 2619

SiO2(am.,s) + 2<H2O> = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0	Inf. Dilution	lgK:-2.947(4SF)	0	CRV	25435
2	25	0	Inf. Dilution	lgK:-2.74(3SF)	5	CRV	403
3	25	0	Inf. Dilution	lgK:-2.89(3SF)	5	CRV	403
4	25	0	Inf. Dilution	lgK:-2.7(0.05SD)	4	ENS	557
5	25	0	Inf. Dilution	lgK:-2.71(3SF)	5	CRV	6405
Note (5): p. 245							
6	25	0	Inf. Dilution	lgK:-2.71(3SF)	4	ENS	6496
7	25	0	Inf. Dilution	lgK:-2.74(3SF)	5	CRV	8740
8	25	0	Inf. Dilution	lgK:-2.714(4SF)	0	CRV	25435
Note (8): All log Ks in source should be ln Ks							
9	25	0	Inf. Dilution	lgK:-2.714(4SF)	4	EOD	29487
10	25	0	Inf. Dilution	lgK:-2.71(3SF)	4	CRV	32224
11	25	0	CHEMVAL2	lgK:-2.74(0.01SD)	0	ENS	5156
12	25	0	CHEMVAL3	lgK:-2.74(0.01SD)	5	ENS	5156

13	25	0	CHEMVAL4	lgK:-1.46(0.01SD)	0	ENS	5156
14	25	0	KCl/m	lgK:-2.654(4SF)	5	MSL	8747
15	25	0	KNO3/m	lgK:-2.658(4SF)	5	MSL	8747
16	25	0	LiCl/m	lgK:-2.674(4SF)	5	MSL	8747
17	25	0	LiNO3/m	lgK:-2.676(4SF)	5	MSL	8747
18	25	0	NaCl/m	lgK:-2.640(4SF)	5	MSL	8747
19	25	0	NaNO3/m	lgK:-2.662(4SF)	0	MSL	8747
20	25	0.28	KNO3/m	lgK:-2.693(4SF)	7	MSL	8747
21	25	0.35	KCl/m	lgK:-2.701(4SF)	7	MSL	8747
22	25	0.41	LiNO3/m	lgK:-2.738(4SF)	7	MSL	8747
23	25	0.51	LiCl/m	lgK:-2.742(4SF)	7	MSL	8747
24	25	0.51	NaCl/m	lgK:-2.708(4SF)	7	MSL	8747
25	25	0.51	NaNO3/m	lgK:-2.730(4SF)	8	MSL	8747
26	25	0.56	KNO3/m	lgK:-2.699(4SF)	7	MSL	8747
27	25	0.72	KCl/m	lgK:-2.714(4SF)	7	MSL	8747
28	25	0.82	LiNO3/m	lgK:-2.796(4SF)	7	MSL	8747
29	25	0.85	KNO3/m	lgK:-2.712(4SF)	7	MSL	8747
30	25	1	NaClO4	lgK:-2.9(0.05SD)	3	MSL	454
31	25	1.02	LiCl/m	lgK:-2.818(4SF)	7	MSL	8747
32	25	1.02	NaCl/m	lgK:-2.754(4SF)	7	MSL	8747
33	25	1.03	NaNO3/m	lgK:-2.767(4SF)	8	MSL	8747
34	25	1.09	KCl/m	lgK:-2.726(4SF)	7	MSL	8747
35	25	1.15	KNO3/m	lgK:-2.708(4SF)	7	MSL	8747
36	25	1.25	LiNO3/m	lgK:-2.848(4SF)	7	MSL	8747
37	25	1.46	KCl/m	lgK:-2.726(4SF)	7	MSL	8747
38	25	1.47	KNO3/m	lgK:-2.726(4SF)	7	MSL	8747
39	25	1.55	LiCl/m	lgK:-2.876(4SF)	3	MSL	8747
40	25	1.55	NaCl/m	lgK:-2.796(4SF)	7	MSL	8747
41	25	1.57	NaNO3/m	lgK:-2.807(4SF)	8	MSL	8747
42	25	1.68	LiNO3/m	lgK:-2.907(4SF)	4	MSL	8747
43	25	1.77	KNO3/m	lgK:-2.724(4SF)	7	MSL	8747
44	25	1.85	KCl/m	lgK:-2.730(4SF)	7	MSL	8747
45	25	2.09	LiCl/m	lgK:-2.959(4SF)	4	MSL	8747
46	25	2.09	NaCl/m	lgK:-2.842(4SF)	7	MSL	8747
47	25	2.1	KNO3/m	lgK:-2.738(4SF)	5	MSL	8747
48	25	2.12	NaNO3/m	lgK:-2.845(4SF)	8	MSL	8747

49	25	2.13	LiNO3/m	lgK:-2.963(4SF)	4	MSL	8747
50	25	2.25	KCl/m	lgK:-2.742(4SF)	5	MSL	8747
51	25	2.43	KNO3/m	lgK:-2.742(4SF)	5	MSL	8747
52	25	2.59	LiNO3/m	lgK:-3.029(4SF)	3	MSL	8747
53	25	2.63	LiCl/m	lgK:-3.030(4SF)	3	MSL	8747
54	25	2.64	NaCl/m	lgK:-2.886(4SF)	7	MSL	8747
55	25	2.66	KCl/m	lgK:-2.752(4SF)	5	MSL	8747
56	25	2.76	NaNO3/m	lgK:-2.900(4SF)	6	MSL	8747
57	25	2.77	KNO3/m	lgK:-2.752(4SF)	5	MSL	8747
58	25	3.07	KCl/m	lgK:-2.767(4SF)	3	MSL	8747
59	25	3.12	KNO3/m	lgK:-2.759(4SF)	3	MSL	8747
60	25	3.2	LiCl/m	lgK:-3.100(4SF)	3	MSL	8747
61	25	3.2	NaCl/m	lgK:-2.928(4SF)	7	MSL	8747
62	25	3.36	NaNO3/m	lgK:-2.932(4SF)	8	MSL	8747
63	25	3.5	KCl/m	lgK:-2.772(4SF)	3	MSL	8747
64	25	3.55	LiNO3/m	lgK:-3.137(4SF)	2	MSL	8747
65	25	3.76	KNO3/m	lgK:-2.757(4SF)	3	MSL	8747
66	25	3.78	NaCl/m	lgK:-2.971(4SF)	7	MSL	8747
67	25	3.94	KCl/m	lgK:-2.775(4SF)	3	MSL	8747
68	25	4.36	LiCl/m	lgK:-3.245(4SF)	2	MSL	8747
69	25	4.37	NaCl/m	lgK:-3.026(4SF)	7	MSL	8747
70	25	4.57	LiNO3/m	lgK:-3.273(4SF)	2	MSL	8747
71	25	4.61	NaNO3/m	lgK:-2.979(4SF)	8	MSL	8747
72	25	4.81	KCl/m	lgK:-2.790(4SF)	3	MSL	8747
73	25	4.98	NaCl/m	lgK:-3.071(4SF)	5	MSL	8747
74	25	5.61	NaCl/m	lgK:-3.116(4SF)	7	MSL	8747
75	25	6.12	NaNO3/m	lgK:-3.068(4SF)	8	MSL	8747
76	25	6.14	NaCl/m	lgK:-3.170(4SF)	7	MSL	8747
77	25	6.79	LiNO3/m	lgK:-3.491(4SF)	7	MSL	8747
78	25	6.86	LiCl/m	lgK:-3.569(4SF)	7	MSL	8747
79	25	8.01	LiNO3/m	lgK:-3.604(4SF)	7	MSL	8747
80	25	8.2	LiCl/m	lgK:-3.693(4SF)	7	MSL	8747
81	25	9.29	LiNO3/m	lgK:-3.706(4SF)	7	MSL	8747
82	25	9.62	LiCl/m	lgK:-3.889(4SF)	7	MSL	8747
83	25	10.65	LiNO3/m	lgK:-3.818(4SF)	7	MSL	8747
84	25	11.11	LiCl/m	lgK:-3.955(4SF)	7	MSL	8747

85	25	12.08	LiNO3/m	lgK:-3.928(4SF)	7	MSL	8747
86	25	12.68	LiCl/m	lgK:-4.071(4SF)	7	MSL	8747
87	25	19.19	LiCl/m	lgK:-4.237(4SF)	7	MSL	8747
88	35	0	Inf. Dilution	lgK:-2.64(3SF)	5	CRV	403
89	60	0	Inf. Dilution	lgK:-2.445(4SF)	0	CRV	25435
90	90	0	Inf. Dilution	lgK:-2.26(3SF)	4	EOD	29487
91	100	0	Inf. Dilution	lgK:-2.202(4SF)	0	CRV	25435
92	100	0	NaNO3/m	lgK:-2.167(4SF)	5	MSL	8747
93	100	0.51	NaNO3/m	lgK:-2.153(4SF)	6	MSL	8747
94	100	1.03	NaNO3/m	lgK:-2.248(4SF)	8	MSL	8747
95	100	2.12	NaNO3/m	lgK:-2.299(4SF)	8	MSL	8747
96	100	5.98	NaNO3/m	lgK:-2.463(4SF)	8	MSL	8747
97	150	0	Inf. Dilution	lgK:-1.971(4SF)	0	CRV	25435
98	150	0	NaNO3/m	lgK:-1.969(4SF)	5	MSL	8747
99	150	0.51	NaNO3/m	lgK:-2.008(4SF)	8	MSL	8747
100	150	1.03	NaNO3/m	lgK:-2.011(4SF)	8	MSL	8747
101	150	2.12	NaNO3/m	lgK:-2.109(4SF)	8	MSL	8747
102	150	3.31	NaNO3/m	lgK:-2.140(4SF)	8	MSL	8747
103	150	4.6	NaNO3/m	lgK:-2.203(4SF)	8	MSL	8747
104	150	5.98	NaNO3/m	lgK:-2.244(4SF)	8	MSL	8747
105	200	0	Inf. Dilution	lgK:-1.802(4SF)	0	CRV	25435
106	200	0	NaNO3/m	lgK:-1.801(4SF)	5	MSL	8747
107	200	0.507	NaNO3/m	lgK:-1.860(4SF)	8	MSL	8747
108	200	1.027	NaNO3/m	lgK:-1.854(4SF)	8	MSL	8747
109	200	2.124	NaNO3/m	lgK:-1.903(4SF)	8	MSL	8747
110	200	3.309	NaNO3/m	lgK:-1.943(4SF)	8	MSL	8747
111	200	4.6	NaNO3/m	lgK:-1.987(4SF)	8	MSL	8747
112	200	5.98	NaNO3/m	lgK:-2.036(4SF)	8	MSL	8747
113	250	0	Inf. Dilution	lgK:-1.684(4SF)	0	CRV	25435
114	250	0	NaNO3/m	lgK:-1.674(4SF)	5	MSL	8747
115	250	0.507	NaNO3/m	lgK:-1.690(4SF)	8	MSL	8747
116	250	1.027	NaNO3/m	lgK:-1.706(4SF)	8	MSL	8747
117	250	2.124	NaNO3/m	lgK:-1.740(4SF)	8	MSL	8747
118	250	3.309	NaNO3/m	lgK:-1.762(4SF)	8	MSL	8747
119	250	4.6	NaNO3/m	lgK:-1.812(4SF)	8	MSL	8747
120	250	5.98	NaNO3/m	lgK:-1.833(4SF)	8	MSL	8747

121	300	0	Inf. Dilution	lgK:-1.605(4SF)	0	CRV	25435
122	300	0	NaNO3/m	lgK:-1.570(4SF)	5	MSL	8747
123	300	0.507	NaNO3/m	lgK:-1.580(4SF)	8	MSL	8747
124	300	1.027	NaNO3/m	lgK:-1.600(4SF)	8	MSL	8747
125	300	2.124	NaNO3/m	lgK:-1.600(4SF)	8	MSL	8747
126	300	3.309	NaNO3/m	lgK:-1.636(4SF)	8	MSL	8747
127	300	5.98	NaNO3/m	lgK:-1.708(4SF)	8	MSL	8747
128	25	0	Inf. Dilution	dH:3.34(3SF)	4	ENS	6496
129	25	0	Inf. Dilution	dH:14.594(5SF)kJ	4	EOD	29487

Reaction No. 2796

Ni+2 + H+1_SiH2O4-2 = Ni+2_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.2	NaClO4	lgK:5.73(0.03SD)	3	MSE	29656
2	25	1	NaClO4	lgK:5.34(0.03SD)	3	MSE	29656

Reaction No. 3025

H+1(2)_SiH2O4-2 + Mn+2 = MnO.SiO2(s) + 2<H+1> + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.3(0.05SD)	3	ENS	557
2	25	0	Inf. Dilution	lgK:-9.72(3SF)	3	CRV	32224

Reaction No. 9803

H+1(2)_SiH2O4-2 + 3<H+1_Tropolone-1> + H+1 = Si+4_Tropolone-1(3) + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:7.1(0.03SD)	4	MGL	1134

Reaction No. 17164

H+1(2)_SiH2O4-2 - 2<H+1> + 3<H+1(2)_Cat-2> = Si+4_Cat-2(3) + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-12.0(3SF)	5	MSL	8740
2	25	0	Inf. Dilution	lgK:-11.5(3SF)	4	EOD	8740
3	25	0	Unknown	lgK:-11.32(4SF)	3	ENS	8687
4	25	0.1	Unknown	lgK:-10.89(4SF)	3	ENS	8687
5	25	0.1	Unknown	lgK:-12.42(4SF)	0	MMR	8688
Note (5): Paper argues against other values but unconvincingly (NMR,SiO2 sat)							
6	25	0.6	NaCl	lgK:-10.44(0.03MD)	6	MPT	1364

Reaction No. 26572

H+1(2)_SiH2O4-2 + 3<H+1(3)_Dopamine-2> = H+1(9)_Dopamine-2(3)_SiH2O4-2 + 2<H+1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-9.70(0.03MD)	6	MGL	2279

Reaction No. 26573

H+1(2)_SiH2O4-2 + 3<H+1(3)_LDopa-3> = H+1(9)_SiH2O4-2_LDopa-3(3) + 2<H+1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-10.08(0.05MD)	6	MGL	2279

Reaction No. 26574

H+1(2)_SiH2O4-2 + 3<H+1(3)_Dopamine-2> = H+1(8)_Dopamine-2(3)_SiH2O4-2 + 3<H+1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-19.33(0.03MD)	6	MGL	2279

Reaction No. 26575

H+1(2)_SiH2O4-2 + 3<H+1(3)_LDopa-3> = H+1(8)_SiH2O4-2_LDopa-3(3) + 3<H+1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-19.35(0.07MD)	6	MGL	2279

Reaction No. 29643

H+1(2)_SiH2O4-2 + 3<H+1(3)_Pyrogallol-3> = Si+4_H+1(3)_Pyrogallol-3(3) + 2<H+1> + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-10.02(0.01SD)	6	MMR	1845

Reaction No. 29644

H+1(2)_SiH2O4-2 + 3<H+1(3)_34DHB-3> = Si+4_34DHB-3(3) + 5<H+1> + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-21.9(0.05SD)	6	MMR	1845

Reaction No. 29645

H+1(2)_SiH2O4-2 + 3<H+1(4)_Gallic-4> = Si+4_H+1(3)_Gallic-4(3) + 5<H+1> + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-21.3(0.03SD)	6	MMR	1845

Reaction No. 47006

H+1 + Ca+2 + Citric-3 + SiH2O4-2 = Ca+2_H+1_SiH2O4-2_Citric-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.8(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:15.15(0.01SD)	3	ETS	94

Reaction No. 47007

H+1 + Ca+2 + Lactic-1 + SiH2O4-2 = Ca+2_H+1_Lactic-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.15(0.01SD)	3	ETS	94

Reaction No. 47008

H+1 + Ca+2 + Malic-2 + SiH2O4-2 = Ca+2_H+1_Malic-2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.8(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.35(0.01SD)	3	ETS	94

Reaction No. 47009

H+1 + Ca+2 + Oxalic-2 + SiH2O4-2 = Ca+2_H+1_Oxalic-2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.8(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.35(0.01SD)	3	ETS	94

Reaction No. 47010

H+1 + Ca+2 + Succinic-2 + SiH2O4-2 = Ca+2_H+1_SiH2O4-2_Succinic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.7(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.30(0.01SD)	3	ETS	94

Reaction No. 47011

2<H+1> + Ca+2 + Ala-1 + SiH2O4-2 = Ca+2_H+1(2)_Ala-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.10(0.01SD)	3	ETS	94

Reaction No. 47012

2<H+1> + Ca+2 + bAla-1 + SiH2O4-2 = Ca+2_H+1(2)_bAla-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.4(3SF)	1	EAE	
2	37	0.15	Blood Plasma	lgK:22.75(0.01SD)	3	ETS	94

Reaction No. 47013

2<H+1> + Ca+2 + Arg-1 + SiH2O4-2 = Ca+2_H+1(2)_Arg-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.7(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.55(0.01SD)	3	ETS	94

Reaction No. 47014

2<H+1> + Ca+2 + Asn-1 + SiH2O4-2 = Ca+2_H+1(2)_Asn-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.5(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.35(0.01SD)	3	ETS	94

Reaction No. 47015

2<H+1> + Ca+2 + Asp-2 + SiH2O4-2 = Ca+2_H+1(2)_Asp-2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.7(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.00(0.01SD)	3	ETS	94

Reaction No. 47016

2<H+1> + Ca+2 + Citrul-1 + SiH2O4-2 = Ca+2_H+1(2)_Citrul-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.5(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.29(0.01SD)	3	ETS	94

Reaction No. 47017

2<H+1> + Ca+2 + Cis-2 + SiH2O4-2 = Ca+2_H+1(2)_Cis-2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.6(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.90(0.01SD)	3	ETS	94

Reaction No. 47018

2<H+1> + Ca+2 + Glu-2 + SiH2O4-2 = Ca+2_H+1(2)_Glu-2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:24.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.20(0.01SD)	3	ETS	94

Reaction No. 47019

2<H+1> + Ca+2 + Gln-1 + SiH2O4-2 = Ca+2_H+1(2)_Gln-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.5(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.35(0.01SD)	3	ETS	94

Reaction No. 47020

2<H+1> + Ca+2 + Gly-1 + SiH2O4-2 = Ca+2_H+1(2)_Gly-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.2(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.05(0.01SD)	3	ETS	94

Reaction No. 47021

2<H+1> + Ca+2 + His-1 + SiH2O4-2 = Ca+2_H+1(2)_His-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.8(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.60(0.01SD)	3	ETS	94

Reaction No. 47022

2<H+1> + Ca+2 + Hyp-1 + SiH2O4-2 = Ca+2_H+1(2)_Hyp-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.0(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.85(0.01SD)	3	ETS	94

Reaction No. 47023

2<H+1> + Ca+2 + Ile-1 + SiH2O4-2 = Ca+2_H+1(2)_Ile-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.1(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.94(0.01SD)	3	ETS	94

Reaction No. 47024

2<H+1> + Ca+2 + Leu-1 + SiH2O4-2 = Ca+2_H+1(2)_Leu-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.1(3SF)	2	EAE	

2	37	0.15	Blood Plasma	lgK:22.94(0.01SD)	3	ETS	94

Reaction No. 47025

2<H+1> + Ca+2 + Met-1 + SiH2O4-2 = Ca+2_H+1(2)_Met-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.7(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.50(0.01SD)	3	ETS	94

Reaction No. 47026

2<H+1> + Ca+2 + Phe-1 + SiH2O4-2 = Ca+2_H+1(2)_Phe-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.6(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.44(0.01SD)	3	ETS	94

Reaction No. 47027

2<H+1> + Ca+2 + Pro-1 + SiH2O4-2 = Ca+2_H+1(2)_Pro-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.2(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:24.05(0.01SD)	3	ETS	94

Reaction No. 47028

2<H+1> + Ca+2 + Ser-1 + SiH2O4-2 = Ca+2_H+1(2)_Ser-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.6(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.44(0.01SD)	3	ETS	94

Reaction No. 47029

2<H+1> + Ca+2 + Thr-1 + SiH2O4-2 = Ca+2_H+1(2)_Thr-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.6(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.40(0.01SD)	3	ETS	94

Reaction No. 47030

2<H+1> + Ca+2 + Trp-1 + SiH2O4-2 = Ca+2_H+1(2)_Trp-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.70(0.01SD)	3	ETS	94

Reaction No. 47031

2<H+1> + Ca+2 + Val-1 + SiH2O4-2 = Ca+2_H+1(2)_Val-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.0(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.85(0.01SD)	3	ETS	94

Reaction No. 47602

H+1 + Mg+2 + Citric-3 + SiH2O4-2 = Mg+2_H+1_SiH2O4-2_Citric-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:15.66(0.01SD)	3	ETS	94

Reaction No. 47603

H+1 + Mg+2 + Lactic-1 + SiH2O4-2 = Mg+2_H+1_Lactic-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.7(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.51(0.01SD)	3	ETS	94

Reaction No. 47604

H+1 + Mg+2 + Malic-2 + SiH2O4-2 = Mg+2_H+1_Malic-2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.0(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.61(0.01SD)	3	ETS	94

Reaction No. 47605

H+1 + Mg+2 + Oxalic-2 + SiH2O4-2 = Mg+2_H+1_Oxalic-2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:15.51(0.01SD)	3	ETS	94

Reaction No. 47606

H+1 + Mg+2 + Succinic-2 + SiH2O4-2 = Mg+2_H+1_SiH2O4-2_Succinic-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:16.1(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:14.66(0.01SD)	3	ETS	94

Reaction No. 47607

2<H+1> + Mg+2 + Ala-1 + SiH2O4-2 = Mg+2_H+1(2)_Ala-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.71(0.01SD)	3	ETS	94

Reaction No. 47608

2<H+1> + Mg+2 + bAla-1 + SiH2O4-2 = Mg+2_H+1(2)_bAla-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.0(3SF)	1	EAE	
2	37	0.15	Blood Plasma	lgK:23.36(0.01SD)	3	ETS	94

Reaction No. 47609

2<H+1> + Mg+2 + Arg-1 + SiH2O4-2 = Mg+2_H+1(2)_Arg-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.11(0.01SD)	3	ETS	94

Reaction No. 47610

2<H+1> + Mg+2 + Asn-1 + SiH2O4-2 = Mg+2_H+1(2)_Asn-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.2(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:22.96(0.01SD)	3	ETS	94

Reaction No. 47611

2<H+1> + Mg+2 + Asp-2 + SiH2O4-2 = Mg+2_H+1(2)_Asp-2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.66(0.01SD)	3	ETS	94

Reaction No. 47612

2<H+1> + Mg+2 + Citrul-1 + SiH2O4-2 = Mg+2_H+1(2)_Citrul-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.2(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.01(0.01SD)	3	ETS	94

Reaction No. 47613

2<H+1> + Mg+2 + Cis-2 + SiH2O4-2 = Mg+2_H+1(2)_Cis-2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.61(0.01SD)	3	ETS	94

Reaction No. 47614

2<H+1> + Mg+2 + Glu-2 + SiH2O4-2 = Mg+2_H+1(2)_Glu-2_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.4(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.76(0.01SD)	3	ETS	94

Reaction No. 47615

2<H+1> + Mg+2 + Gln-1 + SiH2O4-2 = Mg+2_H+1(2)_Gln-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.06(0.01SD)	3	ETS	94

Reaction No. 47616

2<H+1> + Mg+2 + Gly-1 + SiH2O4-2 = Mg+2_H+1(2)_Gly-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.0(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.76(0.01SD)	3	ETS	94

Reaction No. 47617

2<H+1> + Mg+2 + His-1 + SiH2O4-2 = Mg+2_H+1(2)_His-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.5(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.26(0.01SD)	3	ETS	94

Reaction No. 47618

2<H+1> + Mg+2 + Hyp-1 + SiH2O4-2 = Mg+2_H+1(2)_Hyp-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.7(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.51(0.01SD)	3	ETS	94

Reaction No. 47619

2<H+1> + Mg+2 + Ile-1 + SiH2O4-2 = Mg+2_H+1(2)_Ile-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:24.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.66(0.01SD)	3	ETS	94

Reaction No. 47620

2<H+1> + Mg+2 + Leu-1 + SiH2O4-2 = Mg+2_H+1(2)_Leu-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.66(0.01SD)	3	ETS	94

Reaction No. 47621

2<H+1> + Mg+2 + Met-1 + SiH2O4-2 = Mg+2_H+1(2)_Met-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.4(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.16(0.01SD)	3	ETS	94

Reaction No. 47622

2<H+1> + Mg+2 + Phe-1 + SiH2O4-2 = Mg+2_H+1(2)_Phe-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.11(0.01SD)	3	ETS	94

Reaction No. 47623

2<H+1> + Mg+2 + Pro-1 + SiH2O4-2 = Mg+2_H+1(2)_Pro-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.9(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:24.71(0.01SD)	3	ETS	94

Reaction No. 47624

2<H+1> + Mg+2 + Ser-1 + SiH2O4-2 = Mg+2_H+1(2)_Ser-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.3(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.06(0.01SD)	3	ETS	94

Reaction No. 47625

2<H+1> + Mg+2 + Thr-1 + SiH2O4-2 = Mg+2_H+1(2)_Thr-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.2(3SF)	2	EAE	

2	37	0.15	Blood Plasma	lgK:22.96(0.01SD)	3	ETS	94

Reaction No. 47626

2<H+1> + Mg+2 + Trp-1 + SiH2O4-2 = Mg+2_H+1(2)_Trp-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.6(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.36(0.01SD)	3	ETS	94

Reaction No. 47627

2<H+1> + Mg+2 + Val-1 + SiH2O4-2 = Mg+2_H+1(2)_Val-1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:24.7(3SF)	2	EAE	
2	37	0.15	Blood Plasma	lgK:23.51(0.01SD)	3	ETS	94

Reaction No. 49887

CaO.Al2O3.2SiO2(s) + 8<H2O> = Ca+2 + 2<Al+3_OH-1(4)> + 2<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-21.071(5SF)	0	CRV	8798
2	25	0	Inf. Dilution	lgK:-19.1(3SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-20.484(5SF)	0	CRV	8798
4	25	0	Inf. Dilution	lgK:-54.579(5SF)	0	CRV	25435
Note (4): Given as log K, v. prob. should be ln K but -54.6/2.3 = -23.7 still bad							
5	50	0	Inf. Dilution	lgK:-19.857(5SF)	0	CRV	8798
6	100	0	Inf. Dilution	lgK:-18.960(5SF)	0	CRV	8798
7	150	0	Inf. Dilution	lgK:-18.514(5SF)	0	CRV	8798
8	200	0	Inf. Dilution	lgK:-18.485(5SF)	0	CRV	8798
9	250	0	Inf. Dilution	lgK:-18.839(5SF)	0	CRV	8798
10	300	0	Inf. Dilution	lgK:-19.547(5SF)	0	CRV	8798
11	350	0	Inf. Dilution	lgK:-20.586(5SF)	0	CRV	8798

Reaction No. 49888

KAlSi3O8(San.,s) + 8<H2O> = K+1 + Al+3_OH-1(4) + 3<H+1(2)_SiH2O4-2>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-23.764(5SF)	7	CRV	8798
2	25	0	Inf. Dilution	lgK:-22.338(5SF)	7	CRV	8798
3	50	0	Inf. Dilution	lgK:-20.774(5SF)	7	CRV	8798
4	100	0	Inf. Dilution	lgK:-18.267(5SF)	7	CRV	8798

5	150	0	Inf. Dilution	lgK:-16.441(5SF)	7	CRV	8798
6	200	0	Inf. Dilution	lgK:-15.166(5SF)	7	CRV	8798
7	250	0	Inf. Dilution	lgK:-14.351(5SF)	7	CRV	8798
8	300	0	Inf. Dilution	lgK:-13.930(5SF)	7	CRV	8798
9	350	0	Inf. Dilution	lgK:-13.855(5SF)	7	CRV	8798

Reaction No. 49901

KAlSi3O8(Micr.,s) + 8<H2O> = K+1 + Al+3_OH-1(4) + 3<H+1(2)_SiH2O4-2>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-24.446(5SF)	0	CRV	8798
2	25	0	Inf. Dilution	lgK:-21.6(3SF)	1	ELC	
3	25	0	Inf. Dilution	lgK:-22.912(5SF)	0	CRV	8798
4	50	0	Inf. Dilution	lgK:-21.231(5SF)	0	CRV	8798
5	100	0	Inf. Dilution	lgK:-18.535(5SF)	2	CRV	8798
6	150	0	Inf. Dilution	lgK:-16.562(5SF)	0	CRV	8798
7	200	0	Inf. Dilution	lgK:-15.170(5SF)	0	CRV	8798
8	250	0	Inf. Dilution	lgK:-14.260(5SF)	0	CRV	8798
9	300	0	Inf. Dilution	lgK:-13.759(5SF)	0	CRV	8798
10	350	0	Inf. Dilution	lgK:-13.616(5SF)	0	CRV	8798

Reaction No. 50248

NaAlSi3O8(AlbiteHI,s) + 8<H2O> = Na+1 + Al+3_OH-1(4) + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-19.816(5SF)	7	CRV	8798
2	25	0	Inf. Dilution	lgK:-18.779(5SF)	7	CRV	8798
3	50	0	Inf. Dilution	lgK:-17.622(5SF)	7	CRV	8798
4	100	0	Inf. Dilution	lgK:-15.737(5SF)	7	CRV	8798
5	150	0	Inf. Dilution	lgK:-14.359(5SF)	7	CRV	8798
6	200	0	Inf. Dilution	lgK:-13.417(5SF)	7	CRV	8798
7	250	0	Inf. Dilution	lgK:-12.858(5SF)	7	CRV	8798
8	300	0	Inf. Dilution	lgK:-12.640(5SF)	7	CRV	8798
9	350	0	Inf. Dilution	lgK:-12.730(5SF)	7	CRV	8798

Reaction No. 50272

NaAlSi3O8(AlbiteLO,s) + 8<H2O> = Na+1 + Al+3_OH-1(4) + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	5	0	Inf. Dilution	lgK:-21.337(5SF)	7	CRV	8798

2	25	0	Inf. Dilution	lgK:-20.179(5SF)	7	CRV	8798
3	50	0	Inf. Dilution	lgK:-18.848(5SF)	7	CRV	8798
4	100	0	Inf. Dilution	lgK:-16.686(5SF)	7	CRV	8798
5	150	0	Inf. Dilution	lgK:-15.096(5SF)	7	CRV	8798
6	200	0	Inf. Dilution	lgK:-13.987(5SF)	7	CRV	8798
7	250	0	Inf. Dilution	lgK:-13.293(5SF)	7	CRV	8798
8	300	0	Inf. Dilution	lgK:-12.963(5SF)	7	CRV	8798
9	350	0	Inf. Dilution	lgK:-12.959(5SF)	7	CRV	8798

Reaction No. 53218

Si+4_F-1(4) + H+1_F-1 = Si+4_F-1(5) + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	10			lgK:2.46(3SF)	3	MMR	5398
2	25			lgK:2.40(3SF)	3	MMR	5398
3	40			lgK:2.35(3SF)	3	MMR	5398
4	54			lgK:2.30(3SF)	3	MMR	5398
5	10:54			dH:-1.5(2SF)	3	MTD	5398

Reaction No. 53219

Si+4_F-1(5) + H+1_F-1 = Si+4_F-1(6) + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	0.4	LiClO4	lgK:2.10(3SF)	5	MKN	5398
2	10			lgK:1.92(3SF)	3	MMR	5398
3	25	0.4	LiClO4	lgK:1.89(3SF)	5	MKN	5398
4	25			lgK:1.60(3SF)	3	MMR	5398
5	40			lgK:1.33(3SF)	3	MMR	5398
6	54			lgK:1.10(3SF)	3	MMR	5398
7	10:54			dH:-7.8(2SF)	3	MTD	5398

Reaction No. 54154

Si+4_F-1(5) + F-1 = Si+4_F-1(6)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0.5	HCl	lgK:3.96(3SF)	0	MAX	931
2	25	0	Inf. Dilution	lgK:4.5(2SF)	1	ELC	

Reaction No. 54549

SiO2(Quartz,s) + 2<H2O> = H+1(2)_SiH2O4-2

Note: Quartz solubility at low temperature has been well examined by Rimstidt, GCA, 1997, 61, 2553 [9033]. His results indicate that the solubility at 25C is 10.8 ppm, about 80% higher than the widely accepted value of 6 ppm due to Fournier & Potter, GCA, 1982, 46, 1969 [9045] and Walther & Helgeson, 1979, 279, 1078 [9046]. The data are thus being modified (gradually) to relect this.

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.0(0.05SD)	0	ENS	440
Note (1): See reaction comment							
2	25	0	Inf. Dilution	lgK:-4.00(3SF)	0	CRV	6405
Note (2): p. 245							
3	25	0	Inf. Dilution	lgK:-4.0(0.1SD)	0	CRV	6478
4	25	0	Inf. Dilution	lgK:-3.98(3SF)	0	ENS	6496
5	25	0	Inf. Dilution	lgK:-3.72(3SF)	7	MSL	9033
Note (5): See reaction comment							
6	25	0	Inf. Dilution	lgK:-4.000(0.100SD)	0	CRV	14923
7	25	0	Inf. Dilution	lgK:-4.01(3SF)	0	EOD	23102
8	25	0	Inf. Dilution	lgK:-4.000(0.05SD)	0	CRV	27292
9	25	0	Inf. Dilution	lgK:-3.746(4SF)	4	EOD	29487
10	25	0	Inf. Dilution	lgK:-4.18(3SF)	0	CRV	32224
11	25	0	CHEMVAL1	lgK:-3.98(0.01SD)	0	ENS	5156
12	25	0	CHEMVAL2	lgK:-3.98(0.01SD)	0	ENS	5156
13	25	0	CHEMVAL3	lgK:-3.98(0.01SD)	0	ENS	5156
14	25	0	CHEMVAL4	lgK:-3.95(0.01SD)	0	ENS	5156
Note (14): Times are achanging							
15	90	0	Inf. Dilution	lgK:-3.10(3SF)	4	EOD	29487
16	25	0	Inf. Dilution	dG:22.36(4SF)kJ	3	CRV	6421
Note (16): Cited by Von Damm et al, Am. J. Sci., 1991, 291, 977 (p. 997) [8869]							
17	25	0	Inf. Dilution	dG:16.14(4SF)kJ	0	ENS	8869
Note (17): Successful prediction by the density model but not useful data							
18	25	0	Inf. Dilution	dG:22.832(0.3SD)kJ	0	CRV	27292
19	25	0	Inf. Dilution	dH:7.0(0.5SD)	0	CRV	6478
20	25	0	Inf. Dilution	dH:5.99(3SF)	4	ENS	6496
21	25	0	Inf. Dilution	dH:25.400(3.000SD)kJ	4	CRV	14923
22	25	0	Inf. Dilution	dH:25.400(1.5SD)kJ	2	CRV	27292
23	25	0	Inf. Dilution	dH:20.647(5SF)kJ	4	EOD	29487

Reaction No. 54550

SiO2(Cristob.,s) + 2<H2O> = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-3.35(3SF)	0	CRV	403
3	25	0	Inf. Dilution	lgK:-3.5(0.05SD)	0	ENS	440
4	25	0	Inf. Dilution	lgK:-3.45(3SF)	0	CRV	6405
Note (4): p. 245							
5	25	0	Inf. Dilution	lgK:-4.18(3SF)	0	CRV	32224
6	25	0	CHEMVAL1	lgK:-3.55(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL2	lgK:-3.55(0.01SD)	0	ENS	5156
8	25	0	CHEMVAL3	lgK:-3.55(0.01SD)	0	ENS	5156
9	25	0	CHEMVAL4	lgK:-3.52(0.01SD)	0	ENS	5156

Reaction No. 54551

Mg+2 + H+1_SiH2O4-2 = Mg+2_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.0(2SF)	1	EOR	
2	25	1	NaClO4	lgK:0.6(0.05SD)	3	MIS	455

Reaction No. 54552

Mg+2 + 2<H+1_SiH2O4-2> = Mg+2_H+1(2)_SiH2O4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.3(2SF)	1	EOR	
2	25	1	NaClO4	lgK:3.8(0.05SD)	3	MIS	455

Reaction No. 54554

Ca+2 + H+1_SiH2O4-2 = Ca+2_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.0(2SF)	2	EDH	
2	25	1	NaClO4	lgK:0.4(0.05SD)	4	MIS	455

Reaction No. 54555

Ca+2 + 2<H+1_SiH2O4-2> = Ca+2_H+1(2)_SiH2O4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:3.5(2SF)	1	EOR	
2	25	1	NaClO4	lgK:2.9(0.05SD)	2	MIS	455

Reaction No. 54556							
Ca+2_SiH2O4-2_(s) = Ca+2 + SiH2O4-2							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:-7.2(0.1SD)	4	MSL	456

Reaction No. 54557							
Ca+2 + H+1(2)_SiH2O4-2 - H2O = CaO.SiO2(s) + 2<H+1>							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:-13.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-13.6(0.05SD)	0	ENS	440
3	25	0	Inf. Dilution	lgK:-13.64(4SF)	0	CRV	6405
Note (3): p. 245							
4	25	0	Inf. Dilution	lgK:-12.99(4SF)	0	EOD	23102
5	25	0	Inf. Dilution	lgK:-13.73(4SF)	2	CRV	32224
6	25	0	CHEMVAL1	lgK:-14.61(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL2	lgK:-14.61(0.01SD)	0	ENS	5156
8	25	0	CHEMVAL3	lgK:-14.61(0.01SD)	0	ENS	5156
9	25	0	CHEMVAL4	lgK:-13.10(0.01SD)	0	ENS	5156

Reaction No. 54558							
Ca+2 + Mg+2 + 2<H+1(2)_SiH2O4-2> - 2<H2O> = CaO.MgO.2SiO2(s) + 4<H+1 >							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:-21.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-21.2(0.1SD)	0	ENS	440
3	25	0	Inf. Dilution	lgK:-21.20(4SF)	0	CRV	6405
Note (3): p. 246							
4	25	0	Inf. Dilution	lgK:-21.07(4SF)	0	CRV	32224
5	25	0	CHEMVAL1	lgK:-23.70(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL2	lgK:-23.70(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:-23.70(0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:-19.90(0.01SD)	0	ENS	5156

Reaction No. 54559							
2CaO.5MgO.8SiO2.H2O(s) + 14<H+1> = 2<Ca+2> + 5<Mg+2> + 8<H+1(2)_SiH2O4-2> - 8<H2O >							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:66.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:62.4(0.1SD)	0	ENS	440

3	25	0	Inf. Dilution	lgK:62.44(4SF)	0	CRV	6405
Note (3): p. 246							
4	25	0	Inf. Dilution	lgK:61.85(4SF)	0	CRV	32224
5	25	0	CHEMVAL1	lgK:67.61(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL2	lgK:67.61(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:67.61(0.01SD)	0	ENS	5156

Reaction No. 54560

4<H+1> + 6<F-1> + H+1(2)_SiH2O4-2 = SiF6-2 + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	1	NaCl	lgK:31.6(0.05SD)	4	MIS	439
2	25	0	Inf. Dilution	lgK:28.7(3SF)	1	EOR	
3	25	0	CHEMVAL1	lgK:30.18(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:30.18(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:30.18(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:27.80(0.01SD)	0	ENS	5156
Note (6): Low wgt for internal consistency							
7	25	0.1	Na+1 salt	lgK:30.2(0.1SD)	0	MIS	453
8	25	1	NaCl	lgK:30.0(0.05SD)	4	MIS	439
9	60	1	NaCl	lgK:28.2(0.05SD)	4	MIS	439
10	25	0	Inf. Dilution	dH:-16.3(0.1SD)	2	ENS	766

Reaction No. 54561

2<Mg+2> + 4<H2O> + 2<H+1> + 3<SiH2O4-2> = Mg+2(2)_H+1(6)_OH-1(4)_SiH2O4-2(3)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-40.(0.3SD)	3	MSL	441
2	51	0	Inf. Dilution	lgK:-39.(0.3SD)	3	MSL	441
3	70	0	Inf. Dilution	lgK:-38.(0.3SD)	3	MSL	441

Reaction No. 54562

2<Fe+2> + H+1(2)_SiH2O4-2 = 2FeO.SiO2(s) + 4<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-19.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-16.8(0.05SD)	0	ENS	440
3	25	0	Inf. Dilution	lgK:-16.81(4SF)	0	CRV	6405
Note (3): p. 245							
4	25	0	Inf. Dilution	lgK:14.67(4SF)	0	CRV	32224

5	25	0	CHEMVAL1	lgK:-16.71(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL2	lgK:-16.71(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:-16.71(0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:-16.70(0.01SD)	0	ENS	5156

Reaction No. 54563

H+1(2)_SiH2O4-2 + 2<H+1> + 2<F-1> = Si(OH)2F2 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	NaCl	lgK:11.1(0.05SD)	5	MIS	439

Reaction No. 54564

H+1(2)_SiH2O4-2 + 3<H+1> + 4<F-1> = SiOHF4-1 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	0	1	NaCl	lgK:21.4(0.05SD)	5	MIS	439
2	25	0	Inf. Dilution	lgK:22.2(3SF)	2	EAE	

Reaction No. 54594

2<Mn+2> + H+1(2)_SiH2O4-2 = 2MnO.SiO2(s) + 4<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-23.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-24.5(0.05SD)	0	ENS	557
3	25	0	Inf. Dilution	lgK:-23.47(4SF)	4	CRV	32224

Reaction No. 54682

Si(s) + 4<e-1> + 4<H+1> = SiH4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.8(0.05SD)	0	ENS	140
2	25	0	Inf. Dilution	E:-0.147(3SF)	5	CRV	7761
3	25	0	Inf. Dilution	E:-0.143(3SF)	4	CRV	22551
4	25	0	Inf. Dilution	dH:34.2(3SF)kJ	5	CRV	7761

Reaction No. 54838

2<Al(s)> + 2<H2(g)> + 4.5<O2(g)> + 2<Si(s)> = Al2Si2O5(OH)4(Hall.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:662.3(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:657.9(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:479.3(4SF)	3	ENS	6422

4	227	0	Inf. Dilution	lgK:372.2(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:300.8(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-903.(0.4SD)	2	ENS	2171
7	25	0	Inf. Dilution	dG:-903.6(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dG:-898.5(4SF)	0	CRV	7382
9	25	0	Inf. Dilution	dG:-3780.5(5SF)kJ	4	CRV	9015
10	25	0	Inf. Dilution	dG:-898.5(4SF)	0	CRV	12011
11	25	0	Inf. Dilution	dH:-980.(0.3SD)	4	ENS	2171
12	25	0	Inf. Dilution	dH:-980.2(4SF)	4	MTD	6422
13	25	0	Inf. Dilution	dH:-975.0(4SF)	0	CRV	7382
14	25	0	Inf. Dilution	dH:-4101.2(5SF)kJ	4	CRV	9015
15	25	0	Inf. Dilution	dH:-975.1(4SF)	0	CRV	12011
16	127	0	Inf. Dilution	dH:-980.5(4SF)	3	MTD	6422
17	227	0	Inf. Dilution	dH:-980.2(4SF)	2	MTD	6422
18	327	0	Inf. Dilution	dH:-979.6(4SF)	0	MTD	6422

Reaction No. 54839

2<Al(s)> + Si(s) + 3.5<O2(g)> + 2<H2(g)> = Al2O3SiOH(OH)3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-700.(0.5SD)	2	MCO	5479
2	25	0	Inf. Dilution	dG:-2921.80(6SF)kJ	3	CRV	25435
3	25	0	Inf. Dilution	dH:-763.(0.5SD)	0	MCO	5479
4	25	0	Inf. Dilution	dH:-3170.40(6SF)kJ	3	CRV	25435

Reaction No. 54840

2<Al(s)> + 2<H2(g)> + 4.5<O2(g)> + 2<Si(s)> = Al2Si2O5(OH)4(Kaol.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:665.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:665.7(4SF)	0	ENS	6422
Note (2): Changed from 655.7							
3	25	0	Inf. Dilution	lgK:665.3(4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:661.2(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:660.8(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:481.8(4SF)	0	ENS	6422
7	127	0	Inf. Dilution	lgK:481.5(4SF)	0	ENS	6494
8	227	0	Inf. Dilution	lgK:374.2(4SF)	0	ENS	6422

9	227	0	Inf. Dilution	lgK:373.9(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:302.5(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:302.2(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-3789.10(6SF)kJ	0	CRV	440
13	25	0	Inf. Dilution	dG:-908.1(0.2SD)	0	ENS	2171
14	25	0	Inf. Dilution	dG:-3799.40(6SF)kJ	0	CRV	5619
15	25	0	Inf. Dilution	dG:-3777.60(6SF)kJ	0	CRV	6421
16	25	0	Inf. Dilution	dG:-908.1(4SF)	0	EFE	6422
17	25	0	Inf. Dilution	dG:-908.17(5SF)	0	EFE	6482
18	25	0	Inf. Dilution	dG:-907.6(4SF)	0	EFE	6494
19	25	0	Inf. Dilution	dG:-3789.5(5SF)kJ	0	MSL	7380
20	25	0	Inf. Dilution	dG:-903.0(4SF)	0	CRV	7382
21	25	0	Inf. Dilution	dG:-3789.40(6SF)kJ	0	CRV	8872
22	25	0	Inf. Dilution	dG:-3778.20(6SF)kJ	0	EOD	8872
Note (22): Data from Naumov, Ryzhenko & Khodakovsky, 1971; low wgt for consistency							
23	25	0	Inf. Dilution	dG:-3776.90(6SF)kJ	0	CRV	8873
24	25	0	Inf. Dilution	dG:-3780.00(6SF)kJ	0	CRV	8874
25	25	0	Inf. Dilution	dG:-3799.7(5SF)kJ	0	CRV	9015
26	25	0	Inf. Dilution	dG:-908.50(5SF)	0	CRV	10174
27	25	0	Inf. Dilution	dG:-903.0(4SF)	0	CRV	12011
28	25	0	Inf. Dilution	dG:-3801.70(6SF)kJ	0	CRV	25435
29	25	0	Inf. Dilution	dG:-907.88(5SF)	5	CRV	29553
30	25	0	GAMSPATH	dG:-3789.(4SF)kJ	0	CRV	12638
31	50	0	Inf. Dilution	dG:-909.76(5SF)	0	CRV	10174
32	75	0	Inf. Dilution	dG:-911.15(5SF)	0	CRV	10174
33	100	0	Inf. Dilution	dG:-912.64(5SF)	0	CRV	10174
34	125	0	Inf. Dilution	dG:-914.26(5SF)	0	CRV	10174
35	150	0	Inf. Dilution	dG:-915.97(5SF)	0	CRV	10174
36	175	0	Inf. Dilution	dG:-917.80(5SF)	0	CRV	10174
37	200	0	Inf. Dilution	dG:-919.72(5SF)	0	CRV	10174
38	225	0	Inf. Dilution	dG:-921.73(5SF)	0	CRV	10174
39	250	0	Inf. Dilution	dG:-923.83(5SF)	0	CRV	10174
40	300	0	Inf. Dilution	dG:-928.28(5SF)	0	CRV	10174
41	25	0	Inf. Dilution	dH:-985.(0.3SD)	0	ENS	2171
42	25	0	Inf. Dilution	dH:-984.6(4SF)	0	MTD	6422
43	25	0	Inf. Dilution	dH:-984.8(4SF)	0	ENS	6437

44	25	0	Inf. Dilution	dH:-984.78(5SF)	0	EFE	6482
45	25	0	Inf. Dilution	dH:-984.5(4SF)	0	MTD	6494
46	25	0	Inf. Dilution	dH:-979.6(4SF)	0	CRV	7382
47	25	0	Inf. Dilution	dH:-4119.6(5SF)kJ	0	CRV	9015
48	25	0	Inf. Dilution	dH:-976.6(4SF)	0	CRV	12011
49	25	0	Inf. Dilution	dH:-4122.30(6SF)kJ	0	CRV	25435
50	25	0	Inf. Dilution	dH:-984.53(5SF)	5	CRV	29553
51	25	0	GAMSPATH	dH:-4109.60(6SF)kJ	0	CRV	12638
52	127	0	Inf. Dilution	dH:-984.9(4SF)	0	MTD	6422
53	127	0	Inf. Dilution	dH:-984.8(4SF)	0	MTD	6494
54	227	0	Inf. Dilution	dH:-984.6(4SF)	0	MTD	6422
55	227	0	Inf. Dilution	dH:-984.6(4SF)	0	MTD	6494
56	327	0	Inf. Dilution	dH:-983.9(4SF)	0	MTD	6422
57	327	0	Inf. Dilution	dH:-984.1(4SF)	0	MTD	6494

Reaction No. 54916

Cd+2 + H+1(2)_SiH2O4-2 - H2O - 2<H+1> = CdSiO3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.6(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-9.1(0.05SD)	0	ENS	766
3	25	0	Inf. Dilution	lgK:-7.49(3SF)	4	CRV	32224
4	25	0	Inf. Dilution	dH:16.6(0.1SD)	2	ENS	766

Reaction No. 55055

2<K(s)> + Si(s) + 3<F2(g)> = K+1(2)_SiF6-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-668.9(0.1SD)	4	ENS	802
2	25	0	Inf. Dilution	dH:-706.5(0.1SD)	4	ENS	802

Reaction No. 55080

Ca(s) + 1.5<O2(g)> + Si(s) = CaO.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:271.5(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:271.4(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:269.7(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:269.6(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:198.6(4SF)	2	ENS	6422

6	127	0	Inf. Dilution	lgK:198.5(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:155.9(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:155.8(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:127.4(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:127.3(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-370.4(0.05SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-370.4(4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-369.53(5SF)	2	EFE	6482
14	25	0	Inf. Dilution	dG:-370.2(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-1549.0(5SF)kJ	5	CRV	8695
16	25	0	Inf. Dilution	dG:-370.22(5SF)	4	EFE	32152
17	25	0	Inf. Dilution	dH:-390.8(0.05SD)	4	ENS	802
18	25	0	Inf. Dilution	dH:-390.8(4SF)	4	MTD	6422
19	25	0	Inf. Dilution	dH:-390.0(4SF)	3	ENS	6437
20	25	0	Inf. Dilution	dH:-389.94(5SF)	2	EFE	6482
21	25	0	Inf. Dilution	dH:-390.7(4SF)	4	MTD	6494
22	25	0	Inf. Dilution	dH:-1634.90(6SF)kJ	5	CRV	8695
23	127	0	Inf. Dilution	dH:-390.7(4SF)	1	MTD	6422
24	127	0	Inf. Dilution	dH:-390.7(4SF)	1	MTD	6494
25	227	0	Inf. Dilution	dH:-390.5(4SF)	0	MTD	6422
26	227	0	Inf. Dilution	dH:-390.5(4SF)	0	MTD	6494
27	327	0	Inf. Dilution	dH:-390.2(4SF)	0	MTD	6422
28	327	0	Inf. Dilution	dH:-390.3(4SF)	0	MTD	6494

Reaction No. 55081

2<Ca(s)> + 2<O2(g)> + Si(s) = 2CaO.SiO2(beta,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:383.7(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:383.9(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:381.2(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:381.4(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:280.9(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:281.0(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:220.8(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:220.8(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:180.7(4SF)	4	ENS	6422

10	327	0	Inf. Dilution	lgK:180.7(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-524.(0.5SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-523.5(4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-523.7(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-551.(0.5SD)	4	ENS	802
15	25	0	Inf. Dilution	dH:-550.9(4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-551.3(4SF)	4	MTD	6494
17	127	0	Inf. Dilution	dH:-550.7(4SF)	4	MTD	6422
18	127	0	Inf. Dilution	dH:-551.2(4SF)	4	MTD	6494
19	227	0	Inf. Dilution	dH:-550.4(4SF)	4	MTD	6422
20	227	0	Inf. Dilution	dH:-550.9(4SF)	4	MTD	6494
21	327	0	Inf. Dilution	dH:-550.0(4SF)	4	MTD	6422
22	327	0	Inf. Dilution	dH:-550.6(4SF)	4	MTD	6494

Reaction No. 55082

2<Ca(s)> + 2<O2(g)> + Si(s) = 2CaO.SiO2(gamma,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:385.2(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:385.2(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:382.7(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:382.7(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:281.9(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:281.9(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:221.5(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:221.4(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:181.2(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:181.1(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-526.(0.5SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-525.5(4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-525.5(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-554.(0.5SD)	4	ENS	802
15	25	0	Inf. Dilution	dH:-553.3(4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-553.7(4SF)	4	MTD	6494
17	27	0	Inf. Dilution	dH:-553.4(4SF)	0	MTD	6422
18	127	0	Inf. Dilution	dH:-553.3(4SF)	4	MTD	6422
19	127	0	Inf. Dilution	dH:-553.6(4SF)	4	MTD	6494

20	227	0	Inf. Dilution	dH:-553.1(4SF)	4	MTD	6422
21	227	0	Inf. Dilution	dH:-553.4(4SF)	4	MTD	6494
22	327	0	Inf. Dilution	dH:-552.8(4SF)	4	MTD	6422
23	327	0	Inf. Dilution	dH:-553.2(4SF)	4	MTD	6494

Reaction No. 55083

3<Ca(s)> + 2.5<O2(g)> + Si(s) = 3CaO.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:487.7(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:488.2(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:484.6(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:485.0(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:357.1(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:357.3(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:280.6(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:280.8(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:229.7(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:229.8(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-665.(0.5SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-665.4(4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-666.0(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-700.(0.5SD)	4	ENS	802
15	25	0	Inf. Dilution	dH:-700.1(4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-701.0(4SF)	4	MTD	6494
17	127	0	Inf. Dilution	dH:-699.8(4SF)	2	MTD	6422
18	127	0	Inf. Dilution	dH:-700.8(4SF)	2	MTD	6494
19	227	0	Inf. Dilution	dH:-699.2(4SF)	0	MTD	6422
20	227	0	Inf. Dilution	dH:-700.4(4SF)	0	MTD	6494
21	327	0	Inf. Dilution	dH:-698.6(4SF)	0	MTD	6422
22	327	0	Inf. Dilution	dH:-699.8(4SF)	0	MTD	6494

Reaction No. 55084

3<Ca(s)> + 3.5<O2(g)> + 2<Si(s)> = 3CaO.2SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:659.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:659.0(4SF)	0	ENS	6422

3	25	0	Inf. Dilution	lgK:656.6(4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:654.7(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:652.4(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:482.3(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:480.5(4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:378.9(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:377.4(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:310.0(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:308.7(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-899.(0.5SD)	0	ENS	802
13	25	0	Inf. Dilution	dG:-899.0(4SF)	0	EFE	6422
14	25	0	Inf. Dilution	dG:-895.8(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dH:-947.(0.5SD)	0	ENS	802
16	25	0	Inf. Dilution	dH:-946.7(4SF)	0	MTD	6422
17	25	0	Inf. Dilution	dH:-943.8(4SF)	0	MTD	6494
18	127	0	Inf. Dilution	dH:-946.5(4SF)	0	MTD	6422
19	127	0	Inf. Dilution	dH:-943.7(4SF)	0	MTD	6494
20	227	0	Inf. Dilution	dH:-945.9(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-943.3(4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-945.3(4SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-942.9(4SF)	0	MTD	6494

Reaction No. 55097							
Mg(s) + 1.5<O2(g)> + Si(s) = MgO.SiO2(Clinoen.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:256.1(4SF)	2	ENS	6422
2	25	0	Inf. Dilution	lgK:255.4(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:254.5(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:253.8(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:187.0(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:186.5(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:146.6(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:146.2(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:119.6(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:119.3(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-349.5(0.05SD)	0	ENS	802

12	25	0	Inf. Dilution	dG:-349.4(4SF)	2	EFE	6422
13	25	0	Inf. Dilution	dG:-348.61(5SF)	4	EFE	6482
14	25	0	Inf. Dilution	dG:-348.5(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-1458.1(5SF)kJ	3	EOD	12843
16	25	0	Inf. Dilution	dH:-370.2(0.05SD)	4	ENS	802
17	25	0	Inf. Dilution	dH:-370.2(4SF)	4	MTD	6422
18	25	0	Inf. Dilution	dH:-369.49(5SF)	4	EFE	6482
19	25	0	Inf. Dilution	dH:-369.3(4SF)	4	MTD	6494
20	25	0	Inf. Dilution	dH:-1545.0(5SF)kJ	3	EOD	12843
21	127	0	Inf. Dilution	dH:-370.3(4SF)	1	MTD	6422
22	127	0	Inf. Dilution	dH:-369.3(4SF)	1	MTD	6494
23	227	0	Inf. Dilution	dH:-370.2(4SF)	0	MTD	6422
24	227	0	Inf. Dilution	dH:-369.3(4SF)	0	MTD	6494
25	327	0	Inf. Dilution	dH:-370.1(4SF)	0	MTD	6422
26	327	0	Inf. Dilution	dH:-369.2(4SF)	0	MTD	6494

Reaction No. 55098

2<Mg+2> + H+1(2)_SiH2O4-2 = 2MgO.SiO2(s) + 4<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-28.9(0.05SD)	0	ENS	557
3	25	0	Inf. Dilution	lgK:-28.44(4SF)	0	CRV	6405
Note (3): p. 245							
4	25	0	Inf. Dilution	lgK:-28.14(4SF)	4	CRV	32224
5	25	0	CHEMVAL1	lgK:-29.98(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL2	lgK:-29.98(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:-29.98(0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:-29.10(0.01SD)	0	ENS	5156

Reaction No. 55099

Ca(s) + Si(s) + 1.5<O2(g)> = CaO.SiO2(Pseudo.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:270.6(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:270.4(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:268.9(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:268.6(4SF)	0	ENS	6494

5	127	0	Inf. Dilution	lgK:198.0(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:197.8(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:155.5(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:155.3(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:127.2(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:127.0(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-369.(0.5SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-369.2(4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-368.81(5SF)	4	EFE	6482
14	25	0	Inf. Dilution	dG:-368.9(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dH:-389.(0.5SD)	4	ENS	802
16	25	0	Inf. Dilution	dH:-389.2(4SF)	4	MTD	6422
17	25	0	Inf. Dilution	dH:-388.96(5SF)	4	EFE	6482
18	25	0	Inf. Dilution	dH:-389.0(4SF)	4	MTD	6494
19	127	0	Inf. Dilution	dH:-389.1(4SF)	1	MTD	6422
20	127	0	Inf. Dilution	dH:-388.9(4SF)	1	MTD	6494
21	227	0	Inf. Dilution	dH:-388.9(4SF)	0	MTD	6422
22	227	0	Inf. Dilution	dH:-388.8(4SF)	0	MTD	6494
23	327	0	Inf. Dilution	dH:-388.7(4SF)	0	MTD	6422
24	327	0	Inf. Dilution	dH:-388.6(4SF)	0	MTD	6494

Reaction No. 55158

Sr+2 + H+1(2)_SiH2O4-2 = Sr+2_SiO3-2_(s) + 2<H+1> + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-14.2(3SF)	1	EOR	
2	25	0	Inf. Dilution	lgK:-14.1(0.05SD)	3	ENS	5458
3	25	0	Inf. Dilution	lgK:-14.06(4SF)	4	CRV	32224
4	25	0	CHEMVAL1	lgK:-15.64(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL2	lgK:-15.64(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:-15.64(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL4	lgK:-15.60(0.01SD)	0	ENS	5156

Reaction No. 55159

Sr+2_SiH2O4-2 - 2<H+1> - 2<e-1> = SrSiO4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:56.8(0.05SD)	0	ENS	5458

Reaction No. 55240

Mn(s) + Si(s) + 1.5<O2(g)> = MnO.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:217.3(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:218.1(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:215.9(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:216.6(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:158.4(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:159.1(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:123.9(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:124.6(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:101.0(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:101.6(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-296.5(0.1SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-296.5(0.1SD)	5	ENS	802
13	25	0	Inf. Dilution	dG:-296.5(4SF)	4	EFE	6422
14	25	0	Inf. Dilution	dG:-297.5(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dH:-315.7(0.1SD)	4	ENS	802
16	25	0	Inf. Dilution	dH:-315.7(0.1SD)	5	ENS	802
17	25	0	Inf. Dilution	dH:-315.7(4SF)	4	MTD	6422
18	25	0	Inf. Dilution	dH:-315.9(4SF)	4	MTD	6494
19	127	0	Inf. Dilution	dH:-315.7(4SF)	4	MTD	6422
20	127	0	Inf. Dilution	dH:-315.8(4SF)	4	MTD	6494
21	227	0	Inf. Dilution	dH:-315.5(4SF)	4	MTD	6422
22	227	0	Inf. Dilution	dH:-315.6(4SF)	4	MTD	6494
23	327	0	Inf. Dilution	dH:-315.3(4SF)	4	MTD	6422
24	327	0	Inf. Dilution	dH:-315.4(4SF)	4	MTD	6494

Reaction No. 55241

2<Mn(s)> + Si(s) + 2<O2(g)> = 2MnO.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:285.9(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:285.7(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:284.1(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:283.9(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:208.8(4SF)	4	ENS	6422

6	127	0	Inf. Dilution	lgK:208.5(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:163.6(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:163.3(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:133.5(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:133.2(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-390.1(0.1SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-390.1(0.1SD)	5	ENS	802
13	25	0	Inf. Dilution	dG:-390.1(4SF)	4	EFE	6422
14	25	0	Inf. Dilution	dG:-389.8(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dH:-413.6(0.1SD)	4	ENS	802
16	25	0	Inf. Dilution	dH:-413.6(0.1SD)	5	ENS	802
17	25	0	Inf. Dilution	dH:-413.6(4SF)	4	MTD	6422
18	25	0	Inf. Dilution	dH:-413.8(4SF)	4	MTD	6494
19	127	0	Inf. Dilution	dH:-413.5(4SF)	4	MTD	6422
20	127	0	Inf. Dilution	dH:-413.8(4SF)	4	MTD	6494
21	227	0	Inf. Dilution	dH:-413.3(4SF)	4	MTD	6422
22	227	0	Inf. Dilution	dH:-413.6(4SF)	4	MTD	6494
23	327	0	Inf. Dilution	dH:-413.0(4SF)	4	MTD	6422
24	327	0	Inf. Dilution	dH:-413.4(4SF)	4	MTD	6494

Reaction No. 55374

K+1 + 3<Mg+2> + Al+3 + 3<H+1(2)_SiH2O4-2> + 2<F-1> = KMg3AlSi3O10F2(s) + 8<H+1> + 2 <H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-21.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-7.8(0.05SD)	0	ENS	557

Reaction No. 55511

7<Mn(s)> + Si(s) + 6<O2(g)> = Mn7SiO12(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:691.1(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:686.5(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:501.1(4SF)	2	ENS	6494
4	227	0	Inf. Dilution	lgK:389.9(4SF)	1	ENS	6494
5	327	0	Inf. Dilution	lgK:315.9(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-940.(5.SD)	0	ENT	5182
7	25	0	Inf. Dilution	dG:-942.8(4SF)	4	EFE	6494

8	25	0	Inf. Dilution	dH:-1010.(5.SD)	4	ENT	5182
9	25	0	Inf. Dilution	dH:-1018.(4SF)	0	MTD	6494
10	127	0	Inf. Dilution	dH:-1018.(4SF)	1	MTD	6494
11	227	0	Inf. Dilution	dH:-1017.(4SF)	0	MTD	6494
12	327	0	Inf. Dilution	dH:-1016.(4SF)	0	MTD	6494

Reaction No. 55530

3<F2(g)> + Si(s) + 2<e-1> = SiF6-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-511.0(0.05SD)	0	ENS	3006
Note (1): Low wgt for inter-reaction consistency							
2	25	0	Inf. Dilution	dG:-525.7(4SF)	0	CRV	6488
3	25	0	Inf. Dilution	dG:-2138.(4SF)kJ	0	CRV	7421
4	25	0	Inf. Dilution	dG:-525.70(5SF)	1	CRV	10174
5	50	0	Inf. Dilution	dG:-526.39(5SF)	2	CRV	10174
6	75	0	Inf. Dilution	dG:-527.00(5SF)	2	CRV	10174
7	100	0	Inf. Dilution	dG:-527.54(5SF)	1	CRV	10174
8	125	0	Inf. Dilution	dG:-528.02(5SF)	1	CRV	10174
9	150	0	Inf. Dilution	dG:-528.41(5SF)	0	CRV	10174
10	175	0	Inf. Dilution	dG:-528.72(5SF)	0	CRV	10174
11	200	0	Inf. Dilution	dG:-528.93(5SF)	0	CRV	10174
12	225	0	Inf. Dilution	dG:-529.00(5SF)	0	CRV	10174
13	250	0	Inf. Dilution	dG:-528.91(5SF)	0	CRV	10174
14	300	0	Inf. Dilution	dG:-527.95(5SF)	0	CRV	10174
15	25	0	Inf. Dilution	dH:-558.3(0.1SD)	3	ENS	3006
16	25	0	Inf. Dilution	dH:-571.0(4SF)	0	CRV	6488

Reaction No. 55531

H2(g) + 2<O2(g)> + Si(s) + 2<e-1> = SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:206.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-281.3(0.05SD)	0	ENS	557
3	25	0	Inf. Dilution	dG:-1175.7(1.265SD)kJ	0	CRV	14923
4	25	0	Inf. Dilution/m	dG:-1176.80(6SF)kJ	0	CRV	24953
5	25	0	Inf. Dilution	dH:-1382.0(15.330SD)kJ	1	CRV	14923
6	25	0	Inf. Dilution/m	dH:-1396.60(6SF)kJ	1	CRV	24953

Reaction No. 55734							
2<Fe(s)> + 2<O2(g)> + Si(s) = 2FeO.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:241.6(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:241.6(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:240.0(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:240.0(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:175.6(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:175.7(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:137.0(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:137.1(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:111.3(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:111.5(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-329.6(0.1SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-329.6(4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-329.86(5SF)	4	EFE	6482
14	25	0	Inf. Dilution	dG:-329.6(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-1379.1(5SF)kJ	5	CRV	8695
Note (15): p. 169							
16	25	0	Inf. Dilution	dG:-1379.1(5SF)kJ	3	EOD	12843
17	25	0	Inf. Dilution	dG:-1377.70(6SF)kJ	4	CRV	27292
18	25	0	Inf. Dilution	dH:-353.7(0.1SD)	4	ENS	802
19	25	0	Inf. Dilution	dH:-353.7(4SF)	4	MTD	6422
20	25	0	Inf. Dilution	dH:-353.3(4SF)	4	ENS	6437
21	25	0	Inf. Dilution	dH:-353.58(5SF)	4	EFE	6482
22	25	0	Inf. Dilution	dH:-353.3(4SF)	4	MTD	6494
23	25	0	Inf. Dilution	dH:-1478.2(5SF)kJ	3	EOD	12843
24	25	0	Inf. Dilution	dH:-1476.80(6SF)kJ	4	CRV	27292
25	127	0	Inf. Dilution	dH:-353.5(4SF)	2	MTD	6422
26	127	0	Inf. Dilution	dH:-353.0(4SF)	2	MTD	6494
27	227	0	Inf. Dilution	dH:-353.1(4SF)	1	MTD	6422
28	227	0	Inf. Dilution	dH:-352.7(4SF)	1	MTD	6494
29	327	0	Inf. Dilution	dH:-352.7(4SF)	0	MTD	6422
30	327	0	Inf. Dilution	dH:-352.2(4SF)	0	MTD	6494
31	25	0	Inf. Dilution	Cp:132.030(6SF)J/K	4	CRV	27292

Reaction No. 55735							
3<Fe(s)> + Si(s) = Fe3Si(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-22.6(0.1SD)	4	ENS	802
2	25	0	Inf. Dilution	dH:-22.4(0.1SD)	4	ENS	802

Reaction No. 55736							
Fe(s) + 2.33<Si(s)> = FeSi2.33(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.25(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:10.19(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:7.619(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:6.078(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:5.051(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-14.(2SF)	4	ENS	802
7	25	0	Inf. Dilution	dG:-13.98(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-14.(2SF)	4	ENS	802
9	25	0	Inf. Dilution	dH:-14.1(3SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-14.1(3SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-14.1(3SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-14.1(3SF)	0	MTD	6422

Reaction No. 55737							
Fe(s) + 2<Si(s)> = FeSi2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.74(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:13.65(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:10.11(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:7.977(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:6.543(4SF)	3	ENS	6422
6	25	0	Inf. Dilution	dG:-18.7(0.1SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-18.74(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-19.4(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-19.4(3SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-19.5(3SF)	3	MTD	6422
11	227	0	Inf. Dilution	dH:-19.6(3SF)	3	MTD	6422

12	327	0	Inf. Dilution	dH:-19.7(3SF)	3	MTD	6422

Reaction No. 55738

Fe(s) + Si(s) = FeSi(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.74(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:13.66(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:10.22(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:8.156(4SF)	3	ENS	6422
5	327	0	Inf. Dilution	lgK:6.773(4SF)	3	ENS	6422
6	25	0	Inf. Dilution	dG:-17.6(0.1SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-18.75(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-17.6(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-18.8(3SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-18.9(3SF)	3	MTD	6422
11	227	0	Inf. Dilution	dH:-18.9(3SF)	3	MTD	6422
12	327	0	Inf. Dilution	dH:-19.0(3SF)	3	MTD	6422

Reaction No. 55790

Ca+2 + H+1(2)_SiH2O4-2 = Ca+2_SiH2O4-2 + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-18.7(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-19.87(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-19.87(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-19.87(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-17.30(0.01SD)	0	ENS	5156

Reaction No. 55791

Ca+2 + H+1(2)_SiH2O4-2 = Ca+2_H+1_SiH2O4-2 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.8(2SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-9.08(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-9.08(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-9.08(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-8.84(0.01SD)	0	ENS	5156

Reaction No. 55792

Ca+2 + 2<H+1(2)_SiH2O4-2> = Ca+2_H+1(2)_SiH2O4-2(2) + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-16.2(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-16.05(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-16.05(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-16.06(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-15.80(0.01SD)	0	ENS	5156

Reaction No. 55803

Fe+2 + H+1(2)_SiH2O4-2 - H+1 - e-1 = Fe+3_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.6(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-13.97(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-13.97(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-13.97(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-13.20(0.01SD)	0	ENS	5156

Reaction No. 55814

Mg+2 + H+1(2)_SiH2O4-2 = Mg+2_SiH2O4-2 + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-17.7(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-18.79(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-18.79(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-18.79(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-16.20(0.01SD)	0	ENS	5156

Reaction No. 55815

Mg+2 + H+1(2)_SiH2O4-2 = Mg+2_H+1_SiH2O4-2 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.9(2SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-8.83(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-8.83(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-8.83(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-8.59(0.01SD)	0	ENS	5156

Reaction No. 55816

Mg+2 + 2<H+1(2)_SiH2O4-2> = Mg+2_H+1(2)_SiH2O4-2(2) + 2<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-15.4(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:-15.12(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-15.12(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-15.12(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-15.30(0.01SD)	0	ENS	5156

Reaction No. 55927

U+4 + 2<H2O> + H+1(2)_SiH2O4-2 = UO2+2_H+1_SiH2O4-2 + 5<H+1> + 2<e-1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-11.58(4SF)	2	CRV	32224
2	25	0	CHEMVAL1	lgK:-11.62(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:-11.62(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:-11.62(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:-11.00(0.01SD)	4	ENS	5156

Reaction No. 56091

CaO.(Al2O3)7.(SiO2)22.6H2O(s) + 44<H+1> + 16<H2O> = Ca+2 + 14<Al+3> +
22<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:36.84(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:36.84(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:36.84(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:36.80(0.01SD)	4	ENS	5156

Reaction No. 56097

Sr2SiO4(s) + 4<H+1> = 2<Sr+2> + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.61(4SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:42.83(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:42.83(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:42.83(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:42.80(0.01SD)	0	ENS	5156

Reaction No. 56148

2CaO.SiO2(gamma,s) + 4<H+1> = 2<Ca+2> + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:38.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:37.67(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:39.01(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:39.01(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:39.01(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:40.40(0.01SD)	0	ENS	5156

Reaction No. 56149

2CaO.SiO2(beta,s) + 4<H+1> = 2<Ca+2> + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:39.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:39.16(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:40.46(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:40.46(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:40.46(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:39.20(0.01SD)	0	ENS	5156

Reaction No. 56150

3CaO.2SiO2(s) + H2O + 6<H+1> = 3<Ca+2> + 2<H+1(2)_SiH2O4-2>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:50.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:48.62(4SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:51.77(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:51.77(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:51.77(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:51.80(0.01SD)	0	ENS	5156

Reaction No. 56151

3CaO.SiO2(s) + 6<H+1> = 3<Ca+2> + H+1(2)_SiH2O4-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:73.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:73.84(4SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:75.42(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:75.42(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:75.42(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:75.40(0.01SD)	0	ENS	5156

Reaction No. 56152

CaO.MgO.SiO2(s) + 4<H+1> = Ca+2 + Mg+2 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:31.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:29.85(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:30.27(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:30.27(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:30.27(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:30.30(0.01SD)	0	ENS	5156

Reaction No. 56153

2CaO.MgO.2SiO2(s) + H2O + 6<H+1> = 2<H+1(2)_SiH2O4-2> + 2<Ca+2> + Mg+2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:47.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:45.39(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:48.74(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:48.74(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:48.74(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:45.80(0.01SD)	0	ENS	5156

Reaction No. 56154

3CaO.MgO.2SiO2(s) + 8<H+1> = 3<Ca+2> + Mg+2 + 2<H+1(2)_SiH2O4-2>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:69.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:68.55(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:71.54(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:71.54(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:71.54(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:68.70(0.01SD)	0	ENS	5156

Reaction No. 56155

Ca2Al2Si4O12.4H2O(s) + 8<H+1> - 2<e-1> = 2<Ca+2> + 2<Al+3> + 4<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:11.85(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:11.85(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:11.85(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:11.80(0.01SD)	1	ENS	5156

Note (4): query mineral "laumontite"

Reaction No. 56156

CaO.Al2O3.2SiO2(s) + 8<H+1> = Ca+2 + 2<Al+3> + 2<H+1(2)_SiH2O4-2>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:27.19(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:27.06(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:27.06(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:27.06(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:23.00(0.01SD)	0	ENS	5156

Reaction No. 56157

CaO.Al2O3.4SiO2.H2O(s) + 8<H+1> + 3<H2O> = Ca+2 + 2<Al+3> + 4<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:17.46(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:17.46(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:17.46(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:17.50(0.01SD)	3	ENS	5156

Reaction No. 56158

CaO.Al2O3.2SiO2.2H2O(s) + 8<H+1> = Ca+2 + 2<Al+3> + 2<H+1(2)_SiH2O4-2> + 2<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:22.81(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:22.06(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:22.06(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:22.06(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:17.00(0.01SD)	0	ENS	5156

Reaction No. 56159

2CaO.Al2O3.SiO2(s) + 10<H+1> = 2<Ca+2> + 2<Al+3> + H+1(2)_SiH2O4-2 + 3<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:57.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:56.88(4SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:56.42(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:56.42(0.01SD)	0	ENS	5156

5	25	0	CHEMVAL3	lgK:56.42(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:52.00(0.01SD)	0	ENS	5156

Reaction No. 56160

Ca2Al2Si3O10(OH)2(s) + 10<H+1> = 2<Ca+2> + 2<Al+3> + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:33.51(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:32.41(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:32.41(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:32.41(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:32.40(0.01SD)	0	ENS	5156

Reaction No. 56161

Ca2Al3Si3O12(OH)(ortho,s) + 13<H+1> = 2<Ca+2> + 3<Al+3> + 3<H+1(2)_SiH2O4-2 > + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:41.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:44.20(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:39.83(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:39.83(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:39.83(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:39.80(0.01SD)	0	ENS	5156

Reaction No. 56162

Ca2Al4Si8O24.7H2O(s) + 16<H+1> + H2O = 2<Ca+2> + 4<Al+3> + 8<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:25.06(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:25.06(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:25.06(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:16.60(0.01SD)	3	ENS	5156

Reaction No. 56164

3CaO.Al2O3.3SiO2(s) + 12<H+1> = 3<Ca+2> + 2<Al+3> + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:51.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:52.47(4SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:57.21(0.01SD)	0	ENS	5156

4	25	0	CHEMVAL2	lgK:57.21(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:57.21(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:52.60(0.01SD)	0	ENS	5156

Reaction No. 56165

CaO.FeO.2SiO2(s) + 2<H2O> + 4<H+1> = Ca+2 + Fe+2 + 2<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:18.40(4SF)	0	CRV	6405
Note (2): p. 246							
3	25	0	Inf. Dilution	lgK:17.35(4SF)	0	CRV	32224
4	25	0	CHEMVAL1	lgK:20.51(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL2	lgK:20.51(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:20.51(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL4	lgK:20.50(0.01SD)	0	ENS	5156

Reaction No. 56166

3CaO.Fe2O3.3SiO2(s) + 12<H+1> = 3<Ca+2> + 2<Fe+3> + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:28.7(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:59.55(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:59.55(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:59.55(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:59.60(0.01SD)	0	ENS	5156

Reaction No. 56170

CaO.Al2O3.SiO2(s) + 8<H+1> = Ca+2 + 2<Al+3> + H+1(2)_SiH2O4-2 + 2<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:34.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:36.59(4SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:33.71(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:33.71(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:33.71(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:33.60(0.01SD)	0	ENS	5156

Reaction No. 56184

MgO.SiO2(Clinoen.,s) + H2O + 2<H+1> = Mg+2 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:11.64(4SF)	0	ENS	6405
3	25	0	CHEMVAL1	lgK:12.65(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:12.65(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:12.65(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:11.40(0.01SD)	0	ENS	5156

Reaction No. 56185

Mg7Si8O22(OH)2(Anth.,s) + 14<H+1> + 8<H2O> = 7<Mg+2> + 8<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:72.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:67.75(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:80.39(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:80.39(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:80.39(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:80.40(0.01SD)	0	ENS	5156

Reaction No. 56186

Mg3Si2O5(OH)4(Chry.,s) + 6<H+1> = 3<Mg+2> + 2<H+1(2)_SiH2O4-2> + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:32.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:32.9(0.05SD)	0	ENS	557
3	25	0	Inf. Dilution	lgK:32.20(4SF)	0	ENS	6496
4	25	0	Inf. Dilution	lgK:31.52(4SF)	0	CRV	32224
5	25	0	CHEMVAL1	lgK:34.50(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL2	lgK:34.50(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:34.50(0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:34.50(0.01SD)	0	ENS	5156
9	25	0	Inf. Dilution	dH:-46.8(3SF)	2	ENS	6496

Reaction No. 56187

MgAl14Si22O72H12(s) + 16<H2O> + 44<H+1> = Mg+2 + 14<Al+3> + 22<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:36.24(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:36.24(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:36.24(0.01SD)	0	ENS	5156

4	25	0	CHEMVAL4	lgK:36.20(0.01SD)	4	ENS	5156

Reaction No. 56190

Mg2(H3SiO4)4.4H2O(s) + 4<H+1> = 2<Mg+2> + 4<H+1(2)_SiH2O4-2> + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:-9.22(0.01SD)	4	ENS	5156
2	25	0	CHEMVAL2	lgK:-9.22(0.01SD)	4	ENS	5156
3	25	0	CHEMVAL3	lgK:-9.22(0.01SD)	4	ENS	5156
4	25	0	CHEMVAL4	lgK:-9.22(0.01SD)	4	ENS	5156

Reaction No. 56191

Mg3Si4O10(OH)2.3H2O(s) + H2O + 6<H+1> = 3<Mg+2> + 4<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:29.86(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:29.86(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:29.86(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:29.90(0.01SD)	4	ENS	5156

Reaction No. 56192

Mg3Si4O10(OH)2(s) + 6<H+1> + 4<H2O> = 3<Mg+2> + 4<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:21.53(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:24.45(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:24.45(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:24.45(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:19.60(0.01SD)	0	ENS	5156

Reaction No. 56193

2MgO.2Al2O3.5SiO2(s) + 16<H+1> + 2<H2O> = 2<Mg+2> + 4<Al+3> + 5<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:50.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:45.5(0.05SD)	0	ENS	557
3	25	0	Inf. Dilution	lgK:53.88(4SF)	0	CRV	32224
4	25	0	CHEMVAL1	lgK:53.77(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL2	lgK:53.77(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:53.77(0.01SD)	0	ENS	5156

7	25	0	CHEMVAL4	lgK:49.40(0.01SD)	0	ENS	5156

Reaction No. 56198							
KAlSi3O8(Micr.,s) + 4<H2O> + 4<H+1> = K+1 + Al+3 + 3<H+1(2)_SiH2O4-2 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.2(2SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:1.29(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:1.29(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:1.29(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:0.88(0.01SD)	0	ENS	5156

Reaction No. 56199							
KAl7Si11O30(OH)6(s) + 8<H2O> + 22<H+1> = K+1 + 7<Al+3> + 11<H+1(2)_SiH2O4-2 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:17.58(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:17.58(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:17.58(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:17.60(0.01SD)	4	ENS	5156

Reaction No. 56201							
KAlSi2O6(s) + 4<H+1> + 2<H2O> = 2<H+1(2)_SiH2O4-2> + Al+3 + K+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.2(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:6.42(3SF)	0	EOD	23102
3	25	0	CHEMVAL1	lgK:9.78(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:9.78(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:9.78(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:6.61(0.01SD)	0	ENS	5156

Reaction No. 56202							
KAlSiO4(s) + 4<H+1> = K+1 + Al+3 + H+1(2)_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:11.23(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:12.79(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:12.79(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:12.79(0.01SD)	0	ENS	5156

6	25	0	CHEMVAL4	lgK:13.00(0.01SD)	0	ENS	5156

Reaction No. 56203

KAl3Si3O10(OH)2(s) + 10<H+1> = K+1 + 3<Al+3> + 3<H+1(2)_SiH2O4-2>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:14.56(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:13.96(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:13.96(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:13.96(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:13.00(0.01SD)	0	ENS	5156

Reaction No. 56214

NaAlSi3O8(AlbiteHI,s) + 4<H+1> + 4<H2O> = Na+1 + Al+3 + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.0(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:4.41(3SF)	4	CRV	32224
3	25	0	CHEMVAL1	lgK:7.02(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:7.02(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:7.02(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:7.02(0.01SD)	0	ENS	5156

Reaction No. 56215

NaAl7Si11O30(OH)6(s) + 8<H2O> + 22<H+1> = Na+1 + 7<Al+3> + 11<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	CHEMVAL1	lgK:18.36(0.01SD)	0	ENS	5156
2	25	0	CHEMVAL2	lgK:18.36(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL3	lgK:18.36(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL4	lgK:18.70(0.01SD)	4	ENS	5156

Reaction No. 56217

NaAlSi2O6(Dehyd.Anal.,s) + 2<H2O> + 4<H+1> = Na+1 + Al+3 + 2<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.6(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:15.20(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:15.20(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:15.20(0.01SD)	0	ENS	5156

5	25	0	CHEMVAL4	lgK:15.20(0.01SD)	0	ENS	5156

Reaction No. 56218

NaAlSi2O6.H2O(s) + 4<H+1> + H2O = Na+1 + Al+3 + 2<H+1(2)_SiH2O4-2>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.0(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:6.64(3SF)	0	CRV	27383
3	25	0	Inf. Dilution	lgK:7.27(3SF)	0	CRV	32224
4	25	0	Inf. Dilution	lgK:11.98(4SF)	0	CRV	32224
5	25	0	CHEMVAL1	lgK:10.58(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL2	lgK:10.58(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL3	lgK:10.58(0.01SD)	0	ENS	5156
8	25	0	CHEMVAL4	lgK:10.60(0.01SD)	0	ENS	5156

Reaction No. 56219

NaAlSiO4(Neph.,s) + 4<H+1> = Na+1 + Al+3 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:14.13(4SF)	2	CRV	32224
3	25	0	CHEMVAL1	lgK:14.96(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:14.96(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:14.96(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:15.00(0.01SD)	0	ENS	5156

Reaction No. 56220

NaAl3Si3O10(OH)2(s) + 10<H+1> = Na+1 + 3<Al+3> + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:18.49(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:23.23(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:23.23(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:23.23(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:23.20(0.01SD)	0	ENS	5156

Reaction No. 56221

Na2SiO3(s) + H2O + 2<H+1> = 2<Na+1> + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:23.2(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:24.62(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:24.62(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:24.62(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:24.60(0.01SD)	0	ENS	5156

Reaction No. 56222

Na2Si2O5(s) + 3<H2O> + 2<H+1> = 2<Na+1> + 2<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.6(3SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:21.66(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:21.66(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:21.66(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:21.70(0.01SD)	0	ENS	5156

Reaction No. 56227

FeO.SiO2(s) + H2O + 2<H+1> = Fe+2 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.7(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:5.54(3SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:7.12(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:7.12(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:7.12(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:7.12(0.01SD)	0	ENS	5156

Reaction No. 56228

Fe3Si2O5(OH)4(s) + 6<H+1> = 3<Fe+2> + 2<H+1(2)_SiH2O4-2> + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:17.65(4SF)	4	CRV	32224
2	25	0	CHEMVAL1	lgK:23.80(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:23.80(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:23.80(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:23.80(0.01SD)	0	ENS	5156

Reaction No. 56229

Fe3Si4O10(OH)2(s) + 4<H2O> + 6<H+1> = 3<Fe+2> + 4<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:7.34(3SF)	4	CRV	32224
2	25	0	CHEMVAL1	lgK:13.66(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:13.66(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:13.66(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:13.70(0.01SD)	0	ENS	5156

Reaction No. 56236

Al2SiO5(Kyan.,s) + 6<H+1> = 2<Al+3> + H+1(2)_SiH2O4-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:16.326(5SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:13.99(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:13.99(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:13.99(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:14.00(0.01SD)	0	ENS	5156

Reaction No. 56237

Al2Si4O10(OH)2(s) + 4<H2O> = 2<Al+3> + 4<H+1(2)_SiH2O4-2> - 6<H+1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.8(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.081(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:3.46(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:3.46(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:3.46(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:3.42(0.01SD)	0	ENS	5156

Reaction No. 56263

UO2.SiO2(s) + 4<H+1> = U+4 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-1.5(2SF)	0	CRV	32222
3	25	0	Inf. Dilution	lgK:-9.20(3SF)	0	CRV	32224
Note (3): source indicates uncertainty							
4	25	0	CHEMVAL1	lgK:-9.18(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL2	lgK:-9.18(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:-9.18(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL4	lgK:-7.63(0.01SD)	0	ENS	5156

Reaction No. 56331

Mg5Al2Si3O10(OH)8(s) + 16<H+1> = 5<Mg+2> + 2<Al+3> + 3<H+1(2)_SiH2O4-2> + 6<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:61.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:71.92(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:57.02(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:57.02(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:57.02(0.01SD)	0	ENS	5156

Reaction No. 56334

Al2Si2O7.2H2O(Dick.,s) + 6<H+1> = 2<Al+3> + 2<H+1(2)_SiH2O4-2> + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.8(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:5.95(3SF)	0	MGL	6498
3	25	0	CHEMVAL1	lgK:7.78(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:7.78(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:7.78(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:7.74(0.01SD)	0	ENS	5156

Reaction No. 56347

MgO.SiO2(Enstat.,s) + 2<H+1> + H2O = Mg+2 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:11.64(4SF)	0	CRV	6405
3	25	0	Inf. Dilution	lgK:11.46(4SF)	4	CRV	32224
4	25	0	CHEMVAL1	lgK:13.04(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL2	lgK:13.04(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL3	lgK:13.04(0.01SD)	0	ENS	5156
7	25	0	CHEMVAL4	lgK:12.60(0.01SD)	0	ENS	5156

Reaction No. 56348

2<H2(g)> + 3<Mg(s)> + 4.5<O2(g)> + 2<Si(s)> = Mg3Si2O5(OH)4(Chry.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:707.4(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:706.4(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:702.7(4SF)	4	ENS	6422

4	27	0	Inf. Dilution	lgK:701.7(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:512.7(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:511.9(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:398.6(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:398.0(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:322.7(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:322.2(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-965.1(0.05SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-965.1(0.05SD)	5	ENS	802
13	25	0	Inf. Dilution	dG:-965.1(4SF)	4	EFE	6422
14	25	0	Inf. Dilution	dG:-964.48(5SF)	4	EFE	6482
15	25	0	Inf. Dilution	dG:-963.8(4SF)	4	EFE	6494
16	25	0	GAMSPATH	dG:-4037.(4SF)kJ	3	CRV	12638
17	25	0	Inf. Dilution	dH:-1043.4(0.1SD)	4	ENS	802
18	25	0	Inf. Dilution	dH:-1043.4(0.1SD)	5	ENS	802
19	25	0	Inf. Dilution	dH:-1043.(4SF)	4	MTD	6422
20	25	0	Inf. Dilution	dH:-1042.8(5SF)	4	ENS	6437
21	25	0	Inf. Dilution	dH:-1042.9(5SF)	4	EFE	6482
22	25	0	Inf. Dilution	dH:-1042.(4SF)	4	MTD	6494
23	25	0	GAMSPATH	dH:-4364.40(6SF)kJ	3	CRV	12638
24	127	0	Inf. Dilution	dH:-1044.(4SF)	4	MTD	6422
25	127	0	Inf. Dilution	dH:-1042.(4SF)	4	MTD	6494
26	227	0	Inf. Dilution	dH:-1043.(4SF)	4	MTD	6422
27	227	0	Inf. Dilution	dH:-1042.(4SF)	4	MTD	6494
28	327	0	Inf. Dilution	dH:-1042.(4SF)	4	MTD	6422
29	327	0	Inf. Dilution	dH:-1041.(4SF)	4	MTD	6494

Reaction No. 56349							
Ba(s) + Si(s) + 1.5<O2(g)> = BaO.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:269.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:268.1(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:197.4(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:155.0(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:126.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-368.1(0.05SD)	4	ENS	802

7	25	0	Inf. Dilution	dG:-368.1(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-388.0(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-388.0(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-388.0(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-387.9(4SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-388.1(4SF)	4	MTD	6422

Reaction No. 56350

Ba(s) + 2<Si(s)> + 2.5<O2(g)> = BaO.2SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:422.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:419.6(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:308.720(6SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:242.2(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:197.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-576.2(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-576.2(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-609.0(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-609.0(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-608.9(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-608.9(4SF)	4	MTD	6422

Reaction No. 56351

2<Ba(s)> + 2<O2(g)> + Si(s) = 2BaO.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:381.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:378.6(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:279.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:219.3(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:179.5(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-519.8(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-519.9(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-546.8(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-546.8(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-546.7(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-546.7(4SF)	4	MTD	6422

12	327	0	Inf. Dilution	dH:-547.1(4SF)	4	MTD	6422

Reaction No. 56352							
2<Ba(s)> + 4<O2(g)> + 3<Si(s)> = 2BaO.3SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:694.3(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:689.8(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:507.6(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:398.4(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:325.5(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-947.2(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-947.2(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-1000.2(0.1SD)	4	ENS	802
9	25	0	Inf. Dilution	dH:-1000.(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-1000.(4SF)	3	MTD	6422
11	227	0	Inf. Dilution	dH:-1000.(4SF)	2	MTD	6422
12	327	0	Inf. Dilution	dH:-1000.(4SF)	0	MTD	6422

Reaction No. 56355							
6<Al(s)> + 6.5<O2(g)> + 2<Si(s)> = 3Al2O3.2SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1129.(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:1129.(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:1121.(4SF)	3	ENS	6422
4	27	0	Inf. Dilution	lgK:1121.(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:824.5(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:824.3(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:646.3(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:646.1(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:527.5(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:527.4(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-1541.2(0.05SD)	0	ENS	802
12	25	0	Inf. Dilution	dG:-1540.(4SF)	3	EFE	6422
13	25	0	Inf. Dilution	dG:-1540.(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dG:-6432.7(5SF)kJ	0	CRV	9015
15	25	0	Inf. Dilution	dG:-1811.(4SF)	0	CRV	12011

16	25	0	Inf. Dilution	dH:-1632.8(0.1SD)	4	ENS	802
17	25	0	Inf. Dilution	dH:-1637.(4SF)	2	MTD	6422
18	25	0	Inf. Dilution	dH:-1630.(4SF)	2	MTD	6494
19	25	0	Inf. Dilution	dH:-6816.2(5SF)kJ	0	CRV	9015
20	25	0	Inf. Dilution	dH:-1903.(4SF)	0	CRV	12011
21	127	0	Inf. Dilution	dH:-1631.(4SF)	1	MTD	6422
22	127	0	Inf. Dilution	dH:-1630.(4SF)	1	MTD	6494
23	227	0	Inf. Dilution	dH:-1630.(4SF)	0	MTD	6422
24	227	0	Inf. Dilution	dH:-1630.(4SF)	0	MTD	6494
25	327	0	Inf. Dilution	dH:-1630.(4SF)	0	MTD	6422
26	327	0	Inf. Dilution	dH:-1629.(4SF)	0	MTD	6494

Reaction No. 56369

2<Be(s)> + Si(s) + 2<O2(g)> = Be2SiO4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:355.9(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:355.4(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:353.5(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:353.0(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:260.1(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:259.7(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:204.1(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:203.8(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:166.7(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:166.4(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-485.8(0.05SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-485.5(4SF)	4	EFE	6422
13	25	0	Inf. Dilution	dG:-484.8(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-513.7(0.1SD)	4	ENS	802
15	25	0	Inf. Dilution	dH:-512.8(4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-512.2(4SF)	4	MTD	6494
17	127	0	Inf. Dilution	dH:-512.9(4SF)	4	MTD	6422
18	127	0	Inf. Dilution	dH:-512.4(4SF)	4	MTD	6494
19	227	0	Inf. Dilution	dH:-512.9(4SF)	4	MTD	6422
20	227	0	Inf. Dilution	dH:-512.3(4SF)	4	MTD	6494
21	327	0	Inf. Dilution	dH:-512.7(4SF)	4	MTD	6422

22	327	0	Inf. Dilution	dH:-512.0(4SF)	4	MTD	6494

Reaction No. 56373							
H2(g) + 3<Mg(s)> + 6<O2(g)> + 4<Si(s)> = Mg3Si4O10(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:971.0(4SF)	0	ENS	6422
2	25	0	Inf. Dilution	lgK:967.1(4SF)	3	ENS	6494
3	27	0	Inf. Dilution	lgK:964.6(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:960.7(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:706.8(4SF)	0	ENS	6422
6	127	0	Inf. Dilution	lgK:703.8(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:552.1(4SF)	0	ENS	6422
8	227	0	Inf. Dilution	lgK:549.8(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:449.0(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:447.1(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-1324.8(0.05SD)	0	ENS	802
12	25	0	Inf. Dilution	dG:-1325.(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-1318.8(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-1319.(4SF)	4	EFE	6494
15	25	0	GAMSPATH	dG:-5524.(4SF)kJ	0	CRV	12638
16	25	0	Inf. Dilution	dH:-1415.5(0.1SD)	0	ENS	802
17	25	0	Inf. Dilution	dH:-1415.(4SF)	0	MTD	6422
18	25	0	Inf. Dilution	dH:-1410.3(5SF)	4	ENS	6437
19	25	0	Inf. Dilution	dH:-1409.5(5SF)	3	EFE	6482
20	25	0	Inf. Dilution	dH:-1410.(4SF)	4	MTD	6494
21	25	0	GAMSPATH	dH:-5903.3(5SF)kJ	0	CRV	12638
22	127	0	Inf. Dilution	dH:-1416.(4SF)	0	MTD	6422
23	127	0	Inf. Dilution	dH:-1410.(4SF)	1	MTD	6494
24	227	0	Inf. Dilution	dH:-1416.(4SF)	0	MTD	6422
25	227	0	Inf. Dilution	dH:-1410.(4SF)	0	MTD	6494
26	327	0	Inf. Dilution	dH:-1415.(4SF)	0	MTD	6422
27	327	0	Inf. Dilution	dH:-1409.(4SF)	0	MTD	6494

Reaction No. 56374							
2<Mg(s)> + 4<Al(s)> + 5<Si(s)> + 9<O2(g)> = 2MgO.2Al2O3.5SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:1516.(4SF)	5	ENS	6422
2	25	0	Inf. Dilution	lgK:1516.(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:1506.(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:1506.(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:1107.(4SF)	5	ENS	6422
6	127	0	Inf. Dilution	lgK:1107.(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:867.7(4SF)	5	ENS	6422
8	227	0	Inf. Dilution	lgK:867.5(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:708.3(4SF)	5	ENS	6422
10	327	0	Inf. Dilution	lgK:707.9(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-2055.0(0.1SD)	0	ENS	802
12	25	0	Inf. Dilution	dG:-2068.(4SF)	5	EFE	6422
13	25	0	Inf. Dilution	dG:-2067.8(5SF)	4	EFE	6482
14	25	0	Inf. Dilution	dG:-2068.(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dH:-2177.0(0.1SD)	0	ENS	802
Note (15): Low wgt for internal consistency							
16	25	0	Inf. Dilution	dH:-2190.(4SF)	5	MTD	6422
17	25	0	Inf. Dilution	dH:-2190.(4SF)	4	ENS	6437
18	25	0	Inf. Dilution	dH:-2189.0(5SF)	4	EFE	6482
19	25	0	Inf. Dilution	dH:-2190.(4SF)	4	MTD	6494
20	127	0	Inf. Dilution	dH:-2190.(4SF)	5	MTD	6422
21	127	0	Inf. Dilution	dH:-2190.(4SF)	4	MTD	6494
22	227	0	Inf. Dilution	dH:-2189.(4SF)	5	MTD	6422
23	227	0	Inf. Dilution	dH:-2190.(4SF)	4	MTD	6494
24	327	0	Inf. Dilution	dH:-2187.(4SF)	5	MTD	6422
25	327	0	Inf. Dilution	dH:-2190.(4SF)	4	MTD	6494

Reaction No. 56390

1.5<O2(g)> + Si(s) + Sr(s) = Sr+2_SiO3-2_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:271.3(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:269.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:198.5(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:155.8(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:127.4(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-370.4(0.05SD)	4	ENS	802

7	25	0	Inf. Dilution	dG:-370.4(0.05SD)	5	ENS	802
8	25	0	Inf. Dilution	dG:-370.2(4SF)	4	EFE	6422
9	25	0	Inf. Dilution/m	dG:-1549.70(6SF)kJ	3	CRV	24953
10	25	0	Inf. Dilution	dH:-390.5(0.1SD)	4	ENS	802
11	25	0	Inf. Dilution	dH:-390.5(0.1SD)	5	ENS	802
12	25	0	Inf. Dilution	dH:-390.5(4SF)	4	MTD	6422
13	25	0	Inf. Dilution/m	dH:-1633.90(6SF)kJ	3	CRV	24953
14	127	0	Inf. Dilution	dH:-390.4(4SF)	2	MTD	6422
15	227	0	Inf. Dilution	dH:-390.2(4SF)	1	MTD	6422
16	327	0	Inf. Dilution	dH:-389.9(4SF)	0	MTD	6422

Reaction No. 56391							
2<O2(g)> + Si(s) + 2<Sr(s)> = Sr2SiO4(s)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:383.5(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:381.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:280.7(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:220.6(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:180.6(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-523.7(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-523.7(0.05SD)	5	ENS	802
8	25	0	Inf. Dilution	dG:-523.2(4SF)	3	EFE	6422
9	25	0	Inf. Dilution/m	dG:-2191.10(6SF)kJ	3	CRV	24953
10	25	0	Inf. Dilution	dH:-550.8(0.1SD)	4	ENS	802
11	25	0	Inf. Dilution	dH:-550.8(0.1SD)	5	ENS	802
12	25	0	Inf. Dilution	dH:-550.7(4SF)	4	MTD	6422
13	25	0	Inf. Dilution/m	dH:-2304.50(6SF)kJ	3	CRV	24953
14	127	0	Inf. Dilution	dH:-550.4(4SF)	2	MTD	6422
15	227	0	Inf. Dilution	dH:-549.8(4SF)	1	MTD	6422
16	327	0	Inf. Dilution	dH:-549.2(4SF)	0	MTD	6422

Reaction No. 56397							
2<O2(g)> + Si(s) + Zr(s) = ZrSiO4(s)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:334.5(4SF)	2	ENS	6422
2	25	0	Inf. Dilution	lgK:336.3(4SF)	4	ENS	6494

3	27	0	Inf. Dilution	lgK:332.3(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:334.1(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:244.2(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:245.6(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:191.4(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:192.5(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:156.2(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:157.1(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-458.7(0.05SD)	4	ENS	802
12	25	0	Inf. Dilution	dG:-458.7(0.05SD)	5	ENS	802
13	25	0	Inf. Dilution	dG:-456.3(4SF)	0	EFE	6422
14	25	0	Inf. Dilution	dG:-458.8(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-1919.7(3.1MD)kJ	6	CRV	20829
16	25	0	Inf. Dilution	dH:-486.0(0.1SD)	4	ENS	802
17	25	0	Inf. Dilution	dH:-486.0(0.1SD)	5	ENS	802
18	25	0	Inf. Dilution	dH:-483.7(4SF)	2	MTD	6422
19	25	0	Inf. Dilution	dH:-486.2(4SF)	4	MTD	6494
20	25	0	Inf. Dilution	dH:-2034.2(3.1MD)kJ	6	CRV	20829
21	127	0	Inf. Dilution	dH:-483.7(4SF)	1	MTD	6422
22	127	0	Inf. Dilution	dH:-486.1(4SF)	1	MTD	6494
23	227	0	Inf. Dilution	dH:-483.4(4SF)	0	MTD	6422
24	227	0	Inf. Dilution	dH:-485.9(4SF)	0	MTD	6494
25	327	0	Inf. Dilution	dH:-483.1(4SF)	0	MTD	6422
26	327	0	Inf. Dilution	dH:-485.5(4SF)	0	MTD	6494
27	25	0	Inf. Dilution	Cp:98.200(0.6MD)J/K	5	CRV	20829

Reaction No. 56398							
2<Zn(s)> + Si(s) + 2<O2(g)> = Zn2SiO4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:268.2(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:266.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:194.8(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:151.9(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:123.3(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-364.1(0.05SD)	4	ENS	802
7	25	0	Inf. Dilution	dG:-364.1(0.05SD)	5	ENS	802

8	25	0	Inf. Dilution	dG:-365.9(4SF)	4	ENS	6422
9	25	0	Inf. Dilution	dH:-391.2(0.1SD)	4	ENS	802
10	25	0	Inf. Dilution	dH:-391.2(0.1SD)	5	ENS	802
11	25	0	Inf. Dilution	dH:-393.0(4SF)	4	MTD	6422
12	127	0	Inf. Dilution	dH:-393.0(4SF)	4	MTD	6422
13	227	0	Inf. Dilution	dH:-392.8(4SF)	4	MTD	6422
14	327	0	Inf. Dilution	dH:-392.5(4SF)	4	MTD	6422

Reaction No. 56821

V(s) + Si(s) = VSi(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-73.(1.SD)	4	ENS	3006

Reaction No. 56822

2<V(s)> + Si(s) = V2Si(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-37.(1.SD)	4	ENS	3006

Reaction No. 56823

3<V(s)> + Si(s) = V3Si(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-26.(1.SD)	4	ENS	3006

Reaction No. 56824

5<V(s)> + 3<Si(s)> = V5Si3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-94.(1.SD)	4	ENS	3006

Reaction No. 56884

Co(s) + Si(s) = CoSi(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-23.6(0.1SD)	6	ENS	802
2	25	0	Inf. Dilution	dH:-24.0(0.1SD)	6	ENS	802

Reaction No. 56885

Co(s) + 2<Si(s)> = CoSi2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	dH:-24.6(0.1SD)	6	ENS	802

Reaction No. 56886

Co(s) + 3<Si(s)> = CoSi3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-25.6(0.1SD)	6	ENS	802

Reaction No. 56887

2<Co(s)> + Si(s) + 2<O2(g)> = 2CoO.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:229.6(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:229.3(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:228.0(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:227.7(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:166.7(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:166.3(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:130.0(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:129.5(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:105.5(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:105.0(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-313.2(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-312.8(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-353.(0.5SD)	0	ENS	802
Note (13): Low wgt for internal consistency							
14	25	0	Inf. Dilution	dH:-336.7(4SF)	3	MTD	6422
Note (14): Low wgt for internal consistency							
15	25	0	Inf. Dilution	dH:-337.5(4SF)	4	MTD	6494
16	127	0	Inf. Dilution	dH:-336.5(4SF)	4	MTD	6422
17	127	0	Inf. Dilution	dH:-337.2(4SF)	4	MTD	6494
18	227	0	Inf. Dilution	dH:-336.1(4SF)	4	MTD	6422
19	227	0	Inf. Dilution	dH:-336.8(4SF)	4	MTD	6494
20	327	0	Inf. Dilution	dH:-335.7(4SF)	4	MTD	6422
21	327	0	Inf. Dilution	dH:-336.2(4SF)	4	MTD	6494

Reaction No. 57064

KAlSi3O8(Adul.,s) + 4<H2O> + 4<H+1> = K+1 + Al+3 + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:3.4(2SF)	1	ELC	
2	25	0	CHEMVAL1	lgK:1.26(0.01SD)	0	ENS	5156
3	25	0	CHEMVAL2	lgK:1.26(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL3	lgK:1.26(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL4	lgK:1.26(0.01SD)	0	ENS	5156

Reaction No. 57065

KAlSi3O8(San.,s) + 4<H2O> + 4<H+1> = K+1 + Al+3 + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.5(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1.25(3SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:5.20(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:5.20(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:5.20(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:5.19(0.01SD)	0	ENS	5156

Reaction No. 57066

NaAlSi2O6(Jade.,s) + 2<H2O> + 4<H+1> = Na+1 + Al+3 + 2<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.7(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:8.71(3SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:9.42(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:9.42(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:9.42(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:9.41(0.01SD)	0	ENS	5156

Reaction No. 57067

NaAlSi3O8(AlbiteLO,s) + 4<H+1> + 4<H2O> = Na+1 + Al+3 + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.6(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:3.09(3SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:6.43(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:6.43(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:6.43(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:6.43(0.01SD)	0	ENS	5156

Reaction No. 57068

Al2SiO5(Andalu.,s) + 6<H+1> = 2<Al+3> + H+1(2)_SiH2O4-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:16.596(5SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:14.20(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:14.20(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:14.20(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:14.20(0.01SD)	0	ENS	5156

Reaction No. 57069

Al2SiO5(Sillim.,s) + 6<H+1> = 2<Al+3> + H+1(2)_SiH2O4-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:15.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:16.960(5SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:14.50(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:14.50(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:14.50(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:14.50(0.01SD)	0	ENS	5156

Reaction No. 57414

Co+2 + H+1_SiH2O4-2 = Co+2_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	1	Unknown	lgK:4.51(0.06SD)	2	EOD	29656

Note (1): Same authors (in press)

Reaction No. 57469

H+1(2)_SiH2O4-2 + OH-1 = H+1_SiH2O4-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.1(2SF)	1	EOR	
2	25	3	Inert	lgK:5.1(2SF)	1	EOR	
3	50	1	NaCl	lgK:3.96(0.02SD)	7	MHY	5277
4	50	5	NaCl	lgK:4.32(0.02SD)	7	MHY	5277
5	100	0	Inf. Dilution/m	lgK:3.17(0.02SD)	5	MPT	4914
6	100	0.5	NaCl/m	lgK:3.29(0.01SD)	5	MPT	4914
7	100	1	NaCl/m	lgK:3.35(0.01SD)	5	MPT	4914
8	100	3	NaCl/m	lgK:3.52(0.01SD)	5	MPT	4914
9	100	5	NaCl/m	lgK:3.67(0.02SD)	5	MPT	4914

10	125	0	Inf. Dilution/m	lgK:2.94(0.02SD)	5	MPT	4914
11	150	0	Inf. Dilution/m	lgK:2.75(0.02SD)	5	MPT	4914
12	175	0	Inf. Dilution/m	lgK:2.59(0.02SD)	5	MPT	4914
13	200	0	Inf. Dilution/m	lgK:2.45(0.02SD)	5	MPT	4914
14	225	0	Inf. Dilution/m	lgK:2.33(0.02SD)	5	MPT	4914
15	250	0	Inf. Dilution/m	lgK:2.23(0.03SD)	5	MPT	4914
16	275	0	Inf. Dilution/m	lgK:2.15(0.03SD)	5	MPT	4914
17	300	0	Inf. Dilution/m	lgK:2.08(0.04SD)	5	MPT	4914
18	300	1	NaCl	lgK:2.20(0.02SD)	7	MHY	5277
19	300	5	NaCl	lgK:2.26(0.02SD)	7	MHY	5277
20	100	0	Inf. Dilution/m	dH:-26.48(0.46SD)kJ	5	MPT	4914
21	100	0.5	NaCl/m	dH:-26.78(0.46SD)kJ	5	MPT	4914
22	100	1	NaCl/m	dH:-27.11(0.46SD)kJ	5	MPT	4914
23	100	3	NaCl/m	dH:-28.28(0.50SD)kJ	5	MPT	4914
24	100	5	NaCl/m	dH:-29.50(0.63SD)kJ	5	MPT	4914
25	125	0	Inf. Dilution/m	dH:-25.27(0.38SD)kJ	5	MPT	4914
26	150	0	Inf. Dilution/m	dH:-24.02(0.50SD)kJ	5	MPT	4914
27	175	0	Inf. Dilution/m	dH:-22.80(0.67SD)kJ	5	MPT	4914
28	200	0	Inf. Dilution/m	dH:-21.55(0.92SD)kJ	5	MPT	4914
29	225	0	Inf. Dilution/m	dH:-20.33(1.17SD)kJ	5	MPT	4914
30	250	0	Inf. Dilution/m	dH:-19.08(1.42SD)kJ	5	MPT	4914
31	275	0	Inf. Dilution/m	dH:-17.87(1.67SD)kJ	5	MPT	4914
32	300	0	Inf. Dilution/m	dH:-16.61(1.97SD)kJ	5	MPT	4914
33	100	0	Inf. Dilution/m	Cp:49.4(10.9SD)J/K	5	MPT	4914
34	100	0.5	NaCl/m	Cp:46.9(10.9SD)J/K	5	MPT	4914
35	100	1	NaCl/m	Cp:44.4(10.5SD)J/K	5	MPT	4914
36	100	3	NaCl/m	Cp:34.7(10.0SD)J/K	5	MPT	4914
37	100	5	NaCl/m	Cp:25.1(10.0SD)J/K	5	MPT	4914
38	125	0	Inf. Dilution/m	Cp:49.4(10.9SD)J/K	5	MPT	4914
39	150	0	Inf. Dilution/m	Cp:49.4(10.9SD)J/K	5	MPT	4914
40	175	0	Inf. Dilution/m	Cp:49.4(10.9SD)J/K	5	MPT	4914
41	200	0	Inf. Dilution/m	Cp:49.4(10.9SD)J/K	5	MPT	4914
42	225	0	Inf. Dilution/m	Cp:49.4(10.9SD)J/K	5	MPT	4914
43	250	0	Inf. Dilution/m	Cp:49.4(10.9SD)J/K	5	MPT	4914
44	275	0	Inf. Dilution/m	Cp:49.4(10.9SD)J/K	5	MPT	4914
45	300	0	Inf. Dilution/m	Cp:49.4(10.9SD)J/K	5	MPT	4914

Reaction No. 57975							
Si(s) = Si(g)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:411.3(4SF)kJ	3	ENS	7275
2	25	0	Inf. Dilution	dG:405.5(4SF)kJ	5	CRV	8746
3	25	0	Inf. Dilution	dG:405.53(8.000SD)kJ	6	CRV	14923
4	25	0	Inf. Dilution	dH:108.(2.SD)	6	CRV	5605
5	25	0	Inf. Dilution	dH:455.6(4SF)kJ	3	ENS	7275
6	25	0	Inf. Dilution	dH:450.(8.MD)kJ	5	CRV	8746
7	25	0	Inf. Dilution	dH:450.00(8.000SD)kJ	6	CRV	14923
8	25	0	Inf. Dilution	Cp:22.251(0.001SD)J/K	5	CRV	27292

Reaction No. 57976							
O2(g) + Si(s) = SiO2(Quartz,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:150.0(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:150.0(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:149.1(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:149.0(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:109.4(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:109.4(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:85.62(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:85.60(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:69.77(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:69.75(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-856.25(5SF)kJ	4	CRV	440
12	25	0	Inf. Dilution	dG:-856.29(5SF)kJ	4	CRV	5619
13	25	0	Inf. Dilution	dG:-856.25(5SF)kJ	4	CRV	6421
14	25	0	Inf. Dilution	dG:-204.7(4SF)	2	EFE	6422
15	25	0	Inf. Dilution	dG:-204.66(5SF)	5	EFE	6482
16	25	0	Inf. Dilution	dG:-204.7(4SF)	2	EFE	6494
17	25	0	Inf. Dilution	dG:-856.64(5SF)kJ	1	ENS	7275
18	25	0	Inf. Dilution	dG:-856.64(5SF)kJ	1	CRV	8695
19	25	0	Inf. Dilution	dG:-856.3(4SF)kJ	5	CRV	8746
20	25	0	Inf. Dilution	dG:-856.29(5SF)kJ	4	CRV	8872

21	25	0	Inf. Dilution	dG:-856.51(5SF)kJ	2	EOD	8872
Note (21): Data from Naumov, Ryzhenko & Khodakovsky, 1971							
22	25	0	Inf. Dilution	dG:-856.05(5SF)kJ	2	CRV	8873
23	25	0	Inf. Dilution	dG:-856.21(5SF)kJ	4	CRV	8874
24	25	0	Inf. Dilution	dG:-204.65(5SF)	3	CRV	10174
25	25	0	Inf. Dilution	dG:-856.29(1.002SD)kJ	6	CRV	14923
26	25	0	Inf. Dilution	dG:-856.28(5SF)kJ	5	CRV	29482
27	25	0	Inf. Dilution	dG:-856.29(5SF)kJ	5	EOD	29520
28	25	0	Inf. Dilution	dG:-204.48(5SF)	3	CRV	29553
29	25	0	Inf. Dilution	dG:-854.79(5SF)kJ	2	CRV	31308
30	25	0	Inf. Dilution	dG:-204.650(6SF)	5	EFE	32152
31	25	0	Inf. Dilution/m	dG:-856.280(6SF)kJ	3	CRV	24953
32	50	0	Inf. Dilution	dG:-204.90(5SF)	0	CRV	10174
33	75	0	Inf. Dilution	dG:-205.18(5SF)	0	CRV	10174
34	100	0	Inf. Dilution	dG:-205.49(5SF)	0	CRV	10174
35	125	0	Inf. Dilution	dG:-205.81(5SF)	0	CRV	10174
36	150	0	Inf. Dilution	dG:-206.15(5SF)	0	CRV	10174
37	175	0	Inf. Dilution	dG:-206.51(5SF)	0	CRV	10174
38	200	0	Inf. Dilution	dG:-206.88(5SF)	0	CRV	10174
39	225	0	Inf. Dilution	dG:-207.28(5SF)	0	CRV	10174
40	250	0	Inf. Dilution	dG:-207.68(5SF)	0	CRV	10174
41	300	0	Inf. Dilution	dG:-208.54(5SF)	0	CRV	10174
42	25	0	Inf. Dilution	dH:-217.6(0.2SD)	2	CRV	5605
43	25	0	Inf. Dilution	dH:-217.7(4SF)	4	MTD	6422
44	25	0	Inf. Dilution	dH:-217.66(5SF)	5	EFE	6482
45	25	0	Inf. Dilution	dH:-217.7(4SF)	4	MTD	6494
46	25	0	Inf. Dilution	dH:-910.94(5SF)kJ	5	ENS	7275
47	25	0	Inf. Dilution	dH:-910.94(5SF)kJ	5	CRV	8695
48	25	0	Inf. Dilution	dH:-910.7(1.0MD)kJ	5	CRV	8746
49	25	0	Inf. Dilution	dH:-910.70(1.000SD)kJ	6	CRV	14923
50	25	0	Inf. Dilution	dH:-910.70(5SF)kJ	5	CRV	29482
51	25	0	Inf. Dilution	dH:-910.7(4SF)kJ	5	EOD	29520
52	25	0	Inf. Dilution	dH:-217.48(5SF)	3	CRV	29553
53	25	0	Inf. Dilution	dH:-909.(3SF)kJ	2	CRV	31308
54	25	0	GAMSPATH	dH:-910.650(6SF)kJ	3	CRV	12638
55	25	0	Inf. Dilution/m	dH:-910.700(6SF)kJ	3	CRV	24953

56	127	0	Inf. Dilution	dH:-217.7(4SF)	2	MTD	6422
57	127	0	Inf. Dilution	dH:-217.7(4SF)	2	MTD	6494
58	227	0	Inf. Dilution	dH:-217.7(4SF)	2	MTD	6422
59	227	0	Inf. Dilution	dH:-217.6(4SF)	2	MTD	6494
60	327	0	Inf. Dilution	dH:-217.5(4SF)	0	MTD	6422
61	327	0	Inf. Dilution	dH:-217.4(4SF)	0	MTD	6494
62	25	0	Inf. Dilution	Cp:44.602(0.15SD)J/K	5	CRV	27292

Reaction No. 57977

2<F2(g)> + Si(s) = SiF4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1573.(4SF)kJ	5	CRV	8746
2	25	0	Inf. Dilution	dG:-1572.8(0.814SD)kJ	6	CRV	14923
3	25	0	Inf. Dilution	dH:-386.0(0.2SD)	5	CRV	5605
4	25	0	Inf. Dilution	dH:-1615.0(0.8MD)kJ	5	CRV	8746
5	25	0	Inf. Dilution	dH:-1615.0(0.800SD)kJ	6	CRV	14923
6	25	0	Inf. Dilution	Cp:73.622(0.25SD)J/K	5	CRV	27292

Reaction No. 58082

Ca4SO4AlSiF13.10H2O(s) = 4<Ca+2> + Al+3 + 4<H+1> + H+1(2)_SiH2O4-2 + 6<H2O> + SO4-2 + 13 <F-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:-78.77(4SF)	3	ENS	
Note (1): http://www.fipr.state.fl.us/fipr100.htm							
2	25	0	Inf. Dilution	lgK:-78.8(3SF)	2	EAE	

Reaction No. 58127

6<H+1> + SiH2O4-2 + 6<F-1> = H+1(6)_F-1(6)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:53.19(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:52.05(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:51.85(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:51.75(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:51.70(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:51.69(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:51.73(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:51.78(0.01SD)	4	EDH	5390

Reaction No. 58128							
4<H+1> + SiH2O4-2 + 2<F-1> = H+1(4)_F-1(2)_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:34.38(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:33.42(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:33.23(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:32.13(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:32.04(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:32.01(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:32.02(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:32.02(0.01SD)	4	EDH	5390

Reaction No. 58185							
2<H+1> + SiH2O4-2 = SiO2(am.,s) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:25.98(4SF)	0	CRV	9402
3	25	0	Inf. Dilution	dH:-19.40(4SF)	2	CRV	9402

Reaction No. 58186							
2<H+1> + SiH2O4-2 = SiO2(Quartz,s) + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:26.94(4SF)	4	CRV	9402
2	25	0	Inf. Dilution	dH:-21.00(4SF)	4	CRV	9402

Reaction No. 58203							
2<Mn+2> + SiH2O4-2 = 2MnO.SiO2(s) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-0.5(1SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:10.70(4SF)	0	CRV	9402
Note (2): Low wgt for inter-reaction consistency							

Reaction No. 58204							
2<Fe+2> + SiH2O4-2 = 2FeO.SiO2(s) + 2<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.0(2SF)	1	ELC	

2	25	0	Inf. Dilution	lgK:5.74(3SF)	0	CRV	9402
3	25	0	Inf. Dilution	dH:21.70(4SF)	2	CRV	9402
4	25	0	Inf. Dilution	Cp:-34.00(4SF)	2	CRV	9402

Reaction No. 58205

Sr+2 + SiH2O4-2 = Sr+2_SiO3-2_(s) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:2.40(3SF)	0	CRV	9402

Reaction No. 58219

H+1 + UO2+2 + SiH2O4-2 = UO2+2_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	UCT_Sea	lgK:21.06(0.01SD)	4	EDH	5390
2	25	0.1	UCT_Sea	lgK:20.30(0.01SD)	4	EDH	5390
3	25	0.2	UCT_Sea	lgK:20.15(0.01SD)	4	EDH	5390
4	25	0.3	UCT_Sea	lgK:20.07(0.01SD)	4	EDH	5390
5	25	0.4	UCT_Sea	lgK:20.01(0.01SD)	4	EDH	5390
6	25	0.5	UCT_Sea	lgK:19.97(0.01SD)	4	EDH	5390
7	25	0.6	UCT_Sea	lgK:19.95(0.01SD)	4	EDH	5390
8	25	0.7	UCT_Sea	lgK:19.93(0.01SD)	4	EDH	5390

Reaction No. 59987

6<H+1> + 4<SiH2O4-2> = H+1(6)_SiH2O4-2(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:79.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:78.20(4SF)	0	CRV	9402

Reaction No. 65765

Mg+2(2)_H+1(3)_SiH2O4-2(3)_(s) = 2<Mg+2> + 3<H+1_SiH2O4-2>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	50	0	Inf. Dilution	lgK:-38.8(3SF)	0	CRV	552

Note (1): Solid +1 charge! Same problem in S&M hardcopy [76SmM_8]. Sepiolite?

Reaction No. 65766

UO2+2 + H+1_SiH2O4-2 = UO2+2_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:7.97(0.10SD)	5	CRV	25812
2	25	0.1	Unknown	lgK:7.5(2SF)	6	CRV	552

Reaction No. 65767

Fe+3 + H+1_SiH2O4-2 = Fe+3_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:9.3(2SF)	1	ELC	
2	25	0	Inf. Dilution/m	lgK:9.72(0.04SD)	4	MSP	29280
3	25	0	Inf. Dilution/m	lgK:9.7(0.3SD)	2	EOD	29280
Note (3): Taken from Hummel (2012), The PSI/Nagra Chemical Thermo Database							
4	25	0.1	Unknown	lgK:8.9(2SF)	0	CRV	552
5	20:30	0.1	Unknown	dH:4.(1SF)	0	CRV	552

Reaction No. 65965

H+1(2)_SiH2O4-2 + 4<H+1> + 6<F-1> = Si+4_F-1(6) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.1(0.08SD)	4	CRV	552
2	25	0	Inf. Dilution	lgK:27.01(4SF)	0	CRV	32224
3	25	0.1	Unknown	lgK:29.48(4SF)	6	CRV	552
4	25	1	Unknown	lgK:29.96(0.02SD)	6	CRV	552
5	25	2	Unknown	lgK:30.88(4SF)	6	CRV	552
6	25	3	Unknown	lgK:32.33(4SF)	6	CRV	552

Reaction No. 66082

CaO.Al2O3.2SiO2(s) + 2<H+1> + 6<H2O> = 2<Al+3_OH-1(3)_(Gibbsite,s)> + 2<H+1(2)_SiH2O4-2 > + Ca+2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:11.16(4SF)	0	ENS	6405

Reaction No. 66239

Zn(s) + Si(s) + 1.5<O2(g)> = ZnSiO3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:206.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:205.3(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:150.3(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:117.3(4SF)	4	ENS	6422

5	327	0	Inf. Dilution	lgK:95.32(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-281.9(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-301.8(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-301.9(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-301.9(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-302.0(4SF)	4	MTD	6422

Reaction No. 66424

Si(s) + 2<S(s)> = Si+4_S-2(2)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:37.25(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:37.02(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:27.73(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:22.04(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:18.22(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-50.81(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-51.0(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-51.8(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-52.3(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-52.7(3SF)	4	MTD	6422

Reaction No. 66425

3<Si(s)> + 2<N2(g)> = Si3N4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:113.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:112.6(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:80.15(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:60.64(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:47.62(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-154.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-178.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-178.4(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-178.7(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-178.8(4SF)	4	MTD	6422

Reaction No. 66426

2<I2(s)> + Si(s) = Si+4_I-1(4)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:33.57(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:33.37(4SF)	0	ENS	6422
3	121	0	Inf. Dilution	lgK:25.40(4SF)	2	ENS	6422
4	25	0	Inf. Dilution	dG:-45.80(4SF)	3	EFE	6422
5	25	0	Inf. Dilution	dG:-45.802(5SF)	3	CRV	31291
6	25	0	Inf. Dilution	dH:-45.30(4SF)	2	MTD	6422
7	25	0	Inf. Dilution	dH:-45.300(5SF)	3	CRV	31291
8	121	0	Inf. Dilution	dH:-53.36(4SF)	1	MTD	6422

Reaction No. 66427

Si(s) + C(s) = SiC(alpha,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.41(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:12.33(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:9.143(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:7.228(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:5.952(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-16.93(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-17.5(3SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-17.5(3SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-17.5(3SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-17.5(3SF)	4	MTD	6422

Reaction No. 66428

Si(s) + 2<Br2(l)> = Si+4_Br-1(4)_(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:77.77(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:77.27(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:55.82(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:42.46(4SF)	0	ENS	6422
5	25	0	Inf. Dilution	dG:-106.1(4SF)	4	EFE	6422
6	25	0	Inf. Dilution	dH:-109.3(4SF)	4	MTD	6422
7	127	0	Inf. Dilution	dH:-122.8(4SF)	0	MTD	6422
8	227	0	Inf. Dilution	dH:-121.6(4SF)	0	MTD	6422

Reaction No. 66477

2<Al(s)> + 2.5<O2(g)> + Si(s) = Al2SiO5(Kyan.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:427.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:428.2(4SF)	2	ENS	6422
3	25	0	Inf. Dilution	lgK:428.0(4SF)	2	ENS	6494
4	27	0	Inf. Dilution	lgK:425.4(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:425.2(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:312.4(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:312.3(4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:244.6(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:244.5(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:199.5(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:199.4(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-584.1(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-583.98(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-583.9(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-2443.90(6SF)kJ	0	CRV	9015
16	25	0	Inf. Dilution	dG:-583.88(5SF)	0	CRV	10174
17	25	0	Inf. Dilution	dG:-620.5(4SF)	0	CRV	12011
18	50	0	Inf. Dilution	dG:-584.41(5SF)	0	CRV	10174
19	75	0	Inf. Dilution	dG:-585.00(5SF)	0	CRV	10174
20	100	0	Inf. Dilution	dG:-585.65(5SF)	0	CRV	10174
21	125	0	Inf. Dilution	dG:-586.36(5SF)	0	CRV	10174
22	150	0	Inf. Dilution	dG:-587.12(5SF)	0	CRV	10174
23	175	0	Inf. Dilution	dG:-587.94(5SF)	0	CRV	10174
24	200	0	Inf. Dilution	dG:-588.80(5SF)	0	CRV	10174
25	225	0	Inf. Dilution	dG:-589.72(5SF)	0	CRV	10174
26	250	0	Inf. Dilution	dG:-590.67(5SF)	0	CRV	10174
27	300	0	Inf. Dilution	dG:-592.72(5SF)	0	CRV	10174
28	25	0	Inf. Dilution	dH:-620.0(4SF)	0	MTD	6422
29	25	0	Inf. Dilution	dH:-620.5(4SF)	0	ENS	6437
30	25	0	Inf. Dilution	dH:-620.03(5SF)	0	EFE	6482
31	25	0	Inf. Dilution	dH:-619.9(4SF)	0	MTD	6494
32	25	0	Inf. Dilution	dH:-2594.30(6SF)kJ	0	CRV	9015
33	25	0	Inf. Dilution	dH:-656.4(4SF)	0	CRV	12011
34	127	0	Inf. Dilution	dH:-620.2(4SF)	0	MTD	6422

35	127	0	Inf. Dilution	dH:-620.1(4SF)	0	MTD	6494
36	227	0	Inf. Dilution	dH:-620.1(4SF)	0	MTD	6422
37	227	0	Inf. Dilution	dH:-620.0(4SF)	0	MTD	6494
38	327	0	Inf. Dilution	dH:-619.8(4SF)	0	MTD	6422
39	327	0	Inf. Dilution	dH:-619.7(4SF)	0	MTD	6494

Reaction No. 66478

2<Al(s)> + 2.5<O2(g)> + Si(s) = Al2SiO5(Andalu.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:427.(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:428.0(4SF)	0	ENS	6422
3	25	0	Inf. Dilution	lgK:427.8(4SF)	2	ENS	6494
4	27	0	Inf. Dilution	lgK:425.2(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:425.0(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:312.4(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:312.2(4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:244.7(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:244.6(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:199.6(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:199.5(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-583.9(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-583.60(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-583.6(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-2442.70(6SF)kJ	0	CRV	9015
16	25	0	Inf. Dilution	dG:-583.51(5SF)	0	CRV	10174
17	25	0	Inf. Dilution	dG:-620.8(4SF)	0	CRV	12011
18	25	0	Inf. Dilution	dG:-583.16(5SF)	5	CRV	29553
19	50	0	Inf. Dilution	dG:-584.10(5SF)	0	CRV	10174
20	75	0	Inf. Dilution	dG:-584.75(5SF)	0	CRV	10174
21	100	0	Inf. Dilution	dG:-585.45(5SF)	0	CRV	10174
22	125	0	Inf. Dilution	dG:-586.21(5SF)	0	CRV	10174
23	150	0	Inf. Dilution	dG:-587.03(5SF)	0	CRV	10174
24	175	0	Inf. Dilution	dG:-587.90(5SF)	0	CRV	10174
25	200	0	Inf. Dilution	dG:-588.82(5SF)	0	CRV	10174
26	225	0	Inf. Dilution	dG:-589.79(5SF)	0	CRV	10174
27	250	0	Inf. Dilution	dG:-590.80(5SF)	0	CRV	10174

28	300	0	Inf. Dilution	dG:-592.95(5SF)	0	CRV	10174
29	25	0	Inf. Dilution	dH:-619.1(4SF)	0	MTD	6422
30	25	0	Inf. Dilution	dH:-619.4(4SF)	0	ENS	6437
31	25	0	Inf. Dilution	dH:-619.02(5SF)	0	EFE	6482
32	25	0	Inf. Dilution	dH:-619.0(4SF)	0	MTD	6494
33	25	0	Inf. Dilution	dH:-2590.30(6SF)kJ	0	CRV	9015
34	25	0	Inf. Dilution	dH:-655.9(4SF)	0	CRV	12011
35	25	0	Inf. Dilution	dH:-618.57(5SF)	5	CRV	29553
36	127	0	Inf. Dilution	dH:-619.3(4SF)	0	MTD	6422
37	127	0	Inf. Dilution	dH:-619.2(4SF)	0	MTD	6494
38	227	0	Inf. Dilution	dH:-619.2(4SF)	0	MTD	6422
39	227	0	Inf. Dilution	dH:-619.0(4SF)	0	MTD	6494
40	327	0	Inf. Dilution	dH:-618.9(4SF)	0	MTD	6422
41	327	0	Inf. Dilution	dH:-618.7(4SF)	0	MTD	6494

Reaction No. 66479

2<Al(s)> + 2.5<O2(g)> + Si(s) = Al2SiO5(Sillim.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:427.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:427.7(4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:427.3(4SF)	3	ENS	6494
4	27	0	Inf. Dilution	lgK:424.9(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:424.5(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:312.2(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:311.9(4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:244.6(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:244.4(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:199.5(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:199.3(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-583.4(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-583.00(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-582.9(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-2441.00(6SF)kJ	0	CRV	9015
16	25	0	Inf. Dilution	dG:-583.02(5SF)	0	CRV	10174
17	25	0	Inf. Dilution	dG:-627.6(4SF)	0	CRV	12011
18	50	0	Inf. Dilution	dG:-583.63(5SF)	0	CRV	10174

19	75	0	Inf. Dilution	dG:-584.30(5SF)	0	CRV	10174
20	100	0	Inf. Dilution	dG:-585.02(5SF)	0	CRV	10174
21	125	0	Inf. Dilution	dG:-585.81(5SF)	0	CRV	10174
22	150	0	Inf. Dilution	dG:-586.65(5SF)	0	CRV	10174
23	175	0	Inf. Dilution	dG:-587.54(5SF)	0	CRV	10174
24	200	0	Inf. Dilution	dG:-588.48(5SF)	0	CRV	10174
25	225	0	Inf. Dilution	dG:-589.46(5SF)	0	CRV	10174
26	250	0	Inf. Dilution	dG:-590.49(5SF)	0	CRV	10174
27	300	0	Inf. Dilution	dG:-592.68(5SF)	0	CRV	10174
28	25	0	Inf. Dilution	dH:-618.5(4SF)	0	MTD	6422
29	25	0	Inf. Dilution	dH:-618.5(4SF)	0	ENS	6437
30	25	0	Inf. Dilution	dH:-618.09(5SF)	0	EFE	6482
31	25	0	Inf. Dilution	dH:-618.1(4SF)	0	MTD	6494
32	25	0	Inf. Dilution	dH:-2587.80(6SF)kJ	0	CRV	9015
33	25	0	Inf. Dilution	dH:-662.6(4SF)	0	CRV	12011
34	127	0	Inf. Dilution	dH:-618.7(4SF)	0	MTD	6422
35	127	0	Inf. Dilution	dH:-618.3(4SF)	0	MTD	6494
36	227	0	Inf. Dilution	dH:-618.6(4SF)	0	MTD	6422
37	227	0	Inf. Dilution	dH:-618.2(4SF)	0	MTD	6494
38	327	0	Inf. Dilution	dH:-618.4(4SF)	0	MTD	6422
39	327	0	Inf. Dilution	dH:-617.9(4SF)	0	MTD	6494

Reaction No. 66480

2<Al(s)> + 2<H2(g)> + 4.5<O2(g)> + 2<Si(s)> = Al2Si2O7.2H2O(Dick.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:665.0(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:665.0(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:660.6(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:660.6(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:481.2(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:481.3(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:373.6(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:373.7(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:301.9(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:302.0(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-907.2(4SF)	4	EFE	6422

12	25	0	Inf. Dilution	dG:-907.3(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dG:-3795.9(5SF)kJ	4	CRV	9015
14	25	0	GAMSPATH	dG:-3795.(4SF)kJ	3	CRV	12638
15	25	0	Inf. Dilution	dH:-984.3(4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-984.3(4SF)	4	MTD	6494
17	25	0	Inf. Dilution	dH:-4118.3(5SF)kJ	4	CRV	9015
18	27	0	Inf. Dilution	dH:-984.4(4SF)	0	MTD	6494
19	127	0	Inf. Dilution	dH:-984.8(4SF)	3	MTD	6422
20	127	0	Inf. Dilution	dH:-984.7(4SF)	3	MTD	6494
21	227	0	Inf. Dilution	dH:-984.7(4SF)	2	MTD	6422
22	227	0	Inf. Dilution	dH:-984.6(4SF)	2	MTD	6494
23	327	0	Inf. Dilution	dH:-984.2(4SF)	0	MTD	6422
24	327	0	Inf. Dilution	dH:-984.0(4SF)	0	MTD	6494

Reaction No. 66498

4<Ca(s)> + C(s) + 6<Si(s)> + 6<Al(s)> + 13.5<O2(g)> = CaCO3.3CaO.3Al2O3.6SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2301.(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:2286.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:1681.(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:1319.(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:1077.(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-3132.4(5SF)	0	EFE	6482
Note (6): Low wgt for internal consistency							
7	25	0	Inf. Dilution	dG:-3139.(4SF)	4	EFE	6494
8	25	0	Inf. Dilution	dH:-3315.3(5SF)	4	ENS	6437
9	25	0	Inf. Dilution	dH:-3310.2(5SF)	0	EFE	6482
10	25	0	Inf. Dilution	dH:-3318.(4SF)	4	MTD	6494
11	127	0	Inf. Dilution	dH:-3318.(4SF)	4	MTD	6494
12	227	0	Inf. Dilution	dH:-3317.(4SF)	4	MTD	6494
13	327	0	Inf. Dilution	dH:-3315.(4SF)	4	MTD	6494

Reaction No. 66535

2<Na(s)> + 3<Mg(s)> + 2<Al(s)> + 8<Si(s)> + H2(g) + 12<O2(g) > =
Na2Mg3Al2Si8O22(OH)2(Glauc.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1968.(4SF)	4	ENS	6494

2	27	0	Inf. Dilution	lgK:1955.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:1434.(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:1121.(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:912.6(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-2684.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-2857.(4SF)	4	ENS	6437
8	25	0	Inf. Dilution	dH:-2860.(4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-2862.(4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-2861.(4SF)	4	MTD	6494
11	327	0	Inf. Dilution	dH:-2860.(4SF)	4	MTD	6494

Reaction No. 66557

Ca(s) + 2<Al(s)> + Si(s) + 3<O2(g)> = CaO.Al2O3.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:544.7(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:547.0(4SF)	0	ENS	6422
3	25	0	Inf. Dilution	lgK:548.3(4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:543.4(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:544.7(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:399.8(4SF)	0	ENS	6422
7	127	0	Inf. Dilution	lgK:400.8(4SF)	0	ENS	6494
8	227	0	Inf. Dilution	lgK:313.7(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:314.4(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:256.4(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:256.9(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-746.2(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-746.25(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-748.0(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dH:-788.3(4SF)	2	MTD	6422
16	25	0	Inf. Dilution	dH:-788.42(5SF)	0	EFE	6482
17	25	0	Inf. Dilution	dH:-790.2(4SF)	0	MTD	6494
18	127	0	Inf. Dilution	dH:-788.2(4SF)	0	MTD	6422
19	127	0	Inf. Dilution	dH:-790.4(4SF)	0	MTD	6494
20	227	0	Inf. Dilution	dH:-787.8(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-790.2(4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-787.1(4SF)	0	MTD	6422

23	327	0	Inf. Dilution	dH:-789.7(4SF)	0	MTD	6494

Reaction No. 66558							
2<Al(s)> + Ca(s) + 4<O2(g)> + 2<Si(s)> = CaO.Al2O3.2SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:701.2(4SF)	0	ENS	6422
2	25	0	Inf. Dilution	lgK:702.2(4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:696.6(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:697.6(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:512.6(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:513.2(4SF)	0	ENS	6494
7	227	0	Inf. Dilution	lgK:402.1(4SF)	0	ENS	6422
8	227	0	Inf. Dilution	lgK:402.7(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:328.6(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:329.0(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-956.6(4SF)	0	EFE	6422
12	25	0	Inf. Dilution	dG:-956.8(4SF)	0	EFE	6482
13	25	0	Inf. Dilution	dG:-957.9(4SF)	0	EFE	6494
14	25	0	Inf. Dilution	dG:-4002.10(6SF)kJ	6	EFE	8798
Note (14): To avoid confusion note source units (J/mol) & value c.a.= -4 million							
15	25	0	Inf. Dilution	dG:-4002.1(5SF)kJ	4	MTD	12843
16	25	0	Inf. Dilution	dH:-1010.(4SF)	4	MTD	6422
17	25	0	Inf. Dilution	dH:-1011.6(5SF)	0	ENS	6437
18	25	0	Inf. Dilution	dH:-1010.7(5SF)	0	EFE	6482
19	25	0	Inf. Dilution	dH:-1012.(4SF)	0	MTD	6494
20	25	0	Inf. Dilution	dH:-4227.80(6SF)kJ	6	EFE	8798
21	25	0	Inf. Dilution	dH:-4227.8(5SF)kJ	4	MTD	12843
22	25	0	GAMSPATH	dH:-4216.5(5SF)kJ	0	CRV	12638
23	127	0	Inf. Dilution	dH:-1011.(4SF)	2	MTD	6422
24	127	0	Inf. Dilution	dH:-1012.(4SF)	2	MTD	6494
25	227	0	Inf. Dilution	dH:-1010.(4SF)	1	MTD	6422
26	227	0	Inf. Dilution	dH:-1012.(4SF)	1	MTD	6494
27	327	0	Inf. Dilution	dH:-1010.(4SF)	0	MTD	6422
28	327	0	Inf. Dilution	dH:-1011.(4SF)	0	MTD	6494

Reaction No. 66559

2<Ca(s)> + 2<Al(s)> + Si(s) + 3.5<O2(g)> = 2CaO.Al2O3.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:662.7(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:663.2(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:658.4(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:658.9(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:485.1(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:485.4(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:381.2(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:381.3(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:311.9(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:312.0(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-904.1(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-904.86(5SF)	4	EFE	6482
13	25	0	Inf. Dilution	dG:-904.8(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-951.6(4SF)	4	MTD	6422
15	25	0	Inf. Dilution	dH:-950.7(4SF)	4	ENS	6437
16	25	0	Inf. Dilution	dH:-953.19(5SF)	4	EFE	6482
17	25	0	Inf. Dilution	dH:-952.4(4SF)	4	MTD	6494
18	127	0	Inf. Dilution	dH:-951.5(4SF)	4	MTD	6422
19	127	0	Inf. Dilution	dH:-952.6(4SF)	4	MTD	6494
20	227	0	Inf. Dilution	dH:-951.0(4SF)	4	MTD	6422
21	227	0	Inf. Dilution	dH:-952.4(4SF)	4	MTD	6494
22	327	0	Inf. Dilution	dH:-950.4(4SF)	4	MTD	6422
23	327	0	Inf. Dilution	dH:-952.0(4SF)	4	MTD	6494

Reaction No. 66560

3<Ca(s)> + 2<Al(s)> + 3<Si(s)> + 6<O2(g)> = 3CaO.Al2O3.3SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1098.0(5SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1100.(4SF)	0	ENS	6422
3	25	0	Inf. Dilution	lgK:1100.(4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:1093.(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:1093.(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:803.8(4SF)	0	ENS	6422
7	127	0	Inf. Dilution	lgK:803.7(4SF)	0	ENS	6494

8	227	0	Inf. Dilution	lgK:630.2(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:630.3(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:514.6(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:514.7(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-1501.(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-1498.8(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-1501.(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dH:-1588.(4SF)	2	MTD	6422
16	25	0	Inf. Dilution	dH:-1585.6(5SF)	0	ENS	6437
17	25	0	Inf. Dilution	dH:-1585.3(5SF)	0	EFE	6482
18	25	0	Inf. Dilution	dH:-1587.(4SF)	0	MTD	6494
19	127	0	Inf. Dilution	dH:-1589.(4SF)	0	MTD	6422
20	127	0	Inf. Dilution	dH:-1587.(4SF)	0	MTD	6494
21	227	0	Inf. Dilution	dH:-1588.0(5SF)	0	MTD	6422
22	227	0	Inf. Dilution	dH:-1587.(4SF)	0	MTD	6494
23	327	0	Inf. Dilution	dH:-1587.(4SF)	0	MTD	6422
24	327	0	Inf. Dilution	dH:-1586.(4SF)	0	MTD	6494

Reaction No. 66561							
Ca(s) + 2<Al(s)> + 2<Si(s)> + 5<O2(g)> + 2<H2(g)> = CaO.Al2O3.2SiO2.2H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:789.3(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:790.6(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:784.0(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:785.4(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:572.4(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:573.4(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:445.4(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:446.2(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:360.7(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:361.4(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-1077.(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-1077.90(6SF)	4	EFE	6482
13	25	0	Inf. Dilution	dG:-1079.(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-1161.2(5SF)	4	MTD	6422
15	25	0	Inf. Dilution	dH:-1163.5(5SF)	4	ENS	6437

16	25	0	Inf. Dilution	dH:-1162.9(5SF)	4	EFE	6482
17	25	0	Inf. Dilution	dH:-1164.(4SF)	4	MTD	6494
18	127	0	Inf. Dilution	dH:-1162.(4SF)	1	MTD	6422
19	127	0	Inf. Dilution	dH:-1164.(4SF)	1	MTD	6494
20	227	0	Inf. Dilution	dH:-1163.(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-1164.(4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-1163.(4SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-1163.(4SF)	0	MTD	6494

Reaction No. 66571

Ca(s) + Mg(s) + Si(s) + 2<O2(g)> = CaO.MgO.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:374.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:375.9(4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:373.6(4SF)	3	ENS	6494
4	27	0	Inf. Dilution	lgK:373.5(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:371.2(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:275.0(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:273.2(4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:215.9(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:214.4(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:176.5(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:175.3(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-512.8(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-509.61(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-509.8(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dH:-540.9(4SF)	0	MTD	6422
16	25	0	Inf. Dilution	dH:-536.9(4SF)	0	ENS	6437
17	25	0	Inf. Dilution	dH:-537.77(5SF)	0	EFE	6482
18	25	0	Inf. Dilution	dH:-538.0(4SF)	0	MTD	6494
19	127	0	Inf. Dilution	dH:-540.8(4SF)	0	MTD	6422
20	127	0	Inf. Dilution	dH:-538.0(4SF)	0	MTD	6494
21	227	0	Inf. Dilution	dH:-540.6(4SF)	0	MTD	6422
22	227	0	Inf. Dilution	dH:-537.7(4SF)	0	MTD	6494
23	327	0	Inf. Dilution	dH:-540.2(4SF)	0	MTD	6422
24	327	0	Inf. Dilution	dH:-537.4(4SF)	0	MTD	6494

Reaction No. 66572

Ca(s) + Mg(s) + 3<O2(g)> + 2<Si(s)> = CaO.MgO.2SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:531.2(4SF)	2	ENS	6422
2	25	0	Inf. Dilution	lgK:382.6(4SF)	0	ENS	6437
3	25	0	Inf. Dilution	lgK:530.3(4SF)	4	ENS	6494
4	27	0	Inf. Dilution	lgK:527.7(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:526.8(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:388.1(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:387.5(4SF)	3	ENS	6494
8	227	0	Inf. Dilution	lgK:304.4(4SF)	2	ENS	6422
9	227	0	Inf. Dilution	lgK:303.9(4SF)	2	ENS	6494
10	327	0	Inf. Dilution	lgK:248.6(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:248.1(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-724.7(4SF)	2	EFE	6422
13	25	0	Inf. Dilution	dG:-723.28(5SF)	3	EFE	6482
14	25	0	Inf. Dilution	dG:-723.4(4SF)	4	EFE	6494
15	25	0	Inf. Dilution	dG:-3026.8(5SF)kJ	3	EOD	12843
16	25	0	Inf. Dilution	dH:-766.3(4SF)	2	MTD	6422
17	25	0	Inf. Dilution	dH:-764.96(5SF)	4	EFE	6482
18	25	0	Inf. Dilution	dH:-765.2(4SF)	4	MTD	6494
19	25	0	Inf. Dilution	dH:-3201.5(5SF)kJ	3	EOD	12843
20	127	0	Inf. Dilution	dH:-766.4(4SF)	2	MTD	6422
21	127	0	Inf. Dilution	dH:-765.2(4SF)	2	MTD	6494
22	227	0	Inf. Dilution	dH:-766.1(4SF)	2	MTD	6422
23	227	0	Inf. Dilution	dH:-764.9(4SF)	2	MTD	6494
24	327	0	Inf. Dilution	dH:-765.6(4SF)	0	MTD	6422
25	327	0	Inf. Dilution	dH:-764.4(4SF)	0	MTD	6494

Reaction No. 66573

2<Ca(s)> + Mg(s) + 2<Si(s)> + 3.5<O2(g)> = 2CaO.MgO.2SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:643.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:644.7(4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:642.5(4SF)	3	ENS	6494

4	27	0	Inf. Dilution	lgK:640.5(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:638.4(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:471.7(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:470.2(4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:370.5(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:369.3(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:303.0(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:302.1(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-879.5(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-875.66(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-876.6(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dH:-926.7(4SF)	0	MTD	6422
16	25	0	Inf. Dilution	dH:-924.4(4SF)	0	ENS	6437
17	25	0	Inf. Dilution	dH:-922.67(5SF)	0	MTD	6482
18	25	0	Inf. Dilution	dH:-923.7(4SF)	0	MTD	6494
19	127	0	Inf. Dilution	dH:-926.6(4SF)	0	MTD	6422
20	127	0	Inf. Dilution	dH:-923.6(4SF)	0	MTD	6494
21	227	0	Inf. Dilution	dH:-926.2(4SF)	0	MTD	6422
22	227	0	Inf. Dilution	dH:-923.3(4SF)	0	MTD	6494
23	327	0	Inf. Dilution	dH:-925.7(4SF)	0	MTD	6422
24	327	0	Inf. Dilution	dH:-923.0(4SF)	0	MTD	6494

Reaction No. 66574

2<Ca(s)> + 5<Mg(s)> + 8<Si(s)> + 12<O2(g)> + H2(g) = 2CaO.5MgO.8SiO2.H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2038.(4SF)	0	ENS	6422
2	25	0	Inf. Dilution	lgK:2028.(4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:2025.(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:2014.(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:1487.(4SF)	0	ENS	6422
6	127	0	Inf. Dilution	lgK:1479.(4SF)	1	ENS	6494
7	227	0	Inf. Dilution	lgK:1164.(4SF)	0	ENS	6422
8	227	0	Inf. Dilution	lgK:1158.(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:948.6(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:943.5(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-2780.(4SF)	0	EFE	6422

12	25	0	Inf. Dilution	dG:-2767.3(5SF)	3	EFE	6482
13	25	0	Inf. Dilution	dG:-2766.(4SF)	0	EFE	6494
14	25	0	Inf. Dilution	dH:-2954.(4SF)	0	MTD	6422
15	25	0	Inf. Dilution	dH:-2943.(4SF)	2	ENS	6437
16	25	0	Inf. Dilution	dH:-2941.1(5SF)	3	EFE	6482
17	25	0	Inf. Dilution	dH:-2941.(4SF)	0	MTD	6494
18	127	0	Inf. Dilution	dH:-2954.(4SF)	0	MTD	6422
19	127	0	Inf. Dilution	dH:-2941.(4SF)	0	MTD	6494
20	227	0	Inf. Dilution	dH:-2954.(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-2940.(4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-2952.2(5SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-2938.(4SF)	0	MTD	6494

Reaction No. 66575

Ca(s) + 2<Si(s)> + 3.5<O2(g)> + 2<H2(g)> = CaO.2SiO2.2H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:503.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:500.0(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:363.3(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:281.2(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:226.4(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-686.8(4SF)	4	ENS	6422
7	25	0	Inf. Dilution	dH:-750.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-751.3(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-752.1(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-752.8(4SF)	4	MTD	6422

Reaction No. 66576

2<Ca(s)> + 3<Si(s)> + 5.25<O2(g)> + 2.5<H2(g)> = 2CaO.3SiO2.5by2H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:795.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:790.3(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:576.1(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:447.5(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:361.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1085.(4SF)	4	EFE	6422

7	25	0	Inf. Dilution	dH:-1176.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1177.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1176.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1176.(4SF)	4	MTD	6422

Reaction No. 66577							
2<Ca(s)> + Si(s) + 2.585<O2(g)> + 1.165<H2(g)> = 2CaO.SiO2.7by6H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:434.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:431.6(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:315.5(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:245.9(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:199.5(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-592.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-637.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-637.1(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-636.9(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-636.5(4SF)	4	MTD	6422

Reaction No. 66578							
3<Ca(s)> + 2<Si(s)> + 5<O2(g)> + 3<H2(g)> = 3CaO.2SiO2.3H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:771.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:766.5(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:558.3(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:433.4(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:350.2(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1053.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1143.0(5SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1143.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1142.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1141.(4SF)	4	MTD	6422

Reaction No. 66579							
4<Ca(s)> + 3<Si(s)> + 5.75<O2(g)> + 1.5<H2(g)> = 4CaO.3SiO2.3by2H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:988.6(4SF)	4	ENS	6422

2	27	0	Inf. Dilution	lgK:982.1(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:720.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:562.8(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:458.0(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1349.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1439.0(5SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1439.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1439.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1438.(4SF)	4	MTD	6422

Reaction No. 66580

5<Ca(s)> + 6<Si(s)> + 10<O2(g)> + 3<H2(g)> = 5CaO.6SiO2.3H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1622.(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:1611.(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:1179.(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:919.7(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:746.9(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-2213.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-2373.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-2373.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-2372.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-2371.(4SF)	4	MTD	6422

Reaction No. 66581

5<Ca(s)> + 6<Si(s)> + 11.25<O2(g)> + 5.5<H2(g)> = 5CaO.6SiO2.11by2H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1729.(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:1718.(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:1253.(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:973.5(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:787.6(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-2359.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-2554.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-2554.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-2553.(4SF)	4	MTD	6422

10	327	0	Inf. Dilution	dH:-2551.(4SF)	4	MTD	6422

Reaction No. 66582

5<Ca(s)> + 6<Si(s)> + 13.75<O2(g)> + 10.5<H2(g)> = 5CaO.6SiO2.21by2H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1940.(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:1927.(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:1397.(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:1079.(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:866.9(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-2647.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-2911.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-2911.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-2910.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-2906.5(5SF)	4	MTD	6422

Reaction No. 66583

6<Ca(s)> + 6<Si(s)> + 9.5<O2(g)> + H2(g) = 6CaO.6SiO2.H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1655.(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:1644.(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:1208.(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:946.3(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:771.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-2258.(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-2396.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-2396.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-2396.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-2395.(4SF)	4	MTD	6422

Reaction No. 66587

Ca(s) + Si(s) + Ti(s) + 2.5<O2(g)> = CaO.SiO2.TiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:431.3(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:430.0(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:428.5(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:427.2(4SF)	0	ENS	6494

5	127	0	Inf. Dilution	lgK:315.2(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:314.2(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:247.2(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:246.4(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:201.9(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:201.2(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-588.4(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-586.79(5SF)	5	EFE	6482
13	25	0	Inf. Dilution	dG:-586.7(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-622.2(4SF)	4	MTD	6422
15	25	0	Inf. Dilution	dH:-620.61(5SF)	5	EFE	6482
16	25	0	Inf. Dilution	dH:-620.6(4SF)	4	MTD	6494
17	127	0	Inf. Dilution	dH:-622.1(4SF)	1	MTD	6422
18	127	0	Inf. Dilution	dH:-620.5(4SF)	1	MTD	6494
19	227	0	Inf. Dilution	dH:-621.7(4SF)	0	MTD	6422
20	227	0	Inf. Dilution	dH:-620.2(4SF)	0	MTD	6494
21	327	0	Inf. Dilution	dH:-621.3(4SF)	0	MTD	6422
22	327	0	Inf. Dilution	dH:-619.8(4SF)	0	MTD	6494

Reaction No. 66611

Cd(s) + Si(s) + 1.5<O2(g)> = CdSiO3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:193.7(4SF)	3	ENS	6422
2	27	0	Inf. Dilution	lgK:192.4(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:140.6(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:109.6(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:88.87(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-264.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-1105.40(6SF)kJ	4	CRV	9015
8	25	0	Inf. Dilution	dG:-264.20(5SF)	4	CRV	12011
9	25	0	Inf. Dilution	dH:-284.2(4SF)	4	MTD	6422
10	25	0	Inf. Dilution	dH:-1189.10(6SF)kJ	4	CRV	9015
11	25	0	Inf. Dilution	dH:-284.20(5SF)	4	CRV	12011
12	127	0	Inf. Dilution	dH:-284.2(4SF)	1	MTD	6422
13	227	0	Inf. Dilution	dH:-284.0(4SF)	0	MTD	6422
14	327	0	Inf. Dilution	dH:-285.3(4SF)	0	MTD	6422

Reaction No. 66729							
Fe(s) + 1.5<O2(g)> + Si(s) = FeO.SiO2(s)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:195.8(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:195.9(4SF)	3	ENS	6494
3	27	0	Inf. Dilution	lgK:194.5(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:194.6(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:142.5(4SF)	2	ENS	6422
6	127	0	Inf. Dilution	lgK:142.6(4SF)	2	ENS	6494
7	227	0	Inf. Dilution	lgK:111.3(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:111.4(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:90.51(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:90.64(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-267.1(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-267.08(5SF)	4	EFE	6482
13	25	0	Inf. Dilution	dG:-267.2(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dG:-1118.0(5SF)kJ	3	EOD	12843
15	25	0	Inf. Dilution	dH:-285.6(4SF)	4	MTD	6422
16	25	0	Inf. Dilution	dH:-285.24(5SF)	3	ENS	6437
17	25	0	Inf. Dilution	dH:-285.46(5SF)	4	EFE	6482
18	25	0	Inf. Dilution	dH:-285.7(4SF)	4	MTD	6494
19	25	0	Inf. Dilution	dH:-1195.2(5SF)kJ	3	EOD	12843
20	127	0	Inf. Dilution	dH:-285.5(4SF)	1	MTD	6422
21	127	0	Inf. Dilution	dH:-285.4(4SF)	1	MTD	6494
22	227	0	Inf. Dilution	dH:-285.3(4SF)	0	MTD	6422
23	227	0	Inf. Dilution	dH:-285.0(4SF)	0	MTD	6494
24	327	0	Inf. Dilution	dH:-285.1(4SF)	0	MTD	6422
25	327	0	Inf. Dilution	dH:-284.6(4SF)	0	MTD	6494

Reaction No. 66776							
3<Fe(s)> + 2<Al(s)> + 3<Si(s)> + 6<O2(g)> = 3FeO.Al2O3.3SiO2(s)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:865.8(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:860.1(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:630.9(4SF)	4	ENS	6494

4	227	0	Inf. Dilution	lgK:493.5(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:401.9(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1181.1(5SF)	4	EFE	6482
7	25	0	Inf. Dilution	dG:-1181.(4SF)	4	EFE	6494
8	25	0	Inf. Dilution	dH:-1261.1(5SF)	4	ENS	6437
9	25	0	Inf. Dilution	dH:-1258.5(5SF)	4	MTD	6482
10	25	0	Inf. Dilution	dH:-1258.(4SF)	4	MTD	6494
11	127	0	Inf. Dilution	dH:-1258.(4SF)	4	MTD	6494
12	227	0	Inf. Dilution	dH:-1258.(4SF)	4	MTD	6494
13	327	0	Inf. Dilution	dH:-1257.(4SF)	4	MTD	6494

Reaction No. 66777

3<Mg(s)> + 2<Al(s)> + 3<Si(s)> + 6<O2(g)> = 3MgO.Al2O3.3SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1040.(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1033.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:759.3(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:595.1(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:485.7(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1418.7(5SF)	4	EFE	6482
7	25	0	Inf. Dilution	dG:-1418.(4SF)	4	EFE	6494
8	25	0	Inf. Dilution	dH:-1503.7(5SF)	4	ENS	6437
9	25	0	Inf. Dilution	dH:-1502.5(5SF)	4	EFE	6482
10	25	0	Inf. Dilution	dH:-1502.(4SF)	4	MTD	6494
11	127	0	Inf. Dilution	dH:-1502.(4SF)	4	MTD	6494
12	227	0	Inf. Dilution	dH:-1502.(4SF)	4	MTD	6494
13	327	0	Inf. Dilution	dH:-1501.(4SF)	4	MTD	6494

Reaction No. 66778

2<Fe(s)> + 4<Al(s)> + 5<Si(s)> + 9<O2(g)> = 2FeO.2Al2O3.5SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2018.9(5SF)	4	ENS	6437

Reaction No. 66779

K(s) + 3<Fe(s)> + Al(s) + 3<Si(s)> + H2(g) + 6<O2(g)> = KFe3AlSi3O10(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:847.9(4SF)	1	ELC	

2	25	0	Inf. Dilution	lgK:840.6(4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:835.1(4SF)	0	ENS	6494
4	127	0	Inf. Dilution	lgK:610.9(4SF)	0	ENS	6494
5	227	0	Inf. Dilution	lgK:476.5(4SF)	0	ENS	6494
6	327	0	Inf. Dilution	lgK:386.9(4SF)	0	ENS	6494
7	25	0	Inf. Dilution	dG:-1147.(4SF)	0	EFE	6494
8	25	0	Inf. Dilution	dH:-1225.0(5SF)	0	ENS	6437
9	25	0	Inf. Dilution	dH:-1231.(4SF)	1	MTD	6494
10	127	0	Inf. Dilution	dH:-1231.(4SF)	0	MTD	6494
11	227	0	Inf. Dilution	dH:-1230.(4SF)	0	MTD	6494
12	327	0	Inf. Dilution	dH:-1228.(4SF)	0	MTD	6494

Reaction No. 66780							
K(s) + 3<Mg(s)> + Al(s) + 3<Si(s)> + H2(g) + 6<O2(g)> = KMg3AlSi3O10(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1024.0(5SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:1027.(4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:1020.(4SF)	0	ENS	6494
4	127	0	Inf. Dilution	lgK:748.0(4SF)	0	ENS	6494
5	227	0	Inf. Dilution	lgK:584.8(4SF)	0	ENS	6494
6	327	0	Inf. Dilution	lgK:476.0(4SF)	0	ENS	6494
7	25	0	Inf. Dilution	dG:-1392.7(5SF)	0	EFE	6482
8	25	0	Inf. Dilution	dG:-1401.(4SF)	0	EFE	6494
9	25	0	Inf. Dilution	dH:-1482.8(5SF)	2	ENS	6437
10	25	0	Inf. Dilution	dH:-1483.6(5SF)	2	EFE	6482
11	25	0	Inf. Dilution	dH:-1493.(4SF)	0	MTD	6494
12	127	0	Inf. Dilution	dH:-1494.(4SF)	0	MTD	6494
13	227	0	Inf. Dilution	dH:-1493.(4SF)	0	MTD	6494
14	327	0	Inf. Dilution	dH:-1493.(4SF)	0	MTD	6494

Reaction No. 66781							
2<Mg(s)> + 4<Al(s)> + 5<Si(s)> + H2(g) + 9.5<O2(g)> = 2MgO.2Al2O3.5SiO2.H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2253.9(5SF)	4	ENS	6437

Reaction No. 66782
31<H2(g)> + 48<Mg(s)> + 73.5<O2(g)> + 34<Si(s)> = Mg48Si34O85(OH)62(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.1597E+4(5SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1.5792E+4(5SF)	0	EFE	6482
3	25	0	GAMSPATH	dG:-6.614E+4(4SF)kJ	0	CRV	12638
Note (3): Low wgt for internal consistency							
4	25	0	Inf. Dilution	dH:-1.7060E+4(5SF)	0	ENS	6437
5	25	0	Inf. Dilution	dH:-1.7056E+4(5SF)	0	EFE	6482
6	25	0	GAMSPATH	dH:-7.14250E+4(6SF)kJ	0	CRV	12638

Reaction No. 66783							
3<Al(s)> + H2(g) + Na(s) + 6<O2(g)> + 3<Si(s)> = NaAl3Si3O10(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:975.6(4SF)	2	ENS	6494
2	27	0	Inf. Dilution	lgK:969.1(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:710.1(4SF)	0	ENS	6494
4	227	0	Inf. Dilution	lgK:554.6(4SF)	0	ENS	6494
5	327	0	Inf. Dilution	lgK:450.9(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-1329.7(5SF)	0	EFE	6482
7	25	0	Inf. Dilution	dG:-1331.(4SF)	2	EFE	6494
8	25	0	Inf. Dilution	dG:-1331.70(6SF)	0	CRV	10174
9	25	0	Inf. Dilution	dG:-1332.5(5SF)	5	CRV	29553
10	50	0	Inf. Dilution	dG:-1333.40(6SF)	0	CRV	10174
11	75	0	Inf. Dilution	dG:-1335.30(6SF)	0	CRV	10174
12	100	0	Inf. Dilution	dG:-1337.30(6SF)	0	CRV	10174
13	125	0	Inf. Dilution	dG:-1339.50(6SF)	0	CRV	10174
14	150	0	Inf. Dilution	dG:-1341.80(6SF)	0	CRV	10174
15	175	0	Inf. Dilution	dG:-1344.30(6SF)	0	CRV	10174
16	200	0	Inf. Dilution	dG:-1346.90(6SF)	0	CRV	10174
17	225	0	Inf. Dilution	dG:-1349.60(6SF)	0	CRV	10174
18	250	0	Inf. Dilution	dG:-1352.40(6SF)	0	CRV	10174
19	300	0	Inf. Dilution	dG:-1358.30(6SF)	0	CRV	10174
20	25	0	Inf. Dilution	dH:-1422.3(5SF)	4	ENS	6437
21	25	0	Inf. Dilution	dH:-1420.7(5SF)	0	EFE	6482
22	25	0	Inf. Dilution	dH:-1422.(4SF)	4	MTD	6494
23	25	0	Inf. Dilution	dH:-1423.60(6SF)	5	CRV	29553
24	127	0	Inf. Dilution	dH:-1423.(4SF)	0	MTD	6494

25	227	0	Inf. Dilution	dH:-1423.(4SF)	0	MTD	6494
26	327	0	Inf. Dilution	dH:-1422.(4SF)	0	MTD	6494

Reaction No. 66784

7<Fe(s)> + 8<Si(s)> + H2(g) + 12<O2(g)> = Fe7Si8O22(OH)2(Grun.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1571.(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1560.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:1141.(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:890.1(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:722.7(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-2143.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-2302.(4SF)	4	ENS	6437
8	25	0	Inf. Dilution	dH:-2300.(4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-2300.(4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-2298.(4SF)	4	MTD	6494
11	327	0	Inf. Dilution	dH:-2296.(4SF)	4	MTD	6494

Reaction No. 66785

7<Mg(s)> + 8<Si(s)> + H2(g) + 12<O2(g)> = Mg7Si8O22(OH)2(Cumm.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2920.(4SF)	4	ENS	6437

Reaction No. 66786

5<Ca(s)> + C(s) + 2<Si(s)> + 5.5<O2(g)> = CaCO3.4CaO.2SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:968.0(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:961.7(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:707.5(4SF)	3	ENS	6494
4	227	0	Inf. Dilution	lgK:555.1(4SF)	3	ENS	6494
5	327	0	Inf. Dilution	lgK:453.6(4SF)	3	ENS	6494
6	25	0	Inf. Dilution	dG:-1321.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1395.8(5SF)	3	ENS	6437
8	25	0	Inf. Dilution	dH:-1396.(4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-1395.(4SF)	3	MTD	6494
10	227	0	Inf. Dilution	dH:-1394.(4SF)	0	MTD	6494
11	327	0	Inf. Dilution	dH:-1393.(4SF)	0	MTD	6494

Reaction No. 66787

Mg(s) + 2<Al(s)> + 2<Si(s)> + 5<O2(g)> + 2<H2(g)> = MgAl2Si2O6(OH)4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1146.(4SF)	4	ENS	6437

Reaction No. 66788

Mg(s) + 2<Al(s)> + Si(s) + 3.5<O2(g)> + H2(g) = MgAl2SiO5(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-850.4(4SF)	4	ENS	6437

Reaction No. 66789

4<Mg(s)> + 18<Al(s)> + 7.5<Si(s)> + 24<O2(g)> + 2<H2(g) > = Mg4Al18Si7.5O48H4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-6003.5(5SF)	4	ENS	6437

Reaction No. 66790

Na(s) + Fe(s) + 2<Si(s)> + 3<O2(g)> = NaFeSi2O6(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:423.5(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:420.7(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:308.2(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:240.6(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:195.6(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-577.7(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-617.7(4SF)	4	ENS	6437
8	25	0	Inf. Dilution	dH:-617.7(4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-618.4(4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-618.2(4SF)	4	MTD	6494
11	327	0	Inf. Dilution	dH:-617.8(4SF)	4	MTD	6494

Reaction No. 66791

4<Mg(s)> + 4<Al(s)> + 2<Si(s)> + 9<O2(g)> + 4<H2(g)> = Mg4Al4Si2O10(OH)8(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2162.(4SF)	4	ENS	6437

Reaction No. 66792

Al(s) + H2(g) + K(s) + Mg(s) + 6<O2(g)> + 4<Si(s)> = KMgAlSi4O10(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5464.10(6SF)kJ	3	CRV	25435
2	25	0	GAMSPATH	dG:-7971.(4SF)kJ	0	CRV	12638
3	25	0	Inf. Dilution	dH:-1394.4(5SF)	2	ENS	6437
4	25	0	Inf. Dilution	dH:-5844.50(6SF)kJ	3	CRV	25435

Reaction No. 66793

2<Al(s)> + 5<Fe(s)> + 4<H2(g)> + 9<O2(g)> + 3<Si(s)> = Fe5Al2Si3O10(OH)8(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-6535.60(6SF)kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-1708.5(5SF)	2	ENS	6437
3	25	0	Inf. Dilution	dH:-7154.00(6SF)kJ	3	CRV	25435

Reaction No. 66794

18<Fe(s)> + 12<Si(s)> + 25<O2(g)> + 5<H2(g)> = Fe18Si12O40(OH)10(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-4384.4(5SF)	4	ENS	6437

Reaction No. 66795

K(s) + 2<Mg(s)> + 3<Al(s)> + 2<Si(s)> + 6<O2(g)> + H2(g) = KMg2Al3Si2O10(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1514.5(5SF)	4	ENS	6437

Reaction No. 66796

Na(s) + 2<Ca(s)> + 5<Mg(s)> + 7<Si(s)> + Al(s) + 12<O2(g)> + H2(g) = NaCa2Mg5Si7AlO22(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-3006.8(5SF)	4	ENS	6437

Reaction No. 66797

4<Al(s)> + 4<Fe(s)> + 4<H2(g)> + 9<O2(g)> + 2<Si(s)> = Fe4Al4Si2O10(OH)8(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	GAMSPATH	dG:-7167.(4SF)kJ	3	CRV	12638
2	25	0	Inf. Dilution	dH:-1823.5(5SF)	4	ENS	6437
3	25	0	GAMSPATH	dH:-7784.4(5SF)kJ	0	CRV	12638

Note (3): Called '14A-Chamosite'

Reaction No. 66798

7<Fe(s)> + 8<Si(s)> + 12<O2(g)> + H2(g) = Fe7Si8O22(OH)2(Anth.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2300.6(5SF)	4	ENS	6437

Reaction No. 66799

Fe(s) + 2<Al(s)> + 2<Si(s)> + 5<O2(g)> + 2<H2(g)> = FeAl2Si2O6(OH)4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1060.9(5SF)	4	ENS	6437

Reaction No. 66800

K(s) + Fe(s) + Al(s) + 4<Si(s)> + 6<O2(g)> + H2(g) = KFeAlSi4O10(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5127.80(6SF)kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-1310.9(5SF)	2	ENS	6437
3	25	0	Inf. Dilution	dH:-5497.30(6SF)kJ	3	CRV	25435

Reaction No. 66801

2<Na(s)> + 3<Fe(s)> + 2<Al(s)> + 8<Si(s)> + 12<O2(g)> + H2(g) =
Na2Fe3Al2Si8O22(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2605.4(5SF)	4	ENS	6437

Reaction No. 66802

2<Ca(s)> + 4<Fe(s)> + 2<Al(s)> + 7<Si(s)> + 12<O2(g)> + H2(g) =
Ca2Fe4Al2Si7O22(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2629.(4SF)	4	ENS	6437

Reaction No. 66803

2<Ca(s)> + 4<Mg(s)> + 2<Al(s)> + 7<Si(s)> + 12<O2(g)> + H2(g) =
Ca2Mg4Al2Si7O22(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2968.5(5SF)	4	ENS	6437

Reaction No. 66804

4<Fe(s)> + 18<Al(s)> + 7.5<Si(s)> + 24<O2(g)> + 2<H2(g) > = Fe4Al18Si7.5O48H4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-5675.2(5SF)	4	ENS	6437

Reaction No. 66805

2<Ca(s)> + 5<Fe(s)> + 8<Si(s)> + 12<O2(g)> + H2(g) = 2CaO.5FeO.8SiO2.H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2520.(3SF)	4	ENS	6437

Reaction No. 66806

2<Na(s)> + 3<Mg(s)> + 2<Al(s)> + 8<Si(s)> + 12<O2(g)> + H2(g) =
Na2Mg3Al2Si8O22(OH)2(Rieb. ,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-2650.(4SF)	4	ENS	6437

Reaction No. 66807

Na(s) + 2<Ca(s)> + 4<Mg(s)> + 3<Al(s)> + 6<Si(s)> + 12<O2(g) > + H2(g) =
NaCa2Mg4Al3Si6O22(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-3040.1(5SF)	4	ENS	6437

Reaction No. 66808

K(s) + 2<Fe(s)> + 3<Al(s)> + 2<Si(s)> + 6<O2(g)> + H2(g) = KFe2Al3Si2O10(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-1345.2(5SF)	4	ENS	6437

Reaction No. 66809

Ca(s) + 4<Al(s)> + 2<Si(s)> + 6<O2(g)> + H2(g) = CaAl4Si2O10(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1026.(4SF)	1	ENS	6494
2	27	0	Inf. Dilution	lgK:1020.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:747.8(4SF)	0	ENS	6494
4	227	0	Inf. Dilution	lgK:584.7(4SF)	0	ENS	6494
5	327	0	Inf. Dilution	lgK:476.0(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-1398.70(6SF)	1	EFE	6482
7	25	0	Inf. Dilution	dG:-1400.(4SF)	4	EFE	6494
8	25	0	Inf. Dilution	dH:-1492.5(5SF)	4	ENS	6437
9	25	0	Inf. Dilution	dH:-1490.6(5SF)	4	EFE	6482
10	25	0	Inf. Dilution	dH:-1492.(4SF)	1	MTD	6494
11	127	0	Inf. Dilution	dH:-1493.(4SF)	0	MTD	6494
12	227	0	Inf. Dilution	dH:-1493.(4SF)	0	MTD	6494

13	327	0	Inf. Dilution	dH:-1492.(4SF)	0	MTD	6494

Reaction No. 66810							
2<Al(s)> + 2<Ca(s)> + Fe(s) + 0.5<H2(g)> + 6.5<O2(g)> + 3<Si(s) > = Ca2FeAl2Si3O12(OH)(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	GAMSPATH	dG:-6072.(4SF)kJ	1	CRV	12638
2	25	0	Inf. Dilution	dH:-1544.5(5SF)	3	ENS	6437
3	25	0	GAMSPATH	dH:-6461.90(6SF)kJ	2	CRV	12638
Note (3): c.f. dG = -6072 & dH = -6462 kJ for 'Ordered Epidote'							

Reaction No. 66811							
3<Ca(s)> + Mg(s) + 2<Si(s)> + 4<O2(g)> = 3CaO.MgO.2SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:760.4(4SF)	2	ENS	6422
2	25	0	Inf. Dilution	lgK:754.7(4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:755.5(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:749.8(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:556.7(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:552.3(4SF)	0	ENS	6494
7	227	0	Inf. Dilution	lgK:437.4(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:433.9(4SF)	0	ENS	6494
9	327	0	Inf. Dilution	lgK:358.0(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:355.0(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-1037.(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-1030.00(6SF)	0	EFE	6482
13	25	0	Inf. Dilution	dG:-1030.(4SF)	0	EFE	6494
14	25	0	Inf. Dilution	dH:-1092.(4SF)	2	MTD	6422
15	25	0	Inf. Dilution	dH:-1084.4(5SF)	2	ENS	6437
16	25	0	Inf. Dilution	dH:-1084.5(5SF)	0	EFE	6482
17	25	0	Inf. Dilution	dH:-1084.(4SF)	0	MTD	6494
18	127	0	Inf. Dilution	dH:-1092.(4SF)	0	MTD	6422
19	127	0	Inf. Dilution	dH:-1084.(4SF)	0	MTD	6494
20	227	0	Inf. Dilution	dH:-1091.(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-1084.(4SF)	0	MTD	6494
22	327	0	Inf. Dilution	dH:-1090.(4SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-1083.(4SF)	0	MTD	6494

Reaction No. 66812

Fe(s) + 2<Al(s)> + Si(s) + 3.5<O2(g)> + H2(g) = FeAl2SiO5(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-767.54(5SF)	4	ENS	6437

Reaction No. 66814

Mg(s) + 1.5<O2(g)> + Si(s) = MgO.SiO2(Enstat.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:255.5(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:253.8(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:186.5(4SF)	2	ENS	6494
4	227	0	Inf. Dilution	lgK:146.2(4SF)	1	ENS	6494
5	327	0	Inf. Dilution	lgK:119.3(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-348.51(5SF)	4	EFE	6482
7	25	0	Inf. Dilution	dG:-348.5(4SF)	4	EFE	6494
8	25	0	Inf. Dilution	dG:-1458.3(5SF)kJ	3	EOD	12843
9	25	0	GAMSPATH	dG:-1480.(4SF)kJ	0	CRV	12638
10	25	0	Inf. Dilution	dH:-395.57(5SF)	0	ENS	6437
11	25	0	Inf. Dilution	dH:-369.40(5SF)	4	EFE	6482
12	25	0	Inf. Dilution	dH:-369.4(4SF)	4	MTD	6494
13	25	0	Inf. Dilution	dH:-1549.00(6SF)kJ	2	CRV	8695
14	25	0	Inf. Dilution	dH:-1545.6(5SF)kJ	3	EOD	12843
15	25	0	GAMSPATH	dH:-1546.80(6SF)kJ	3	CRV	12638
16	127	0	Inf. Dilution	dH:-369.4(4SF)	1	MTD	6494
17	227	0	Inf. Dilution	dH:-369.2(4SF)	0	MTD	6494
18	327	0	Inf. Dilution	dH:-369.0(4SF)	0	MTD	6494

Reaction No. 66815

K(s) + Al(s) + 2<Si(s)> + 3<O2(g)> = KAlSi2O6(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:503.1(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:503.7(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:499.8(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:500.4(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:367.6(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:368.1(4SF)	3	ENS	6494

7	227	0	Inf. Dilution	lgK:288.2(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:288.7(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:235.3(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:235.7(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-686.3(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-687.2(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-725.2(4SF)	4	MTD	6422
14	25	0	Inf. Dilution	dH:-722.(3SF)	4	ENS	6437
15	25	0	Inf. Dilution	dH:-726.1(4SF)	4	MTD	6494
16	127	0	Inf. Dilution	dH:-726.2(4SF)	3	MTD	6422
17	127	0	Inf. Dilution	dH:-726.9(4SF)	3	MTD	6494
18	227	0	Inf. Dilution	dH:-726.6(4SF)	2	MTD	6422
19	227	0	Inf. Dilution	dH:-726.8(4SF)	2	MTD	6494
20	327	0	Inf. Dilution	dH:-726.8(4SF)	0	MTD	6422
21	327	0	Inf. Dilution	dH:-726.4(4SF)	0	MTD	6494

Reaction No. 66816

2<Al(s)> + 4<H2(g)> + 5<Mg(s)> + 9<O2(g)> + 3<Si(s)> = Mg5Al2Si3O10(OH)8(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1446.3(5SF)	3	CRV	6494
2	25	0	Inf. Dilution	dG:-1971.9(5SF)	3	EFE	6482
3	25	0	Inf. Dilution	dG:-8263.40(6SF)kJ	1	CRV	25435
4	25	0	GAMSPATH	dG:-8186.(4SF)kJ	0	CRV	12638
Note (4): Low wgt for internal consistency							
5	25	0	Inf. Dilution	dH:-2133.9(5SF)	1	ENS	6437
6	25	0	Inf. Dilution	dH:-2129.4(5SF)	3	EFE	6482
7	25	0	Inf. Dilution	dH:-8929.90(6SF)kJ	1	CRV	25435
8	25	0	GAMSPATH	dH:-8841.6(5SF)kJ	0	CRV	12638

Reaction No. 66817

3<Al(s)> + 2<Ca(s)> + 0.5<H2(g)> + 6.5<O2(g)> + 3<Si(s) > = Ca2Al3Si3O12(OH)(ortho,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1140.(4SF)	1	EDH	
2	25	0	Inf. Dilution	lgK:1140.(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:1132.(4SF)	0	ENS	6494
4	127	0	Inf. Dilution	lgK:831.7(4SF)	2	ENS	6494

5	227	0	Inf. Dilution	lgK:651.4(4SF)	1	ENS	6494
6	327	0	Inf. Dilution	lgK:531.3(4SF)	0	ENS	6494
7	25	0	Inf. Dilution	dG:-1552.1(5SF)	0	EFE	6482
8	25	0	Inf. Dilution	dG:-1555.(4SF)	4	EFE	6494
9	25	0	GAMSPATH	dG:-6502.(4SF)kJ	0	CRV	12638
10	25	0	Inf. Dilution	dH:-1648.9(5SF)	2	ENS	6437
11	25	0	Inf. Dilution	dH:-1646.6(5SF)	0	EFE	6482
12	25	0	Inf. Dilution	dH:-1649.(4SF)	4	MTD	6494
13	25	0	GAMSPATH	dH:-6879.00(6SF)kJ	0	CRV	12638
14	127	0	Inf. Dilution	dH:-1650.(4SF)	1	MTD	6494
15	227	0	Inf. Dilution	dH:-1649.(4SF)	0	MTD	6494
16	327	0	Inf. Dilution	dH:-1649.(4SF)	0	MTD	6494

Reaction No. 66818

3<Fe(s)> + H2(g) + 6<O2(g)> + 4<Si(s)> = Fe3Si4O10(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:791.3(4SF)	1	ELC	
2	25	0	GAMSPATH	dG:-4513.(4SF)kJ	0	CRV	12638
Note (2): Low wgt for inter-reaction consistency							
3	25	0	Inf. Dilution	dH:-1161.8(5SF)	2	ENS	6437

Reaction No. 66819

2<Al(s)> + 2<Ca(s)> + H2(g) + 6<O2(g)> + 3<Si(s)> = Ca2Al2Si3O10(OH)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1020.(4SF)	3	ENS	6494
2	27	0	Inf. Dilution	lgK:1014.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:743.7(4SF)	2	ENS	6494
4	227	0	Inf. Dilution	lgK:581.7(4SF)	1	ENS	6494
5	327	0	Inf. Dilution	lgK:473.8(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-1391.1(5SF)	5	EFE	6482
7	25	0	Inf. Dilution	dG:-1392.(4SF)	3	EFE	6494
8	25	0	GAMSPATH	dG:-5813.(4SF)kJ	2	CRV	12638
9	25	0	Inf. Dilution	dH:-1481.8(5SF)	3	ENS	6437
10	25	0	Inf. Dilution	dH:-1481.5(5SF)	5	EFE	6482
11	25	0	Inf. Dilution	dH:-1483.(4SF)	3	MTD	6494
12	127	0	Inf. Dilution	dH:-1483.(4SF)	1	MTD	6494

13	227	0	Inf. Dilution	dH:-1482.(4SF)	0	MTD	6494
14	327	0	Inf. Dilution	dH:-1481.(4SF)	0	MTD	6494

Reaction No. 66820							
Na(s) + Al(s) + Si(s) + 2<O2(g)> = NaAlSiO4(Neph.,s)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:347.2(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:346.9(4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:346.1(4SF)	2	ENS	6494
4	27	0	Inf. Dilution	lgK:344.6(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:343.9(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:253.4(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:252.8(4SF)	1	ENS	6494
8	194	0	Inf. Dilution	lgK:214.1(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:198.4(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:161.8(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-473.2(4SF)	0	EFE	6422
12	25	0	Inf. Dilution	dG:-472.2(4SF)	0	EFE	6494
13	25	0	Inf. Dilution	dG:-475.50(5SF)	0	CRV	10174
14	50	0	Inf. Dilution	dG:-476.27(5SF)	0	CRV	10174
15	75	0	Inf. Dilution	dG:-477.11(5SF)	0	CRV	10174
16	100	0	Inf. Dilution	dG:-478.00(5SF)	0	CRV	10174
17	125	0	Inf. Dilution	dG:-478.94(5SF)	0	CRV	10174
18	150	0	Inf. Dilution	dG:-479.94(5SF)	0	CRV	10174
19	175	0	Inf. Dilution	dG:-480.98(5SF)	0	CRV	10174
20	200	0	Inf. Dilution	dG:-482.07(5SF)	0	CRV	10174
21	225	0	Inf. Dilution	dG:-483.21(5SF)	0	CRV	10174
22	250	0	Inf. Dilution	dG:-484.38(5SF)	0	CRV	10174
23	300	0	Inf. Dilution	dG:-486.84(5SF)	0	CRV	10174
24	25	0	Inf. Dilution	dH:-500.6(4SF)	0	MTD	6422
25	25	0	Inf. Dilution	dH:-503.21(5SF)	0	ENS	6437
26	25	0	Inf. Dilution	dH:-499.6(4SF)	0	MTD	6494
27	127	0	Inf. Dilution	dH:-501.4(4SF)	0	MTD	6422
28	127	0	Inf. Dilution	dH:-500.4(4SF)	0	MTD	6494
29	194	0	Inf. Dilution	dH:-501.1(4SF)	0	MTD	6422
30	227	0	Inf. Dilution	dH:-500.6(4SF)	0	MTD	6494

31	327	0	Inf. Dilution	dH:-499.9(4SF)	0	MTD	6494

Reaction No. 66822							
K(s) + Al(s) + Si(s) + 2<O2(g)> = KAlSiO4(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:351.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:351.3(4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:351.9(4SF)	3	ENS	6494
4	27	0	Inf. Dilution	lgK:349.0(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:349.6(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:256.6(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:257.1(4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:201.1(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:201.5(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:164.1(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:164.5(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-479.3(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-480.1(4SF)	0	EFE	6494
14	25	0	Inf. Dilution	dG:-482.98(5SF)	0	CRV	10174
15	50	0	Inf. Dilution	dG:-483.81(5SF)	0	CRV	10174
16	75	0	Inf. Dilution	dG:-484.69(5SF)	0	CRV	10174
17	100	0	Inf. Dilution	dG:-485.63(5SF)	0	CRV	10174
18	125	0	Inf. Dilution	dG:-486.62(5SF)	0	CRV	10174
19	150	0	Inf. Dilution	dG:-487.66(5SF)	0	CRV	10174
20	175	0	Inf. Dilution	dG:-488.75(5SF)	0	CRV	10174
21	200	0	Inf. Dilution	dG:-489.89(5SF)	0	CRV	10174
22	225	0	Inf. Dilution	dG:-491.06(5SF)	0	CRV	10174
23	250	0	Inf. Dilution	dG:-492.28(5SF)	0	CRV	10174
24	300	0	Inf. Dilution	dG:-494.81(5SF)	0	CRV	10174
25	25	0	Inf. Dilution	dH:-507.0(4SF)	0	MTD	6422
26	25	0	Inf. Dilution	dH:-505.4(4SF)	0	ENS	6437
27	25	0	Inf. Dilution	dH:-507.8(4SF)	0	MTD	6494
28	127	0	Inf. Dilution	dH:-507.9(4SF)	0	MTD	6422
29	127	0	Inf. Dilution	dH:-508.6(4SF)	0	MTD	6494
30	227	0	Inf. Dilution	dH:-508.2(4SF)	0	MTD	6422
31	227	0	Inf. Dilution	dH:-508.5(4SF)	0	MTD	6494

32	327	0	Inf. Dilution	dH:-508.4(4SF)	0	MTD	6422
33	327	0	Inf. Dilution	dH:-508.2(4SF)	0	MTD	6494

Reaction No. 66823							
2<Al(s)> + H2(g) + 6<O2(g)> + 4<Si(s)> = Al2Si4O10(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:853.0(4SF)	0	ELC	
2	25	0	Inf. Dilution	lgK:922.6(4SF)	2	ENS	6494
3	27	0	Inf. Dilution	lgK:916.5(4SF)	0	ENS	6494
4	127	0	Inf. Dilution	lgK:670.9(4SF)	0	ENS	6494
5	227	0	Inf. Dilution	lgK:523.6(4SF)	0	ENS	6494
6	327	0	Inf. Dilution	lgK:425.4(4SF)	0	ENS	6494
7	25	0	Inf. Dilution	dG:-1258.8(5SF)	0	EFE	6482
8	25	0	Inf. Dilution	dG:-1259.(4SF)	2	EFE	6494
9	25	0	Inf. Dilution	dG:-1259.00(6SF)	0	CRV	10174
10	25	0	Inf. Dilution	dG:-1258.4(5SF)	3	CRV	29553
11	25	0	GAMSPATH	dG:-5253.(4SF)kJ	0	CRV	12638
12	50	0	Inf. Dilution	dG:-1260.50(6SF)	0	CRV	10174
13	75	0	Inf. Dilution	dG:-1262.20(6SF)	0	CRV	10174
14	100	0	Inf. Dilution	dG:-1264.00(6SF)	0	CRV	10174
15	125	0	Inf. Dilution	dG:-1265.90(6SF)	0	CRV	10174
16	150	0	Inf. Dilution	dG:-1267.90(6SF)	0	CRV	10174
17	175	0	Inf. Dilution	dG:-1270.10(6SF)	0	CRV	10174
18	200	0	Inf. Dilution	dG:-1272.40(6SF)	0	CRV	10174
19	225	0	Inf. Dilution	dG:-1274.80(6SF)	0	CRV	10174
20	250	0	Inf. Dilution	dG:-1277.30(6SF)	0	CRV	10174
21	300	0	Inf. Dilution	dG:-1282.50(6SF)	0	CRV	10174
22	25	0	Inf. Dilution	dH:-1348.2(5SF)	0	ENS	6437
23	25	0	Inf. Dilution	dH:-1348.2(5SF)	0	EFE	6482
24	25	0	Inf. Dilution	dH:-1348.(4SF)	0	MTD	6494
25	25	0	Inf. Dilution	dH:-1347.7(5SF)	3	CRV	29553
26	25	0	GAMSPATH	dH:-5626.60(6SF)kJ	0	CRV	12638
27	127	0	Inf. Dilution	dH:-1348.(4SF)	0	MTD	6494
28	227	0	Inf. Dilution	dH:-1348.(4SF)	0	MTD	6494
29	327	0	Inf. Dilution	dH:-1347.(4SF)	0	MTD	6494

Reaction No. 66825							
7<Mg(s)> + 8<Si(s)> + 12<O2(g)> + H2(g) = Mg7Si8O22(OH)2(Anth.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1991.(4SF)	0	ENS	6422
2	25	0	Inf. Dilution	lgK:1987.(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:1978.(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:1974.(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:1452.(4SF)	0	ENS	6422
6	127	0	Inf. Dilution	lgK:1449.(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:1136.(4SF)	0	ENS	6422
8	227	0	Inf. Dilution	lgK:1134.(4SF)	4	ENS	6494
9	327	0	Inf. Dilution	lgK:926.0(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:923.7(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-2717.(4SF)	0	EFE	6422
12	25	0	Inf. Dilution	dG:-2710.9(5SF)	3	EFE	6482
13	25	0	Inf. Dilution	dG:-2711.(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dH:-2889.(4SF)	0	MTD	6422
15	25	0	Inf. Dilution	dH:-2885.(4SF)	4	ENS	6437
16	25	0	Inf. Dilution	dH:-2884.6(5SF)	3	EFE	6482
17	25	0	Inf. Dilution	dH:-2885.(4SF)	4	MTD	6494
18	127	0	Inf. Dilution	dH:-2890.(4SF)	0	MTD	6422
19	127	0	Inf. Dilution	dH:-2885.(4SF)	4	MTD	6494
20	227	0	Inf. Dilution	dH:-2889.(4SF)	0	MTD	6422
21	227	0	Inf. Dilution	dH:-2883.(4SF)	4	MTD	6494
22	327	0	Inf. Dilution	dH:-2888.(4SF)	0	MTD	6422
23	327	0	Inf. Dilution	dH:-2881.(4SF)	4	MTD	6494

Reaction No. 66826							
3<Al(s)> + H2(g) + K(s) + 6<O2(g)> + 3<Si(s)> = KAl3Si3O10(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:983.(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:982.6(4SF)	0	ENS	6422
3	25	0	Inf. Dilution	lgK:982.6(4SF)	0	ENS	6494
4	27	0	Inf. Dilution	lgK:976.1(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:976.1(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:715.5(4SF)	0	ENS	6422

7	127	0	Inf. Dilution	lgK:715.2(4SF)	0	ENS	6494
8	227	0	Inf. Dilution	lgK:559.1(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:558.7(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:454.8(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:454.4(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-1340.(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-1337.6(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-1340.(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-1340.30(6SF)	0	CRV	10174
16	25	0	Inf. Dilution	dG:-1339.0(5SF)	5	CRV	29553
17	25	0	GAMSPATH	dG:-5588.(4SF)kJ	0	CRV	12638
18	50	0	Inf. Dilution	dG:-1342.10(6SF)	0	CRV	10174
19	75	0	Inf. Dilution	dG:-1344.10(6SF)	0	CRV	10174
20	100	0	Inf. Dilution	dG:-1346.20(6SF)	0	CRV	10174
21	125	0	Inf. Dilution	dG:-1348.50(6SF)	0	CRV	10174
22	150	0	Inf. Dilution	dG:-1350.90(6SF)	0	CRV	10174
23	175	0	Inf. Dilution	dG:-1353.40(6SF)	0	CRV	10174
24	200	0	Inf. Dilution	dG:-1356.00(6SF)	0	CRV	10174
25	225	0	Inf. Dilution	dG:-1358.80(6SF)	0	CRV	10174
26	250	0	Inf. Dilution	dG:-1361.70(6SF)	0	CRV	10174
27	300	0	Inf. Dilution	dG:-1367.80(6SF)	0	CRV	10174
28	25	0	Inf. Dilution	dH:-1430.(4SF)	0	MTD	6422
29	25	0	Inf. Dilution	dH:-1428.4(5SF)	0	ENS	6437
30	25	0	Inf. Dilution	dH:-1428.5(5SF)	0	EFE	6482
31	25	0	Inf. Dilution	dH:-1432.(4SF)	0	MTD	6494
32	25	0	Inf. Dilution	dH:-1429.9(5SF)	5	CRV	29553
33	25	0	GAMSPATH	dH:-5969.40(6SF)kJ	0	CRV	12638
34	127	0	Inf. Dilution	dH:-1431.(4SF)	0	MTD	6422
Note (34): Low Wgt for internal consistency							
35	127	0	Inf. Dilution	dH:-1433.(4SF)	0	MTD	6494
36	227	0	Inf. Dilution	dH:-1431.(4SF)	0	MTD	6422
37	227	0	Inf. Dilution	dH:-1432.(4SF)	0	MTD	6494
38	327	0	Inf. Dilution	dH:-1430.(4SF)	0	MTD	6422
39	327	0	Inf. Dilution	dH:-1431.(4SF)	0	MTD	6494

Reaction No. 66828

Al(s) + K(s) + 4<O2(g)> + 3<Si(s)> = KAlSi3O8(San.,s)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:657.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:655.2(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:650.9(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:478.5(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:375.0(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:306.1(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-893.9(4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dG:-893.60(5SF)	0	EFE	6482
9	25	0	Inf. Dilution	dG:-3746.00(6SF)kJ	0	EFE	8798
10	25	0	Inf. Dilution	dG:-895.43(5SF)	0	CRV	10174
11	25	0	GAMSPATH	dG:-3739.(4SF)kJ	0	CRV	12638
12	50	0	Inf. Dilution	dG:-896.85(5SF)	0	CRV	10174
13	75	0	Inf. Dilution	dG:-898.36(5SF)	0	CRV	10174
14	100	0	Inf. Dilution	dG:-899.97(5SF)	0	CRV	10174
15	125	0	Inf. Dilution	dG:-901.67(5SF)	0	CRV	10174
16	150	0	Inf. Dilution	dG:-903.46(5SF)	0	CRV	10174
17	175	0	Inf. Dilution	dG:-905.33(5SF)	0	CRV	10174
18	200	0	Inf. Dilution	dG:-907.29(5SF)	0	CRV	10174
19	225	0	Inf. Dilution	dG:-909.31(5SF)	0	CRV	10174
20	250	0	Inf. Dilution	dG:-911.41(5SF)	0	CRV	10174
21	300	0	Inf. Dilution	dG:-915.78(5SF)	0	CRV	10174
22	25	0	Inf. Dilution	dH:-946.4(4SF)	0	MTD	6422
23	25	0	Inf. Dilution	dH:-946.4(4SF)	0	ENS	6437
24	25	0	Inf. Dilution	dH:-945.91(5SF)	2	EFE	6482
25	25	0	Inf. Dilution	dH:-3965.70(6SF)kJ	2	EFE	8798
26	25	0	GAMSPATH	dH:-3960.30(6SF)kJ	0	CRV	12638
27	127	0	Inf. Dilution	dH:-947.1(4SF)	0	MTD	6422
28	227	0	Inf. Dilution	dH:-946.6(4SF)	0	MTD	6422
29	327	0	Inf. Dilution	dH:-945.7(4SF)	0	MTD	6422

Reaction No. 66829							
3<Ca(s)> + 2<Fe(s)> + 3<Si(s)> + 6<O2(g)> = 3CaO.Fe2O3.3SiO2(s)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:955.5(4SF)	1	ELC	

2	25	0	Inf. Dilution	lgK:950.8(4SF)	0	ENS	6494
3	27	0	Inf. Dilution	lgK:944.5(4SF)	0	ENS	6494
4	127	0	Inf. Dilution	lgK:693.3(4SF)	0	ENS	6494
5	227	0	Inf. Dilution	lgK:542.7(4SF)	0	ENS	6494
6	327	0	Inf. Dilution	lgK:442.3(4SF)	0	ENS	6494
7	25	0	Inf. Dilution	dG:-1297.(4SF)	0	EFE	6494
8	25	0	Inf. Dilution	dH:-1379.0(5SF)	2	ENS	6437
9	25	0	Inf. Dilution	dH:-1379.(4SF)	0	MTD	6494
10	127	0	Inf. Dilution	dH:-1379.(4SF)	0	MTD	6494
11	227	0	Inf. Dilution	dH:-1378.(4SF)	0	MTD	6494
12	327	0	Inf. Dilution	dH:-1377.(4SF)	0	MTD	6494

Reaction No. 66830							
Na(s) + Al(s) + 2<Si(s)> + 3<O2(g)> = NaAlSi2O6(Jade.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:499.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:500.0(4SF)	3	ENS	6422
3	25	0	Inf. Dilution	lgK:499.4(4SF)	3	ENS	6494
4	27	0	Inf. Dilution	lgK:496.7(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:496.1(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:364.7(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:364.2(4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:285.4(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:285.0(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:232.5(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:232.2(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-682.1(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-680.33(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-681.3(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-682.07(5SF)	0	CRV	10174
16	50	0	Inf. Dilution	dG:-682.91(5SF)	0	CRV	10174
17	75	0	Inf. Dilution	dG:-683.82(5SF)	0	CRV	10174
18	100	0	Inf. Dilution	dG:-684.82(5SF)	0	CRV	10174
19	125	0	Inf. Dilution	dG:-685.88(5SF)	0	CRV	10174
20	150	0	Inf. Dilution	dG:-687.01(5SF)	0	CRV	10174
21	175	0	Inf. Dilution	dG:-688.21(5SF)	0	CRV	10174

22	200	0	Inf. Dilution	dG:-689.47(5SF)	0	CRV	10174
23	225	0	Inf. Dilution	dG:-690.79(5SF)	0	CRV	10174
24	250	0	Inf. Dilution	dG:-692.17(5SF)	0	CRV	10174
25	300	0	Inf. Dilution	dG:-695.07(5SF)	0	CRV	10174
26	25	0	Inf. Dilution	dH:-725.8(4SF)	0	MTD	6422
27	25	0	Inf. Dilution	dH:-724.(3SF)	0	ENS	6437
28	25	0	Inf. Dilution	dH:-723.02(5SF)	0	EFE	6482
29	25	0	Inf. Dilution	dH:-724.0(4SF)	0	MTD	6494
30	127	0	Inf. Dilution	dH:-725.7(4SF)	0	MTD	6422
31	127	0	Inf. Dilution	dH:-724.9(4SF)	0	MTD	6494
32	227	0	Inf. Dilution	dH:-725.6(4SF)	0	MTD	6422
33	227	0	Inf. Dilution	dH:-724.8(4SF)	0	MTD	6494
34	327	0	Inf. Dilution	dH:-725.4(4SF)	0	MTD	6422
35	327	0	Inf. Dilution	dH:-724.6(4SF)	0	MTD	6494

Reaction No. 66831

Ca(s) + Fe(s) + 3<O2(g)> + 2<Si(s)> = CaO.FeO.2SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:468.9(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:465.8(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:342.2(4SF)	2	ENS	6494
4	227	0	Inf. Dilution	lgK:268.1(4SF)	1	ENS	6494
5	327	0	Inf. Dilution	lgK:218.7(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-639.7(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dG:-2676.3(5SF)kJ	3	EOD	12843
8	25	0	Inf. Dilution	dH:-339.2(4SF)	0	ENS	6437
9	25	0	Inf. Dilution	dH:-678.8(4SF)	4	MTD	6494
10	25	0	Inf. Dilution	dH:-2839.9(5SF)kJ	3	EOD	12843
11	127	0	Inf. Dilution	dH:-678.6(4SF)	1	MTD	6494
12	227	0	Inf. Dilution	dH:-678.2(4SF)	0	MTD	6494
13	327	0	Inf. Dilution	dH:-677.7(4SF)	0	MTD	6494

Reaction No. 66948

Li(s) + Al(s) + Si(s) + 2<O2(g)> = LiAlSiO4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:352.2(4SF)	4	ENS	6422

2	25	0	Inf. Dilution	lgK:352.0(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:349.9(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:349.7(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:257.4(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:257.3(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:201.9(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:201.8(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:164.8(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:164.8(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-480.4(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-480.2(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-507.7(4SF)	4	MTD	6422
14	25	0	Inf. Dilution	dH:-507.5(4SF)	4	MTD	6494
15	127	0	Inf. Dilution	dH:-507.8(4SF)	3	MTD	6422
16	127	0	Inf. Dilution	dH:-507.6(4SF)	3	MTD	6494
17	227	0	Inf. Dilution	dH:-508.4(4SF)	2	MTD	6422
18	227	0	Inf. Dilution	dH:-508.2(4SF)	2	MTD	6494
19	327	0	Inf. Dilution	dH:-508.1(4SF)	0	MTD	6422
20	327	0	Inf. Dilution	dH:-508.0(4SF)	0	MTD	6494

Reaction No. 66949							
Li(s) + Al(s) + 2<Si(s)> + 3<O2(g)> = LiAlSi2O6(alpha,s)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:504.8(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:504.6(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:501.5(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:501.3(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:368.5(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:368.4(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:288.7(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:288.6(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:235.4(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:235.3(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-688.7(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-688.4(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-730.1(4SF)	3	MTD	6422

14	25	0	Inf. Dilution	dH:-729.8(4SF)	3	MTD	6494
15	127	0	Inf. Dilution	dH:-730.3(4SF)	1	MTD	6422
16	127	0	Inf. Dilution	dH:-730.0(4SF)	1	MTD	6494
17	227	0	Inf. Dilution	dH:-730.9(4SF)	0	MTD	6422
18	227	0	Inf. Dilution	dH:-730.7(4SF)	0	MTD	6494
19	327	0	Inf. Dilution	dH:-730.7(4SF)	0	MTD	6422
20	327	0	Inf. Dilution	dH:-730.5(4SF)	0	MTD	6494

Reaction No. 66950

Li(s) + Al(s) + 2<Si(s)> + 3<O2(g)> = LiAlSi2O6(beta,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:501.2(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:501.0(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:498.0(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:497.7(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:366.2(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:366.0(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:287.1(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:286.9(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:234.4(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:234.2(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-683.8(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-683.4(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-723.4(4SF)	3	MTD	6422
14	25	0	Inf. Dilution	dH:-723.1(4SF)	3	MTD	6494
15	127	0	Inf. Dilution	dH:-723.5(4SF)	1	MTD	6422
16	127	0	Inf. Dilution	dH:-723.2(4SF)	1	MTD	6494
17	227	0	Inf. Dilution	dH:-724.0(4SF)	0	MTD	6422
18	227	0	Inf. Dilution	dH:-723.8(4SF)	0	MTD	6494
19	327	0	Inf. Dilution	dH:-723.6(4SF)	0	MTD	6422
20	327	0	Inf. Dilution	dH:-732.4(4SF)	0	MTD	6494

Reaction No. 66955

2<Li(s)> + 1.5<O2(g)> + Si(s) = Li2SiO3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:273.1(4SF)	4	ENS	6422

2	27	0	Inf. Dilution	lgK:271.3(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:199.5(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:156.3(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:127.4(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-1557.2(5SF)kJ	5	EFE	6330
7	25	0	Inf. Dilution	dG:-372.6(4SF)	4	EFE	6422
8	25	0	Inf. Dilution	dH:-1648.1(5SF)kJ	3	CRV	6330
9	25	0	Inf. Dilution	dH:-394.2(4SF)	3	MTD	6422
10	127	0	Inf. Dilution	dH:-394.5(4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-396.0(4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-396.0(4SF)	0	MTD	6422

Reaction No. 66956

2<Li(s)> + 2<Si(s)> + 2.5<O2(g)> = Li2Si2O5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:423.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:420.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:309.2(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:242.2(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:197.5(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-577.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-612.1(4SF)	3	MTD	6422
8	127	0	Inf. Dilution	dH:-612.3(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-613.7(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-613.3(4SF)	0	MTD	6422

Reaction No. 66957

4<Li(s)> + Si(s) + 2<O2(g)> = Li4SiO4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:386.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:383.5(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:282.1(4SF)	3	ENS	6422
4	227	0	Inf. Dilution	lgK:220.9(4SF)	2	ENS	6422
5	327	0	Inf. Dilution	lgK:180.0(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-526.7(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-556.9(4SF)	3	MTD	6422

8	127	0	Inf. Dilution	dH:-557.7(4SF)	1	MTD	6422
9	227	0	Inf. Dilution	dH:-561.5(4SF)	0	MTD	6422
10	327	0	Inf. Dilution	dH:-562.3(4SF)	0	MTD	6422

Reaction No. 66974							
2<Mg(s)> + 2<O2(g)> + Si(s) = 2MgO.SiO2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:360.5(4SF)	0	ENS	6422
2	25	0	Inf. Dilution	lgK:359.8(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:358.2(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:357.4(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:263.4(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:262.8(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:206.6(4SF)	1	ENS	6422
8	227	0	Inf. Dilution	lgK:206.1(4SF)	1	ENS	6494
9	327	0	Inf. Dilution	lgK:168.7(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:168.3(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-491.8(4SF)	0	EFE	6422
12	25	0	Inf. Dilution	dG:-491.16(5SF)	4	EFE	6482
13	25	0	Inf. Dilution	dG:-490.8(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dG:-2053.6(5SF)kJ	5	CRV	8695
Note (14): p. 169. Sign changed							
15	25	0	Inf. Dilution	dG:-2055.1(5SF)kJ	5	CRV	8695
Note (15): p. 167. Sign changed							
16	25	0	Inf. Dilution	dG:-2053.6(5SF)kJ	3	EOD	12843
17	25	0	Inf. Dilution	dG:-2053.5(5SF)kJ	5	CRV	29482
18	25	0	Inf. Dilution	dH:-520.3(4SF)	4	MTD	6422
19	25	0	Inf. Dilution	dH:-519.70(5SF)	4	EFE	6482
20	25	0	Inf. Dilution	dH:-519.4(4SF)	4	MTD	6494
21	25	0	Inf. Dilution	dH:-2174.0(5SF)kJ	5	CRV	8695
22	25	0	Inf. Dilution	dH:-2173.0(5SF)kJ	3	EOD	12843
23	25	0	Inf. Dilution	dH:-2172.6(5SF)kJ	5	CRV	29482
24	127	0	Inf. Dilution	dH:-520.4(4SF)	2	MTD	6422
25	127	0	Inf. Dilution	dH:-519.4(4SF)	2	MTD	6494
26	227	0	Inf. Dilution	dH:-520.2(4SF)	1	MTD	6422
27	227	0	Inf. Dilution	dH:-519.3(4SF)	1	MTD	6494

28	327	0	Inf. Dilution	dH:-519.9(4SF)	0	MTD	6422
29	327	0	Inf. Dilution	dH:-519.0(4SF)	0	MTD	6494

Reaction No. 67042

Al(s) + Na(s) + 3<O2(g)> + 2<Si(s)> = NaAlSi2O6(Dehyd.Anal.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:493.9(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:493.3(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:490.7(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:490.1(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:360.7(4SF)	3	ENS	6422
6	127	0	Inf. Dilution	lgK:360.2(4SF)	3	ENS	6494
7	227	0	Inf. Dilution	lgK:282.6(4SF)	2	ENS	6422
8	227	0	Inf. Dilution	lgK:282.3(4SF)	2	ENS	6494
9	327	0	Inf. Dilution	lgK:230.6(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:230.3(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-673.8(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-673.0(4SF)	4	EFE	6494
13	25	0	Inf. Dilution	dH:-713.5(4SF)	4	MTD	6422
14	25	0	Inf. Dilution	dH:-713.0(4SF)	4	MTD	6494
15	127	0	Inf. Dilution	dH:-714.3(4SF)	3	MTD	6422
16	127	0	Inf. Dilution	dH:-713.7(4SF)	3	MTD	6494
17	227	0	Inf. Dilution	dH:-714.0(4SF)	2	MTD	6422
18	227	0	Inf. Dilution	dH:-713.5(4SF)	2	MTD	6494
19	327	0	Inf. Dilution	dH:-713.7(4SF)	0	MTD	6422
20	327	0	Inf. Dilution	dH:-713.0(4SF)	0	MTD	6494

Reaction No. 67043

Al(s) + Na(s) + 4<O2(g)> + 3<Si(s)> = NaAlSi3O8(AlbiteLO,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:651.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:650.0(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:645.7(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:474.7(4SF)	0	ENS	6422
Note (4): Low Wgt for internal consistency							
5	227	0	Inf. Dilution	lgK:372.1(4SF)	0	ENS	6422

6	327	0	Inf. Dilution	lgK:303.7(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-886.7(4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dG:-887.12(5SF)	0	EFE	6482
9	25	0	Inf. Dilution	dG:-3713.00(6SF)kJ	0	EFE	8798
10	25	0	Inf. Dilution	dG:-888.94(5SF)	0	CRV	10174
11	25	0	Inf. Dilution	dG:-887.37(5SF)	5	CRV	29553
12	50	0	Inf. Dilution	dG:-890.22(5SF)	0	CRV	10174
13	75	0	Inf. Dilution	dG:-891.61(5SF)	0	CRV	10174
14	100	0	Inf. Dilution	dG:-893.10(5SF)	0	CRV	10174
15	125	0	Inf. Dilution	dG:-894.67(5SF)	0	CRV	10174
16	150	0	Inf. Dilution	dG:-896.34(5SF)	0	CRV	10174
17	175	0	Inf. Dilution	dG:-898.09(5SF)	0	CRV	10174
18	200	0	Inf. Dilution	dG:-899.91(5SF)	0	CRV	10174
19	225	0	Inf. Dilution	dG:-901.81(5SF)	0	CRV	10174
20	250	0	Inf. Dilution	dG:-903.78(5SF)	0	CRV	10174
21	300	0	Inf. Dilution	dG:-907.90(5SF)	0	CRV	10174
22	25	0	Inf. Dilution	dH:-938.7(4SF)	0	MTD	6422
23	25	0	Inf. Dilution	dH:-940.51(5SF)	0	MTD	6482
24	25	0	Inf. Dilution	dH:-3936.20(6SF)kJ	0	EFE	8798
25	25	0	Inf. Dilution	dH:-940.77(5SF)	5	CRV	29553
26	127	0	Inf. Dilution	dH:-939.5(4SF)	0	MTD	6422
Note (26): Low Wgt for internal consistency							
27	227	0	Inf. Dilution	dH:-939.1(4SF)	0	MTD	6422
28	327	0	Inf. Dilution	dH:-938.4(4SF)	0	MTD	6422

Reaction No. 67044							
Al(s) + Na(s) + 4<O2(g)> + 3<Si(s)> = NaAlSi3O8(AlbiteHI,s)							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:649.(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:650.6(4SF)	0	ENS	6422
3	27	0	Inf. Dilution	lgK:646.3(4SF)	0	ENS	6422
4	127	0	Inf. Dilution	lgK:474.9(4SF)	0	ENS	6422
5	227	0	Inf. Dilution	lgK:372.0(4SF)	0	ENS	6422
6	327	0	Inf. Dilution	lgK:303.4(4SF)	0	ENS	6422
7	25	0	Inf. Dilution	dG:-887.6(4SF)	0	EFE	6422
8	25	0	Inf. Dilution	dG:-885.11(5SF)	0	EFE	6482

9	25	0	Inf. Dilution	dG:-3705.10(6SF)kJ	0	EFE	8798
10	25	0	Inf. Dilution	dG:-887.14(5SF)	0	CRV	10174
11	25	0	Inf. Dilution	dG:-3705.1(5SF)kJ	3	EOD	12843
12	50	0	Inf. Dilution	dG:-888.50(5SF)	0	CRV	10174
13	75	0	Inf. Dilution	dG:-889.95(5SF)	0	CRV	10174
14	100	0	Inf. Dilution	dG:-891.51(5SF)	0	CRV	10174
15	125	0	Inf. Dilution	dG:-893.15(5SF)	0	CRV	10174
16	150	0	Inf. Dilution	dG:-894.89(5SF)	0	CRV	10174
17	175	0	Inf. Dilution	dG:-896.71(5SF)	0	CRV	10174
18	200	0	Inf. Dilution	dG:-898.60(5SF)	0	CRV	10174
19	225	0	Inf. Dilution	dG:-900.57(5SF)	0	CRV	10174
20	250	0	Inf. Dilution	dG:-902.61(5SF)	0	CRV	10174
21	300	0	Inf. Dilution	dG:-906.87(5SF)	0	CRV	10174
22	25	0	Inf. Dilution	dH:-940.9(4SF)	0	MTD	6422
23	25	0	Inf. Dilution	dH:-937.29(5SF)	0	MTD	6482
24	25	0	Inf. Dilution	dH:-3923.40(6SF)kJ	0	EFE	8798
25	25	0	Inf. Dilution	dH:-3923.4(5SF)kJ	3	EOD	12843
26	127	0	Inf. Dilution	dH:-941.8(4SF)	0	MTD	6422
Note (26): Low Wgt for internal consistency							
27	227	0	Inf. Dilution	dH:-941.7(4SF)	0	MTD	6422
28	327	0	Inf. Dilution	dH:-941.4(4SF)	0	MTD	6422

Reaction No. 67045

Al(s) + H2(g) + Na(s) + 3.5<O2(g)> + 2<Si(s)> = NaAlSi2O6.H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:539.2(4SF)	1	ENS	6422
2	25	0	Inf. Dilution	lgK:541.3(4SF)	1	ENS	6494
3	27	0	Inf. Dilution	lgK:535.6(4SF)	0	ENS	6422
4	27	0	Inf. Dilution	lgK:537.8(4SF)	0	ENS	6494
5	127	0	Inf. Dilution	lgK:391.8(4SF)	0	ENS	6422
6	127	0	Inf. Dilution	lgK:393.6(4SF)	0	ENS	6494
7	227	0	Inf. Dilution	lgK:305.5(4SF)	0	ENS	6422
8	227	0	Inf. Dilution	lgK:307.1(4SF)	0	ENS	6494
9	327	0	Inf. Dilution	lgK:248.0(4SF)	0	ENS	6422
10	327	0	Inf. Dilution	lgK:249.5(4SF)	0	ENS	6494
11	25	0	Inf. Dilution	dG:-735.5(4SF)	1	EFE	6422

12	25	0	Inf. Dilution	dG:-738.5(4SF)	1	EFE	6494
13	25	0	Inf. Dilution	dG:-740.73(5SF)	0	CRV	10174
14	25	0	Inf. Dilution	dG:-3078.90(6SF)kJ	3	CRV	25435
15	25	0	Inf. Dilution	dG:-3084.(4SF)kJ	3	EFE	29327
16	50	0	Inf. Dilution	dG:-742.18(5SF)	0	CRV	10174
17	75	0	Inf. Dilution	dG:-743.73(5SF)	0	CRV	10174
18	100	0	Inf. Dilution	dG:-745.38(5SF)	0	CRV	10174
19	125	0	Inf. Dilution	dG:-747.13(5SF)	0	CRV	10174
20	150	0	Inf. Dilution	dG:-748.96(5SF)	0	CRV	10174
21	175	0	Inf. Dilution	dG:-750.87(5SF)	0	CRV	10174
22	200	0	Inf. Dilution	dG:-752.86(5SF)	0	CRV	10174
23	225	0	Inf. Dilution	dG:-754.93(5SF)	0	CRV	10174
24	250	0	Inf. Dilution	dG:-757.06(5SF)	0	CRV	10174
25	300	0	Inf. Dilution	dG:-761.51(5SF)	0	CRV	10174
26	25	0	Inf. Dilution	dH:-789.4(4SF)	3	MTD	6422
27	25	0	Inf. Dilution	dH:-791.1(4SF)	1	MTD	6494
28	25	0	Inf. Dilution	dH:-3296.9(5SF)kJ	3	CRV	25435
29	25	0	Inf. Dilution	dH:-3308.00(6SF)kJ	3	CRV	27383
30	25	0	Inf. Dilution	dH:-3303.2(2.9SD)kJ	4	ETR	29327
31	127	0	Inf. Dilution	dH:-790.0(4SF)	0	MTD	6422
32	127	0	Inf. Dilution	dH:-791.9(4SF)	0	MTD	6494
33	227	0	Inf. Dilution	dH:-789.4(4SF)	0	MTD	6422
34	227	0	Inf. Dilution	dH:-791.6(4SF)	0	MTD	6494
35	327	0	Inf. Dilution	dH:-788.5(4SF)	0	MTD	6422
36	327	0	Inf. Dilution	dH:-790.9(4SF)	0	MTD	6494

Reaction No. 67047

2<Na(s)> + 1.5<O2(g)> + Si(s) = Na2SiO3(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:257.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:255.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:187.4(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:146.4(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:119.1(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-1462.8(5SF)kJ	5	EFE	6330
7	25	0	Inf. Dilution	dG:-350.7(4SF)	4	EFE	6422

8	25	0	Inf. Dilution	dH:-1554.9(5SF)kJ	5	CRV	6330
9	25	0	Inf. Dilution	dH:-373.2(4SF)	4	MTD	6422
10	127	0	Inf. Dilution	dH:-374.6(4SF)	4	MTD	6422
11	227	0	Inf. Dilution	dH:-374.5(4SF)	4	MTD	6422
12	327	0	Inf. Dilution	dH:-374.2(4SF)	4	MTD	6422

Reaction No. 67048

2<Na(s)> + 2.5<O2(g)> + 2<Si(s)> = Na2Si2O5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:407.2(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:404.5(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:296.9(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:232.3(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:189.2(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-555.5(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-590.4(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-591.8(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-591.5(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-591.0(4SF)	4	MTD	6422

Reaction No. 67049

4<Na(s)> + 2<O2(g)> + Si(s) = Na4SiO4(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:346.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:343.9(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:252.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:196.7(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:159.8(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-472.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-503.5(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-506.3(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-506.6(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-506.7(4SF)	4	MTD	6422

Reaction No. 67050

6<Na(s)> + 3.5<O2(g)> + 2<Si(s)> = Na6Si2O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	0	Inf. Dilution	lgK:596.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:593.0(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:434.7(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:339.4(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:275.9(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-814.3(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-868.1(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-872.0(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-871.9(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-871.5(4SF)	4	MTD	6422

Reaction No. 67106

2<Ni(s)> + 2<O2(g)> + Si(s) = 2NiO.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:226.1(4SF)	2	ENS	6422
2	27	0	Inf. Dilution	lgK:224.6(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:163.5(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:126.9(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:102.6(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-308.4(4SF)	2	EFE	6422
7	25	0	Inf. Dilution	dG:-1288.40(2.5SD)kJ	5	CRV	20828
8	25	0	Inf. Dilution	dH:-335.4(4SF)	2	MTD	6422
9	25	0	Inf. Dilution	dH:-1396.00(2.5SD)kJ	5	CRV	20828
10	127	0	Inf. Dilution	dH:-335.1(4SF)	0	MTD	6422
11	227	0	Inf. Dilution	dH:-334.6(4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-334.2(4SF)	0	MTD	6422
13	25	0	Inf. Dilution	Cp:123.200(0.25SD)J/K	5	CRV	20828

Reaction No. 67146

1.5<O2(g)> + Pb(s) + Si(s) = PbO.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:185.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:184.7(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:134.8(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:104.9(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:85.04(4SF)	0	ENS	6422

6	25	0	Inf. Dilution	dG:-253.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-253.86(5SF)	4	CRV	12011
8	25	0	Inf. Dilution	dH:-273.7(4SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-273.83(5SF)	4	CRV	12011
10	127	0	Inf. Dilution	dH:-273.6(4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-273.3(4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-272.9(4SF)	0	MTD	6422

Reaction No. 67147

2<O2(g)> + 2<Pb(s)> + Si(s) = 2PbO.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:219.4(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:218.0(4SF)	0	ENS	6422
3	127	0	Inf. Dilution	lgK:158.6(4SF)	2	ENS	6422
4	227	0	Inf. Dilution	lgK:123.1(4SF)	1	ENS	6422
5	327	0	Inf. Dilution	lgK:99.38(4SF)	0	ENS	6422
6	25	0	Inf. Dilution	dG:-299.3(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dG:-299.4(4SF)	4	CRV	12011
8	25	0	Inf. Dilution	dH:-325.9(4SF)	4	MTD	6422
9	25	0	Inf. Dilution	dH:-325.8(4SF)	4	CRV	12011
10	127	0	Inf. Dilution	dH:-325.6(4SF)	1	MTD	6422
11	227	0	Inf. Dilution	dH:-325.2(4SF)	0	MTD	6422
12	327	0	Inf. Dilution	dH:-324.7(4SF)	0	MTD	6422

Reaction No. 67148

3<O2(g)> + 4<Pb(s)> + Si(s) = 4PbO.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:286.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:284.9(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:206.4(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:159.3(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:128.0(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-391.39(5SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-431.5(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-431.0(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-430.4(4SF)	4	MTD	6422

10	327	0	Inf. Dilution	dH:-429.6(4SF)	4	MTD	6422

Reaction No. 67209

1.5<O2(g)> + 2<Rb(s)> + Si(s) = Rb2O.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:252.5(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:250.9(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:183.8(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:143.5(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:116.7(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-344.5(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-367.2(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-368.4(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-368.3(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-368.1(4SF)	4	MTD	6422

Reaction No. 67210

2.5<O2(g)> + 2<Rb(s)> + 2<Si(s)> = Rb2O.2SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:406.9(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:404.2(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:296.4(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:231.6(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:188.5(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-555.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-591.4(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-592.5(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-592.2(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-591.7(4SF)	4	MTD	6422

Reaction No. 67211

4.5<O2(g)> + 2<Rb(s)> + 4<Si(s)> = Rb2O.4SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:710.7(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:706.0(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:518.0(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:405.1(4SF)	4	ENS	6422

5	327	0	Inf. Dilution	lgK:329.9(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-969.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1032.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1033.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1033.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1032.(4SF)	4	MTD	6422

Reaction No. 67229							
O2(g) + Si(s) = SiO2(Cristob.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:149.7(4SF)	4	ENS	6422
2	25	0	Inf. Dilution	lgK:149.7(4SF)	4	ENS	6494
3	27	0	Inf. Dilution	lgK:148.7(4SF)	4	ENS	6422
4	27	0	Inf. Dilution	lgK:148.7(4SF)	4	ENS	6494
5	127	0	Inf. Dilution	lgK:109.2(4SF)	4	ENS	6422
6	127	0	Inf. Dilution	lgK:109.2(4SF)	4	ENS	6494
7	227	0	Inf. Dilution	lgK:85.46(4SF)	4	ENS	6422
8	227	0	Inf. Dilution	lgK:85.46(4SF)	4	ENS	6494
9	270	0	Inf. Dilution	lgK:77.94(4SF)	4	ENS	6422
10	327	0	Inf. Dilution	lgK:69.71(4SF)	4	ENS	6494
11	25	0	Inf. Dilution	dG:-204.2(4SF)	4	EFE	6422
12	25	0	Inf. Dilution	dG:-204.09(5SF)	3	EFE	6482
13	25	0	Inf. Dilution	dG:-204.3(4SF)	4	EFE	6494
14	25	0	Inf. Dilution	dG:-855.43(5SF)kJ	3	CRV	8695
15	25	0	GAMSPATH	dG:-842.2(4SF)kJ	0	CRV	12638
Note (15): dH not in GAMSPATH database; dH = -217kcal/mol assumed							
16	25	0	Inf. Dilution/m	dG:-854.020(6SF)kJ	3	CRV	24953
17	25	0	Inf. Dilution	dH:-217.1(4SF)	4	MTD	6422
18	25	0	Inf. Dilution	dH:-216.96(5SF)	4	EFE	6482
19	25	0	Inf. Dilution	dH:-217.1(4SF)	4	MTD	6494
20	25	0	Inf. Dilution	dH:-909.48(5SF)kJ	3	CRV	8695
21	25	0	Inf. Dilution/m	dH:-907.860(6SF)kJ	3	CRV	24953
22	127	0	Inf. Dilution	dH:-217.1(4SF)	4	MTD	6422
23	127	0	Inf. Dilution	dH:-217.1(4SF)	4	MTD	6494
24	227	0	Inf. Dilution	dH:-217.1(4SF)	4	MTD	6422
25	227	0	Inf. Dilution	dH:-217.1(4SF)	4	MTD	6494

26	270	0	Inf. Dilution	dH:-217.0(4SF)	4	MTD	6422
27	327	0	Inf. Dilution	dH:-216.7(4SF)	4	MTD	6494

Reaction No. 67383

Al(s) + K(s) + 4<O2(g)> + 3<Si(s)> = KAlSi3O8(Micr.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:656.(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:655.7(4SF)	2	ENS	6422
3	25	0	Inf. Dilution	lgK:656.8(4SF)	2	ENS	6494
4	27	0	Inf. Dilution	lgK:651.4(4SF)	0	ENS	6422
5	27	0	Inf. Dilution	lgK:652.5(4SF)	0	ENS	6494
6	127	0	Inf. Dilution	lgK:478.6(4SF)	1	ENS	6422
7	127	0	Inf. Dilution	lgK:479.4(4SF)	1	ENS	6494
8	227	0	Inf. Dilution	lgK:374.9(4SF)	0	ENS	6422
9	227	0	Inf. Dilution	lgK:375.5(4SF)	0	ENS	6494
10	327	0	Inf. Dilution	lgK:305.8(4SF)	0	ENS	6422
11	327	0	Inf. Dilution	lgK:306.3(4SF)	0	ENS	6494
12	25	0	Inf. Dilution	dG:-894.5(4SF)	0	EFE	6422
13	25	0	Inf. Dilution	dG:-895.18(5SF)	0	EFE	6482
14	25	0	Inf. Dilution	dG:-896.1(4SF)	0	EFE	6494
15	25	0	Inf. Dilution	dG:-3749.50(6SF)kJ	0	EFE	8798
16	25	0	Inf. Dilution	dG:-897.07(5SF)	0	CRV	10174
17	50	0	Inf. Dilution	dG:-898.40(5SF)	0	CRV	10174
18	75	0	Inf. Dilution	dG:-899.82(5SF)	0	CRV	10174
19	100	0	Inf. Dilution	dG:-901.35(5SF)	0	CRV	10174
20	125	0	Inf. Dilution	dG:-902.97(5SF)	0	CRV	10174
21	150	0	Inf. Dilution	dG:-904.67(5SF)	0	CRV	10174
22	175	0	Inf. Dilution	dG:-906.46(5SF)	0	CRV	10174
23	200	0	Inf. Dilution	dG:-908.33(5SF)	0	CRV	10174
24	225	0	Inf. Dilution	dG:-910.27(5SF)	0	CRV	10174
25	250	0	Inf. Dilution	dG:-912.28(5SF)	0	CRV	10174
26	300	0	Inf. Dilution	dG:-916.48(5SF)	0	CRV	10174
27	25	0	Inf. Dilution	dH:-948.4(4SF)	0	MTD	6422
28	25	0	Inf. Dilution	dH:-949.04(5SF)	0	EFE	6482
29	25	0	Inf. Dilution	dH:-950.0(4SF)	0	MTD	6494
30	25	0	Inf. Dilution	dH:-3974.90(6SF)kJ	0	EFE	8798

31	127	0	Inf. Dilution	dH:-949.1(4SF)	0	MTD	6422
Note (31): Low Wgt for internal consistency							
32	127	0	Inf. Dilution	dH:-950.9(4SF)	0	MTD	6494
33	227	0	Inf. Dilution	dH:-948.7(4SF)	0	MTD	6422
34	227	0	Inf. Dilution	dH:-950.9(4SF)	0	MTD	6494
35	327	0	Inf. Dilution	dH:-947.9(4SF)	0	MTD	6422
36	327	0	Inf. Dilution	dH:-950.6(4SF)	0	MTD	6494

Reaction No. 67384

Al(s) + K(s) + 4<O2(g)> + 3<Si(s)> = KAlSi3O8(Adul.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:654.3(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:650.0(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:477.8(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:374.4(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:305.5(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-892.6(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-945.1(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-946.1(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-945.9(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-945.4(4SF)	4	MTD	6422

Reaction No. 67388

2<K(s)> + 1.5<O2(g)> + Si(s) = K2O.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:255.0(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:253.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:185.9(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:145.3(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:118.3(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-347.9(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-370.0(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-371.1(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-370.9(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-370.4(4SF)	4	MTD	6422

Reaction No. 67389

2<K(s)> + 2.5<O2(g)> + 2<Si(s)> = K2O.2SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:413.6(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:410.9(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:301.6(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:235.9(4SF)	4	ENS	6422
5	250	0	Inf. Dilution	lgK:224.4(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-564.2(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-599.7(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-600.6(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-600.2(4SF)	4	MTD	6422
10	250	0	Inf. Dilution	dH:-600.0(4SF)	4	MTD	6422

Reaction No. 67390

2<K(s)> + 4.5<O2(g)> + 4<Si(s)> = K2O.4SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:711.1(4SF)	4	ENS	6422
2	27	0	Inf. Dilution	lgK:706.4(4SF)	4	ENS	6422
3	127	0	Inf. Dilution	lgK:518.5(4SF)	4	ENS	6422
4	227	0	Inf. Dilution	lgK:405.8(4SF)	4	ENS	6422
5	327	0	Inf. Dilution	lgK:330.7(4SF)	4	ENS	6422
6	25	0	Inf. Dilution	dG:-970.1(4SF)	4	EFE	6422
7	25	0	Inf. Dilution	dH:-1032.(4SF)	4	MTD	6422
8	127	0	Inf. Dilution	dH:-1032.(4SF)	4	MTD	6422
9	227	0	Inf. Dilution	dH:-1031.(4SF)	4	MTD	6422
10	327	0	Inf. Dilution	dH:-1030.(4SF)	4	MTD	6422

Reaction No. 67397

0.5<O2(g)> + Si(s) = SiO(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.30(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:22.19(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:17.81(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:15.17(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:13.39(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-30.43(4SF)	4	EFE	6494

7	25	0	Inf. Dilution	dH:-24.0(3SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-24.14(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-24.28(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-24.45(4SF)	4	MTD	6494

Reaction No. 67398

O2(g) + Si(s) = SiO2(Coesite,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:149.4(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:148.4(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:108.8(4SF)	2	ENS	6494
4	227	0	Inf. Dilution	lgK:85.14(4SF)	1	ENS	6494
5	327	0	Inf. Dilution	lgK:69.34(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-203.77(5SF)	4	EFE	6482
7	25	0	Inf. Dilution	dG:-203.8(4SF)	4	EFE	6494
8	25	0	Inf. Dilution/m	dG:-850.850(6SF)kJ	3	CRV	24953
9	25	0	Inf. Dilution	dH:-216.92(5SF)	4	EFE	6482
10	25	0	Inf. Dilution	dH:-217.0(4SF)	4	MTD	6494
11	25	0	Inf. Dilution/m	dH:-905.580(6SF)kJ	3	CRV	24953
12	127	0	Inf. Dilution	dH:-217.0(4SF)	2	MTD	6494
13	227	0	Inf. Dilution	dH:-216.9(4SF)	1	MTD	6494
14	327	0	Inf. Dilution	dH:-216.8(4SF)	0	MTD	6494

Reaction No. 67399

O2(g) + Si(s) = SiO2(Stish.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:140.7(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:139.7(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:102.2(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:79.72(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:64.73(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-191.9(4SF)	4	EFE	6494
7	25	0	Inf. Dilution/m	dG:-802.830(6SF)kJ	3	CRV	24953
8	25	0	Inf. Dilution	dH:-205.9(4SF)	4	MTD	6494
9	25	0	Inf. Dilution/m	dH:-861.320(6SF)kJ	3	CRV	24953
10	127	0	Inf. Dilution	dH:-205.9(4SF)	4	MTD	6494

11	227	0	Inf. Dilution	dH:-205.8(4SF)	4	MTD	6494
12	327	0	Inf. Dilution	dH:-205.7(4SF)	4	MTD	6494

Reaction No. 67400

O2(g) + Si(s) = SiO2(glass,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:148.8(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:147.8(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:108.6(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:85.00(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:69.31(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-203.0(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-215.5(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-215.6(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-215.5(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-215.3(4SF)	4	MTD	6494

Reaction No. 67407

Al(s) + F2(g) + K(s) + 3<Mg(s)> + 5<O2(g)> + 3<Si(s)> = KMg3AlSi3O10F2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1056.(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1050.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:771.9(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:605.4(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:494.4(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1441.(4SF)	4	EFE	6494
7	25	0	GAMSPATH	dG:-6031.(4SF)kJ	3	CRV	12638
8	25	0	Inf. Dilution	dH:-1524.(4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-1525.(4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-1524.(4SF)	4	MTD	6494
11	327	0	Inf. Dilution	dH:-1523.(4SF)	4	MTD	6494

Reaction No. 67408

Al(s) + Na(s) + 2<O2(g)> + Si(s) = NaAlSiO4(glass,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:346.4(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:344.2(4SF)	0	ENS	6494

3	127	0	Inf. Dilution	lgK:253.2(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:198.6(4SF)	3	ENS	6494
5	327	0	Inf. Dilution	lgK:162.2(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-472.6(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-499.3(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-500.0(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-499.9(4SF)	3	MTD	6494
10	327	0	Inf. Dilution	dH:-499.6(4SF)	0	MTD	6494

Reaction No. 67409

Al(s) + Na(s) + 2<O2(g)> + Si(s) = NaAlSiO4(Carn.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:345.7(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:343.4(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:252.4(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:197.8(4SF)	3	ENS	6494
5	327	0	Inf. Dilution	lgK:161.3(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-471.6(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dG:-1958.(4SF)kJ	2	EFE	24802
8	25	0	Inf. Dilution	dH:-499.4(4SF)	4	MTD	6494
9	127	0	Inf. Dilution	dH:-500.1(4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-500.0(4SF)	3	MTD	6494
11	327	0	Inf. Dilution	dH:-499.8(4SF)	0	MTD	6494

Reaction No. 67410

Al(s) + Na(s) + 4<O2(g)> + 3<Si(s)> = NaAlSi3O8(glass,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:642.1(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:637.9(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:469.2(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:367.9(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:300.4(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-876.0(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-926.3(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-927.1(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-926.8(4SF)	4	MTD	6494

10	327	0	Inf. Dilution	dH:-926.3(4SF)	4	MTD	6494

Reaction No. 67411							
Al(s) + Li(s) + 5<O2(g)> + 4<Si(s)> = LiAlSi4O10(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:807.7(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:802.5(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:589.7(4SF)	3	ENS	6494
4	227	0	Inf. Dilution	lgK:462.1(4SF)	2	ENS	6494
5	327	0	Inf. Dilution	lgK:376.9(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-1102.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1168.(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-1168.(4SF)	3	MTD	6494
9	227	0	Inf. Dilution	dH:-1169.(4SF)	2	MTD	6494
10	327	0	Inf. Dilution	dH:-1168.(4SF)	0	MTD	6494

Reaction No. 67412							
2<Al(s)> + 2<Ca(s)> + H2(g) + 4<O2(g)> + Si(s) = Ca2Al2SiO6(OH)2(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:713.6(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:708.9(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:519.9(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:406.5(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:330.9(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-973.5(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1038.(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-1038.(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-1038.(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-1037.(4SF)	4	MTD	6494

Reaction No. 67413							
2<Al(s)> + Ca(s) + 4<O2(g)> + 2<Si(s)> = CaO.Al2O3.2SiO2(glass,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:691.7(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:687.2(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:506.0(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:397.3(4SF)	4	ENS	6494

5	327	0	Inf. Dilution	lgK:324.8(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-943.7(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-995.0(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-995.2(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-994.9(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-994.4(4SF)	4	MTD	6494

Reaction No. 67414

Mg(s) + 1.5<O2(g)> + Si(s) = MgO.SiO2(Ilmen.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:244.8(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:243.2(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:178.5(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:139.8(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:113.9(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-334.0(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-355.3(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-355.0(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-354.7(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-354.4(4SF)	4	MTD	6494

Reaction No. 67415

Ca(s) + Mg(s) + 3<O2(g)> + 2<Si(s)> = CaO.MgO.2SiO2(glass,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:523.5(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:520.1(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:382.7(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:300.3(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:245.4(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-714.2(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-754.3(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-754.3(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-754.0(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-753.5(4SF)	4	MTD	6494

Reaction No. 67416

Ca(s) + 1.5<O2(g)> + Si(s) = CaO.SiO2(glass,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:267.5(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:265.8(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:195.7(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:153.7(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:125.7(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-364.9(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-384.5(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-384.4(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-384.3(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-384.1(4SF)	4	MTD	6494

Reaction No. 67417

2<Al(s)> + 3<Mg(s)> + 6<O2(g)> + 3<Si(s)> = 3MgO.Al2O3.3SiO2(glass,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1022.(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1016.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:747.6(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:586.8(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:479.7(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1395.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1473.(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-1472.(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-1471.(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-1470.(4SF)	4	MTD	6494

Reaction No. 67418

2<C(s)> + 5<Ca(s)> + 6.5<O2(g)> + 2<Si(s)> = 2CaCO3.3CaO.2SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1054.(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1047.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:769.3(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:603.0(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:492.2(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1437.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1523.(4SF)	4	MTD	6494

8	127	0	Inf. Dilution	dH:-1522.(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-1521.(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-1520.(4SF)	4	MTD	6494

Reaction No. 67419

3<Ca(s)> + H2(g) + 5<O2(g)> + 3<Si(s)> = 3CaO.3SiO2.H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:855.3(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:849.7(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:623.4(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:487.7(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:397.2(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1167.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1242.(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-1243.(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-1242.(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-1242.(4SF)	4	MTD	6494

Reaction No. 67420

Ca(s) + 3<Fe(s)> + 0.5<H2(g)> + 4.5<O2(g)> + 2<Si(s)> = CaO.3FeO.2SiO2.OH(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:602.1(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:598.1(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:437.5(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:341.1(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:277.0(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-821.5(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-882.6(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-881.8(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-880.9(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-879.8(4SF)	4	MTD	6494

Reaction No. 67421

B(s) + Ca(s) + 0.5<H2(g)> + 2.5<O2(g)> + Si(s) = CaBSiO4OH(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:406.1(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:403.4(4SF)	0	ENS	6494

3	127	0	Inf. Dilution	lgK:295.6(4SF)	3	ENS	6494
4	227	0	Inf. Dilution	lgK:230.8(4SF)	2	ENS	6494
5	327	0	Inf. Dilution	lgK:187.7(4SF)	0	ENS	6494
6	25	0	Inf. Dilution	dG:-554.0(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-592.2(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-592.4(4SF)	3	MTD	6494
9	227	0	Inf. Dilution	dH:-592.3(4SF)	2	MTD	6494
10	327	0	Inf. Dilution	dH:-592.0(4SF)	0	MTD	6494

Reaction No. 67422

4<Be(s)> + H2(g) + 4.5<O2(g)> + 2<Si(s)> = 4BeO.2SiO2.H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:752.5(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:747.5(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:548.1(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:428.4(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:348.6(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-1027.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-1095.(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-1095.(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-1095.(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-1095.(4SF)	4	MTD	6494

Reaction No. 67423

2<Al(s)> + 3<Be(s)> + 9<O2(g)> + 6<Si(s)> = 3BeO.Al2O3.6SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1489.(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:1479.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:1087.(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:852.1(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:695.2(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-2032.(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-2153.(4SF)	4	MTD	6494
8	27	0	Inf. Dilution	dH:-2153.(4SF)	0	MTD	6494
9	127	0	Inf. Dilution	dH:-2153.(4SF)	4	MTD	6494
10	227	0	Inf. Dilution	dH:-2153.(4SF)	4	MTD	6494

11	327	0	Inf. Dilution	dH:-2152.(4SF)	4	MTD	6494

Reaction No. 67424

Al(s) + Be(s) + 0.5<H2(g)> + 2.5<O2(g)> + Si(s) = BeAlSiO4(OH)(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:415.2(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:412.5(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:302.2(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:236.0(4SF)	4	ENS	6494
5	327	0	Inf. Dilution	lgK:191.9(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-566.5(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-605.4(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-605.6(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-605.5(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-605.1(4SF)	4	MTD	6494

Reaction No. 67425

6.75<Al(s)> + B(s) + 0.375<H2(g)> + 9<O2(g)> + 3<Si(s)> = Al6.75Si3BO17.25(OH).75(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1501.(4SF)	3	ENS	6494
2	27	0	Inf. Dilution	lgK:1491.(4SF)	0	ENS	6494
3	127	0	Inf. Dilution	lgK:1095.(4SF)	3	ENS	6494
4	227	0	Inf. Dilution	lgK:856.7(4SF)	3	ENS	6494
5	327	0	Inf. Dilution	lgK:698.1(4SF)	3	ENS	6494
6	25	0	Inf. Dilution	dG:-2048.(4SF)	3	EFE	6494
7	25	0	Inf. Dilution	dH:-2177.(4SF)	3	MTD	6494
8	127	0	Inf. Dilution	dH:-2178.(4SF)	3	MTD	6494
9	227	0	Inf. Dilution	dH:-2178.(4SF)	3	MTD	6494
10	327	0	Inf. Dilution	dH:-2177.(4SF)	3	MTD	6494

Reaction No. 67426

2<Al(s)> + F2(g) + 2<O2(g)> + Si(s) = Al2SiO4F2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:509.9(4SF)	4	ENS	6494
2	27	0	Inf. Dilution	lgK:506.6(4SF)	4	ENS	6494
3	127	0	Inf. Dilution	lgK:372.3(4SF)	4	ENS	6494
4	227	0	Inf. Dilution	lgK:291.8(4SF)	4	ENS	6494

5	327	0	Inf. Dilution	lgK:238.1(4SF)	4	ENS	6494
6	25	0	Inf. Dilution	dG:-695.7(4SF)	4	EFE	6494
7	25	0	Inf. Dilution	dH:-737.2(4SF)	4	MTD	6494
8	127	0	Inf. Dilution	dH:-737.3(4SF)	4	MTD	6494
9	227	0	Inf. Dilution	dH:-737.0(4SF)	4	MTD	6494
10	327	0	Inf. Dilution	dH:-736.5(4SF)	4	MTD	6494

Reaction No. 67429

Mg(s) + 1.5<O2(g)> + Si(s) = MgO.SiO2(Proto.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-348.2(4SF)	4	EFE	6482
2	25	0	Inf. Dilution	dH:-369.0(4SF)	4	EFE	6482

Reaction No. 67430

Al(s) + K(s) + 4<O2(g)> + 3<Si(s)> = KAlSi3O8(Feld.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:658.(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-895.18(5SF)	0	EFE	6482
3	25	0	Inf. Dilution	dG:-897.07(5SF)	0	CRV	10174
4	25	0	Inf. Dilution	dG:-896.44(5SF)	5	CRV	29553
5	25	0	GAMSPATH	dG:-3737.(4SF)kJ	0	CRV	12638
6	50	0	Inf. Dilution	dG:-898.39(5SF)	0	CRV	10174
7	75	0	Inf. Dilution	dG:-899.81(5SF)	0	CRV	10174
8	100	0	Inf. Dilution	dG:-901.33(5SF)	0	CRV	10174
9	125	0	Inf. Dilution	dG:-902.93(5SF)	0	CRV	10174
10	150	0	Inf. Dilution	dG:-904.63(5SF)	0	CRV	10174
11	175	0	Inf. Dilution	dG:-906.42(5SF)	0	CRV	10174
12	200	0	Inf. Dilution	dG:-908.28(5SF)	0	CRV	10174
13	225	0	Inf. Dilution	dG:-910.23(5SF)	0	CRV	10174
14	250	0	Inf. Dilution	dG:-912.25(5SF)	0	CRV	10174
15	300	0	Inf. Dilution	dG:-916.48(5SF)	0	CRV	10174
16	25	0	Inf. Dilution	dH:-949.04(5SF)	0	EFE	6482
17	25	0	Inf. Dilution	dH:-950.30(5SF)	5	CRV	29553
18	25	0	GAMSPATH	dH:-3962.40(6SF)kJ	0	CRV	12638

Reaction No. 67431

O2(g) + Si(s) = SiO2(Tridy.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-204.11(5SF)	4	EFE	6482
2	25	0	Inf. Dilution	dG:-855.26(5SF)kJ	3	CRV	8695
3	25	0	Inf. Dilution/m	dG:-853.610(6SF)kJ	3	CRV	24953
4	25	0	Inf. Dilution	dH:-216.96(5SF)	4	EFE	6482
5	25	0	Inf. Dilution	dH:-909.06(5SF)kJ	5	CRV	8695
6	25	0	Inf. Dilution/m	dH:-907.490(6SF)kJ	3	CRV	24953

Reaction No. 67432

3<Al(s)> + 2<Ca(s)> + 0.5<H2(g)> + 6.5<O2(g)> + 3<Si(s) > = Ca2Al3Si3O12(OH)(clino,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1137.(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1552.7(5SF)	0	EFE	6482
3	25	0	GAMSPATH	dG:-6484.(4SF)kJ	0	CRV	12638
4	25	0	Inf. Dilution	dH:-1647.9(5SF)	2	EFE	6482
5	25	0	GAMSPATH	dH:-6879.40(6SF)kJ	0	CRV	12638

Reaction No. 67434

Al2Si2O5(OH)4(Kaol.,s) + 6<H+1> = 2<Al+3> + 2<H+1(2)_SiH2O4-2> + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:7.0(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:7.428(4SF)	0	CRV	440
3	25	0	Inf. Dilution	lgK:5.706(4SF)	0	CRV	5619
4	25	0	Inf. Dilution	lgK:11.122(5SF)	0	CRV	6421
5	25	0	Inf. Dilution	lgK:7.435(4SF)	0	ENS	6496
6	25	0	Inf. Dilution	lgK:5.45(3SF)	0	MGL	6498
Note (6): Possibly for reaction with halved stoichiometry, possibly not!							
7	25	0	Inf. Dilution	lgK:7.41(3SF)	0	CRV	8872
8	25	0	Inf. Dilution	lgK:11.122(5SF)	0	EOD	8872
Note (8): Data for Naumov, Ryzhenko & KKhodakovsky, 1971							
9	25	0	Inf. Dilution	lgK:7.028(4SF)	0	CRV	8873
10	25	0	Inf. Dilution	lgK:7.604(4SF)	0	CRV	8874
11	25	0	Inf. Dilution	lgK:7.435(4SF)	3	EOD	29487
Note (11): Presumed reference to Kaolinite							
12	25	0	Inf. Dilution	lgK:6.5(2SF)	3	MSL	31760
Note (12): Slightly impure solid & average							

13	25	0	Inf. Dilution	lgK:7.446(4SF)	0	CRV	32224
14	40	0	Inf. Dilution	lgK:6.(1SF)	5	MSL	31760
Note (14): Slightly impure solid & average							
15	90	0	Inf. Dilution	lgK:2.80(3SF)	3	EOD	29487
Note (15): Presumed reference to Kaolinite							
16	25	0	Inf. Dilution	dH:-35.3(3SF)	2	ENS	6496
17	25	0	Inf. Dilution	dH:-147.7(4SF)kJ	3	EOD	29487
Note (17): Presumed reference to Kaolinite							

Reaction No. 67435

Mg3Si4O10(OH)2.H2O(s) + 6<H+1> + 3<H2O> = 3<Mg+2> + 4<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:25.79(4SF)	1	ENS	6496
2	25	0	Inf. Dilution	lgK:23.54(4SF)	1	CRV	32224
Note (2): trihydrate							

Reaction No. 67455

Al2Si2O5(OH)4(Hall.,s) + 6<H+1> = 2<Al+3> + 2<H+1(2)_SiH2O4-2> + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:10.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:8.72(3SF)	0	MGL	6498

Reaction No. 67456

Al2O3SiOH(OH)3(s) + 6<H+1> = 2<Al+3> + H+1(2)_SiH2O4-2 + 3<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:14.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:12.00(4SF)	0	MGL	6498

Reaction No. 69044

4<H+1> + 4<SiH2O4-2> = H+1(4)_SiH2O4-2(4)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:55.90(4SF)	5	CRV	9402

Reaction No. 70514

C(s) + Si(s) = SiC(beta,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-62.8(3SF)kJ	4	EFE	7381

2	25	0	Inf. Dilution	dH:-65.3(3SF)kJ	4	ENS	7381

Reaction No. 70515

2<Cl2(g)> + Si(s) = SiCl4(g)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-617.0(4SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dG:-616.98(5SF)kJ	4	EFE	7381
3	25	0	Inf. Dilution	dH:-657.0(4SF)kJ	3	CRV	6330
4	25	0	Inf. Dilution	dH:-657.01(5SF)kJ	4	ENS	7381

Reaction No. 71000

SiO3-2 + 3<H2O> + 4<e-1> = Si(s) + 6<OH-1>

Note: The species SiO3-2 is an artefact; it should be SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:-1.697(4SF)	0	CRV	6330
2	25	0	Inf. Dilution	E:-1.697(4SF)	0	CRV	22519

Reaction No. 71001

SiO2(Quartz,s) + 4<e-1> + 4<H+1> = Si(s) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-66.9(3SF)	1	EOR	
2	25	0	Inf. Dilution	E:-0.857(3SF)	0	CRV	6330
3	25	0	Inf. Dilution	E:-0.990(3SF)	5	CRV	7761
4	25	0	Inf. Dilution	E:0.857(3SF)	0	CRV	22519
Note (4): Wrong sign							
5	25	0	Inf. Dilution	E:-0.909(3SF)	0	CRV	22551
6	25	0	Inf. Dilution	dH:339.0(4SF)kJ	5	CRV	7761

Reaction No. 71002

SiO(s) + 2<H+1> + 2<e-1> = Si(s) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.8(1SF)	0	CRV	6330
3	25	0	Inf. Dilution	E:-0.8(1SF)	0	CRV	7761
4	25	0	Inf. Dilution	E:-0.8(1SF)	0	CRV	22519
5	25	0	Inf. Dilution	dH:131.4(4SF)kJ	0	CRV	7761

Reaction No. 71003

SiF6-2 + 4<e-1> = Si(s) + 6<F-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-91.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.24(3SF)	0	CRV	6330

Note (2): Low wgt for inter-reaction consistency

Reaction No. 71432

SiO2(Quartz,s) + 8<H+1> + 8<e-1> = SiH4(g) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-76.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.569(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:373.6(4SF)kJ	2	CRV	7761

Reaction No. 71433

H+1(2)_SiH2O4-2 + 4<e-1> + 4<H+1> = Si(s) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-63.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.931(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	E:-0.848(3SF)	0	CRV	22551
4	25	0	Inf. Dilution	dH:313.9(4SF)kJ	2	CRV	7761

Reaction No. 71434

SiO2(am.,s) + 4<H+1> + 4<e-1> = Si(s) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-65.9(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.973(3SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:329.5(4SF)kJ	2	CRV	7761

Reaction No. 71435

SiO2(Quartz,s) + 2<H+1> + 2<e-1> = SiO(s) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-38.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.2(2SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:214.3(4SF)kJ	2	CRV	7761

Reaction No. 71448

Si(s) + 4<H2O> + 4<e-1> = SiH4(g) + 4<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	3	Inert	lgK:-68.9(3SF)	1	EOR	
2	25	0	Inf. Dilution	E:-0.975(3SF)	5	CRV	7761
3	25	0	Inf. Dilution	dH:257.5(4SF)kJ	5	CRV	7761

Reaction No. 71449

SiH2O4-2 + 6<H2O> + 8<e-1> = SiH4(g) + 10<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-189.9(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.405(4SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:847.5(4SF)kJ	2	CRV	7761

Reaction No. 71450

SiO(s) + H2O + 2<e-1> = Si(s) + 2<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-56.5(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.6(2SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:239.7(4SF)kJ	2	CRV	7761

Reaction No. 71451

H+1_SiH2O4-2 + H2O + 4<e-1> = Si(s) + 5<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-123.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.820(4SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:565.5(4SF)kJ	2	CRV	7761

Reaction No. 71452

SiH2O4-2 + 2<H2O> + 4<e-1> = Si(s) + 6<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-124.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.834(4SF)	0	CRV	7761
3	25	0	Inf. Dilution	dH:589.3(4SF)kJ	2	CRV	7761

Reaction No. 71453

SiH2O4-2 + H2O + 2<e-1> = SiO(s) + 4<OH-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	25	3	Inert	lgK:-69.(2SF)	2	EES	
2	25	0	Inf. Dilution	E:-2.0(2SF)	3	CRV	7761
3	25	0	Inf. Dilution	dH:339.9(4SF)kJ	3	CRV	7761

Reaction No. 71739

O2(g) + Si(s) = SiO2(am.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:149.0(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-849.9(4SF)kJ	0	EOD	23101
3	25	0	Inf. Dilution	dG:-848.90(5SF)kJ	5	CRV	31308
4	25	0	GAMSPATH	dG:-848.9(4SF)kJ	0	CRV	12638
5	25	0	Inf. Dilution/m	dG:-849.230(6SF)kJ	0	CRV	24953
6	25	0	Inf. Dilution	dH:-903.(3SF)kJ	0	EOD	23101
7	25	0	Inf. Dilution	dH:-903.(3SF)kJ	5	CRV	31308
8	25	0	GAMSPATH	dH:-897.750(6SF)kJ	0	CRV	12638
9	25	0	Inf. Dilution/m	dH:-901.550(6SF)kJ	2	CRV	24953

Reaction No. 71746

2<H2(g)> + 3<Mg(s)> + 4.5<O2(g)> + 2<Si(s)> = Mg3Si2O5(OH)4(Serp.,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	GAMSPATH	dG:-4040.(4SF)kJ	3	CRV	12638
2	25	0	GAMSPATH	dH:-4367.70(6SF)kJ	3	CRV	12638

Reaction No. 71748

2<O2(g)> + Si(s) + U(s) = UO2.SiO2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:329.7(4SF)	1	ELC	
2	25	0	GAMSPATH	dG:-1084.(4SF)kJ	0	CRV	12638
3	25	0	GAMSPATH	dH:-1991.30(6SF)kJ	0	CRV	12638

Reaction No. 71754

1.5<H2(g)> + 2<O2(g)> + Si(s) + e-1 = H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:219.6(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1251.7(1.162SD)kJ	0	CRV	14923
3	25	0	GAMSPATH	dG:-1252.(4SF)kJ	0	CRV	12638
4	25	0	Inf. Dilution	dH:-1431.4(3.743SD)kJ	1	CRV	14923

5	25	0	GAMSPATH	dH:-1430.70(6SF)kJ	1	CRV	12638

Reaction No. 71819

2<H2(g)> + 2<O2(g)> + Si(s) = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:229.5(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-1307.5(5SF)kJ	0	CRV	6494
3	25	0	Inf. Dilution	dG:-1309.20(6SF)kJ	0	CRV	8798
4	25	0	Inf. Dilution	dG:-1316.6(5SF)kJ	0	CRV	9015
5	25	0	Inf. Dilution	dG:-1309.2(6SF)kJ	0	CRV	9033
6	25	0	Inf. Dilution	dG:-1309.3(5SF)kJ	4	EOD	12843
7	25	0	Inf. Dilution	dG:-1307.7(1.156SD)kJ	0	CRV	14923
8	25	0	Inf. Dilution	dG:-312.92(5SF)	5	CRV	29553
9	25	0	Inf. Dilution	dG:-1239.70(0.6SD)kJ	0	CRV	30883
10	25	0	Inf. Dilution	dH:-1460.0(5SF)kJ	0	CRV	6494
11	25	0	Inf. Dilution	dH:-1460.20(6SF)kJ	0	CRV	8798
12	25	0	Inf. Dilution	dH:-1468.6(5SF)kJ	0	CRV	9015
13	25	0	Inf. Dilution	dH:-1460.9(6SF)kJ	0	MSL	9033
14	25	0	Inf. Dilution	dH:-1458.9(5SF)kJ	4	MTD	12843
15	25	0	Inf. Dilution	dH:-1457.0(3.163SD)kJ	0	CRV	14923
16	25	0	Inf. Dilution	dH:-348.68(5SF)	5	CRV	29553
17	25	0	GAMSPATH	dH:-1456.30(6SF)kJ	0	CRV	12638

Reaction No. 71820

2.7<Al(s)> + H2(g) + 0.7<K(s)> + 6<O2(g)> + 3.3<Si(s)> = Illite(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	GAMSPATH	dG:-5491.(4SF)kJ	3	CRV	12638
2	25	0	GAMSPATH	dH:-5870.(4SF)kJ	3	CRV	12638

Reaction No. 71872

Al+3_OH-1(3)_(Gibbsite,s) + H+1(2)_SiH2O4-2 = 0.5<Al2Si2O5(OH)4(Kaol.,s)> + 2.5<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.1(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:4.25(3SF)	0	CRV	6405

Note (2): p. 246

Reaction No. 71873

Al2Si2O5(OH)4(Kaol.,s) + 2<K+1> + 6<Mg+2> + 4<H+1(2)_SiH2O4-2> - H2O = 2 <KMg3AlSi3O10(OH)2(s)> + 14<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-69.2(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:70.14(4SF)	0	CRV	6405
Note (2): p. 246							

Reaction No. 71874							
KAl3Si3O10(OH)2(s) + H+1 + 1.5<H2O> = 1.5<Al2Si2O5(OH)4(Kaol.,s)> + K+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.4(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:3.51(3SF)	2	CRV	6405

Reaction No. 71875							
CaO.Al2O3.2SiO2(s) + 2<H+1> + H2O = Al2Si2O5(OH)4(Kaol.,s) + Ca+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:19.66(4SF)	0	CRV	6405

Reaction No. 71876							
2<KAlSi3O8(Feld.,s)> + 2<H+1> + 9<H2O> = Al2Si2O5(OH)4(Kaol.,s) + 4<H+1(2)_SiH2O4-2 > + 2<K+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-7.0(2SF)	3	CRV	6405

Reaction No. 71877							
Na2O.(Al2O3)7.(SiO2)22.6H2O(s) + 2<H+1> + 23<H2O> = 7<Al2Si2O5(OH)4(Kaol.,s)> + 8 <H+1(2)_SiH2O4-2> + 2<Na+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-18.22(4SF)	5	CRV	6405
Note (1): p. 246 - comment re inconsistency with Stumm & Morgan data							

Reaction No. 71878							
CaO.(Al2O3)7.(SiO2)22.6H2O(s) + 2<H+1> + 23<H2O> = 7<Al2Si2O5(OH)4(Kaol.,s)> + 8 <H+1(2)_SiH2O4-2> + Ca+2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-15.4(3SF)	0	CRV	6405
Note (2): p. 246 - comment re inconsistency with Stumm & Morgan data							

Reaction No. 71947

H+1(2)_SiH2O4-2 + 3<H+1(2)_45DiClCat-2> = Si+4_45DiClCat-2(3) + 2<H+1> + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-4.27(3SF)	0	ENS	7271
Note (1): Incorrect citation of value for 3,4-Dinitrocatechol							
2	25	0.1	Unknown	lgK:-8.5(2SF)	3	MMR	8688
Note (2): Suspect other values							

Reaction No. 71949

H+1(2)_SiH2O4-2 + 3<H+1(2)_34DiNitrCat-2> = Si+4_34DiNitrCat-2(3) + 2<H+1> + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-4.27(3SF)	4	ENS	8688

Reaction No. 71951

H+1(2)_SiH2O4-2 + 3<H+1(2)_4NitrCat-2> = Si+4_4NitrCat-2(3) + 2<H+1> + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	Unknown	lgK:-7.74(3SF)	4	MMR	8688
Note (1): Suspect other values							

Reaction No. 73370

Al+3 + H+1(2)_SiH2O4-2 = Al+3_H+1_SiH2O4-2 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-2.50(0.05SD)	4	MGL	7023

Reaction No. 73538

2<H2(g)> + 3.5<O2(g)> + 2<Si(s)> + 2<e-1> = Si2O3(OH)4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:398.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2269.9(2.878SD)kJ	0	CRV	14923

Reaction No. 73539

2.5<H2(g)> + 3.5<O2(g)> + 2<Si(s)> + e-1 = Si2O2(OH)5-1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:409.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-2332.1(2.878SD)kJ	0	CRV	14923

Reaction No. 73540

1.5<H2(g)> + 4.5<O2(g)> + 3<Si(s)> + 3<e-1> = Si3O6(OH)3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:535.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-3048.5(3.870SD)kJ	0	CRV	14923

Reaction No. 73541

2.5<H2(g)> + 5<O2(g)> + 3<Si(s)> + 3<e-1> = Si3O5(OH)5-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:576.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-3292.0(3.896SD)kJ	0	CRV	14923

Reaction No. 73542

2<H2(g)> + 6<O2(g)> + 4<Si(s)> + 4<e-1> = Si4O8(OH)4-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:715.3(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-4075.2(5.437SD)kJ	0	CRV	14923

Reaction No. 73543

2.5<H2(g)> + 6<O2(g)> + 4<Si(s)> + 3<e-1> = Si4O7(OH)5-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:726.1(4SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-4136.8(4.934SD)kJ	0	CRV	14923

Reaction No. 73546

2<H+1(2)_SiH2O4-2> = 2<H+1> + H2O + Si2O3(OH)4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-19.000(0.300SD)	3	CRV	14923
2	25	0	Inf. Dilution	lgK:-19.000(0.15SD)	2	CRV	27292
3	25	0	Inf. Dilution	dG:108.450(0.9SD)kJ	2	CRV	27292

Reaction No. 73547

2<H+1(2)_SiH2O4-2> = H+1 + Si2O2(OH)5-1 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-8.100(0.300SD)	3	CRV	14923
2	25	0	Inf. Dilution	lgK:-8.100(0.15SD)	2	CRV	27292
3	25	0	Inf. Dilution	lgK:-8.50(3SF)	4	EOD	30201
4	50	0	Inf. Dilution	lgK:-8.14(0.25SD)	4	EOD	30201

5	75	0	Inf. Dilution	lgK:-7.86(0.25SD)	4	EOD	30201
6	100	0	Inf. Dilution	lgK:-7.65(0.25SD)	4	EOD	30201
7	25	0	Inf. Dilution	dG:46.235(0.9SD)kJ	2	CRV	27292

Reaction No. 73548

3<H+1(2)_SiH2O4-2> = 3<H+1> + 3<H2O> + Si3O6(OH)3-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.600(0.300SD)	3	CRV	14923
2	25	0	Inf. Dilution	lgK:-28.600(0.15SD)	2	CRV	27292
3	25	0	Inf. Dilution	dG:163.250(0.9SD)kJ	2	CRV	27292

Reaction No. 73549

3<H+1(2)_SiH2O4-2> = 3<H+1> + Si3O5(OH)5-3 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-27.500(0.300SD)	2	CRV	14923
2	25	0	Inf. Dilution	lgK:-27.500(0.15SD)	2	CRV	27292
3	25	0	Inf. Dilution	lgK:-29.40(4SF)	3	EOD	30201
4	50	0	Inf. Dilution	lgK:-28.75(0.25SD)	4	EOD	30201
5	75	0	Inf. Dilution	lgK:-28.34(4SF)	4	EOD	30201
6	100	0	Inf. Dilution	lgK:-28.11(0.25SD)	4	EOD	30201
7	25	0	Inf. Dilution	dG:156.970(0.9SD)kJ	1	CRV	27292

Reaction No. 73550

4<H+1(2)_SiH2O4-2> = 4<H+1> + 4<H2O> + Si4O8(OH)4-4

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-36.300(0.500SD)	3	CRV	14923
2	25	0	Inf. Dilution	lgK:-36.300(0.25SD)	2	CRV	27292
3	25	0	Inf. Dilution	dG:207.200(1.4SD)kJ	2	CRV	27292

Reaction No. 73551

4<H+1(2)_SiH2O4-2> = 3<H+1> + 4<H2O> + Si4O7(OH)5-3

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-25.500(0.300SD)	3	CRV	14923
2	25	0	Inf. Dilution	lgK:-25.500(0.15SD)	3	CRV	27292
3	25	0	Inf. Dilution	dG:145.560(0.9SD)kJ	3	CRV	27292

Reaction No. 74156

Si+4_F-1(6) + 4<e-1> = Si(s) + 6<F-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-93.0(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-1.24(3SF)	0	CRV	22519

Reaction No. 74236

SiO2(Cristob.,s) + 4<e-1> + 4<H+1> = Si(s) + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-66.6(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.900(3SF)	0	CRV	22551

Reaction No. 74237

H+1_SiH2O4-2 + 4<e-1> + 5<H+1> = Si(s) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-53.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.712(3SF)	0	CRV	22551

Reaction No. 74238

SiO(g) + 2<e-1> + 2<H+1> = Si(s) + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:19.3(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.808(3SF)	0	CRV	22551

Reaction No. 74239

H+1(2)_SiH2O4-2 + 8<e-1> + 8<H+1> = SiH4(g) + 4<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-73.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	E:-0.495(3SF)	0	CRV	22551

Reaction No. 74240

H+1(2)_SiO3-2_(s) = H+1_SiO3-2 + H+1

Note: The species SiO3-2 is an artefact; it should be SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.(2SF)	0	CRV	22551

Reaction No. 74597

2<Ca(s)> + 6<O2(g)> + 3<Si(s)> + Zr(s) = Ca2ZrSi3O12(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5951.5(15.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-6283.0(15.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:252.500(2.5MD)J/K	6	CRV	20829

Reaction No. 74598

3<Ca(s)> + 4.5<O2(g)> + 2<Si(s)> + Zr(s) = Ca3ZrSi2O9(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-4769.3(10.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-5029.0(10.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:238.100(1.5MD)J/K	6	CRV	20829

Reaction No. 74599

3.5<O2(g)> + 2<Si(s)> + Sr(s) + Zr(s) = SrZrSi2O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3444.0(4.2MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3640.8(4.2MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:187.810(0.4MD)J/K	6	CRV	20829

Reaction No. 74600

9<O2(g)> + 5<Si(s)> + 6<Sr(s)> + Zr(s) = Sr6ZrSi5O18(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-7800.(4SF)kJ	1	EES	
2	25	0	Inf. Dilution	dH:-9867.9(8.6MD)kJ	6	CRV	20829

Reaction No. 74601

2<Na(s)> + 2.5<O2(g)> + Si(s) + Zr(s) = Na2ZrSiO5(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2523.5(20.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-2670.0(20.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:172.400(2.0MD)J/K	6	CRV	20829

Reaction No. 74602

2<Na(s)> + 3.5<O2(g)> + 2<Si(s)> + Zr(s) = Na2ZrSi2O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3412.4(10.0MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-3606.0(10.0MD)kJ	6	CRV	20829

3	25	0	Inf. Dilution	Cp:211.000(5.0MD)J/K	6	CRV	20829

Reaction No. 74603

2<H2(g)> + 2<Na(s)> + 5.5<O2(g)> + 3<Si(s)> + Zr(s) = Na2ZrSi3O9.2H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-4654.0(20.0MD)kJ	6	CRV	20829

Reaction No. 74605

2<Na(s)> + 5.5<O2(g)> + 4<Si(s)> + Zr(s) = Na2ZrSi4O11(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5181.0(20.000MD)kJ	6	CRV	20829

Reaction No. 74606

3<H2(g)> + 2<Na(s)> + 9<O2(g)> + 6<Si(s)> + Zr(s) = Na2ZrSi6O15.3H2O(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-7931.0(30.000MD)kJ	6	CRV	20829

Reaction No. 74607

4<Na(s)> + 6<O2(g)> + 3<Si(s)> + 2<Zr(s)> = Na4Zr2Si3O12(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5944.0(20.057MD)kJ	6	CRV	20829
2	25	0	Inf. Dilution	dH:-6258.0(20.0MD)kJ	6	CRV	20829
3	25	0	Inf. Dilution	Cp:365.000(5.0MD)J/K	6	CRV	20829

Reaction No. 74609

2<Cs(s)> + 3.5<O2(g)> + 2<Si(s)> + Zr(s) = Cs2ZrSi2O7(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dH:-3587.0(10.0MD)kJ	6	CRV	20829

Reaction No. 75172

Ca(s) + H2(g) + 6<O2(g)> + 2<Si(s)> + 2<U(s)> = Ca(UO2)2.(SiO4H)2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1189.(4SF)	5	EOD	13532

Reaction No. 75220

CaAl2Si4O12.2H2O(s) + 8<H+1> + 2<H2O> = Ca+2 + 2<Al+3> + 4<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.87(4SF)	0	EOD	23102

2	25	0	Inf. Dilution	lgK:14.42(4SF)	3	CRV	27383
3	25	0	Inf. Dilution	lgK:18.68(4SF)	0	CRV	32224

Reaction No. 75223

Zn2SiO4(s) + 4<H+1> = 2<Zn+2> + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.40(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:15.33(4SF)	0	EOD	23102
3	25	0	Inf. Dilution	lgK:13.89(4SF)	4	CRV	32224

Reaction No. 75224

ZnSiO3(s) + 2<H+1> + H2O = Zn+2 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:6.9(2SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:2.93(3SF)	0	EOD	23102
3	25	0	Inf. Dilution	lgK:-19.85(4SF)	0	CRV	32224

Reaction No. 75559

6<Al(s)> + Cl2(g) + 8<Na(s)> + 12<O2(g)> + 6<Si(s)> = Na8Al6Si6O24Cl2(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1.2697E+4(5SF)kJ	2	EFE	24802

Reaction No. 75627

2<Cl2(g)> + Si(s) = SiCl4(l)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-619.8(4SF)kJ	3	CRV	6330
2	25	0	Inf. Dilution	dH:-687.0(4SF)kJ	3	CRV	6330

Reaction No. 75630

H+1(2)_SiO3-2 = H+1_SiO3-2 + H+1

Note: The species SiO3-2 is an artefact; it should be SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	23	0	Inf. Dilution	lgK:9.86(3SF)	0	CRV	6405

Note (1): p. 285

Reaction No. 75631

Mg+2(2)_H+1(3)_SiH2O4-2(3)_H2O(4)_(s) = 2<Mg+2> + 3<H+1_SiH2O4-2> + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	50	0	Inf. Dilution	lgK:-38.8(3SF)	0	CRV	8

Note (1): Solid +1 charge! Same problem in S&M database [95SMM_552]. Sepiolite?

Reaction No. 75633

1.7<Al(s)> + 0.7<Ca(s)> + 0.3<Na(s)> + 4<O2(g)> + Si(s) = Plagioclase(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3906.5(5SF)kJ	5	CRV	25435
2	25	0	Inf. Dilution	dH:-4108.2(5SF)kJ	5	CRV	25435

Reaction No. 75634

0.35<Ca(s)> + 0.23<Fe(s)> + 0.42<Mg(s)> + 1.5<O2(g)> + Si(s) = Augite(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-2835.3(5SF)kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-2888.5(5SF)kJ	3	CRV	25435

Reaction No. 75635

0.4<Fe(s)> + 1.6<Mg(s)> + 2<O2(g)> + Si(s) = Olivine(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1919.7(5SF)kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-2022.9(5SF)kJ	3	CRV	25435

Reaction No. 75636

2<Al(s)> + 2<H2(g)> + 3.72<O2(g)> + 1.22<Si(s)> = Allophane(s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3227.90(6SF)kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-3512.20(6SF)kJ	3	CRV	25435

Reaction No. 75637

O2(g) + Si(s) = SiO2(Moganite,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-851.310(6SF)kJ	3	CRV	25435
2	25	0	Inf. Dilution	dH:-900.720(6SF)kJ	3	CRV	25435

Reaction No. 75638

SiO2(Chalcedony,s) + 2<H2O> = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #

1	0	0	Inf. Dilution	lgK:-3.857(4SF)	0	CRV	25435
2	25	0	Inf. Dilution	lgK:-3.472(4SF)	0	CRV	25435
3	25	0	Inf. Dilution	lgK:-3.73(3SF)	4	CRV	32224
4	60	0	Inf. Dilution	lgK:-3.106(4SF)	0	CRV	25435
5	100	0	Inf. Dilution	lgK:-2.790(4SF)	0	CRV	25435
6	150	0	Inf. Dilution	lgK:-2.470(4SF)	0	CRV	25435
7	200	0	Inf. Dilution	lgK:-2.205(4SF)	0	CRV	25435
8	250	0	Inf. Dilution	lgK:-1.988(4SF)	0	CRV	25435
9	300	0	Inf. Dilution	lgK:-1.835(4SF)	0	CRV	25435

Reaction No. 76501							
2<Al(s)> + 2<H2(g)> + 2<Na(s)> + 6<O2(g)> + 3<Si(s)> = Na2Al2Si3O10.2H2O(Natr.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:931.4(4SF)	3	CRV	6494
2	25	0	Inf. Dilution	dG:-5316.5(5SF)kJ	3	CRV	6494
3	25	0	Inf. Dilution	dG:-5316.6(5.0SD)kJ	5	MCL	29317
4	25	0	Inf. Dilution	dH:-5718.6(5SF)kJ	3	CRV	6494
5	25	0	Inf. Dilution	dH:-5718.6(5.0SD)kJ	5	CRV	29317

Reaction No. 76502							
2<B(s)> + Ca(s) + 4<O2(g)> + 2<Si(s)> = CaB2Si2O8(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:644.2(4SF)	3	CRV	6494
2	25	0	Inf. Dilution	dG:-3677.0(5SF)kJ	3	CRV	6494
3	25	0	Inf. Dilution	dH:-3902.8(5SF)kJ	3	CRV	6494

Reaction No. 76503							
2<Al(s)> + Ca(s) + 3<H2(g)> + 6.5<O2(g)> + 3<Si(s)> = CaAl2Si3O10.3H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:980.7(4SF)	3	CRV	6494
2	25	0	Inf. Dilution	dG:-5597.6(5SF)kJ	3	CRV	6494
3	25	0	Inf. Dilution	dG:-5597.9(5.0SD)kJ	5	MCL	29317
4	25	0	Inf. Dilution	dH:-6049.0(5SF)kJ	3	CRV	6494
5	25	0	Inf. Dilution	dH:-6049.00(5.5SD)kJ	4	CRV	27383
6	25	0	Inf. Dilution	dH:-6049.0(5.0SD)kJ	5	CRV	29317

Reaction No. 77352

Am(s) + 1.5<H2(g)> + 2<O2(g)> + Si(s) - 2<e-1> = Am+3_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1896.80(6SF)kJ	5	CRV	20827

Reaction No. 77366

Am+3 + H+1(2)_SiH2O4-2 = Am+3_H+1_SiH2O4-2 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.680(4SF)	5	CRV	20827
2	25	0	Inf. Dilution	dG:9.590(4SF)kJ	5	CRV	20827

Reaction No. 77619

2<SiH2O4-2> + 2<H+1> = H+1(2)_SiH2O4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:27.32(4SF)	1	ELC	

Reaction No. 79081

SiH2O4-2 + H+1 + Am+3 = Am+3_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:21.47(4SF)	1	ELC	

Reaction No. 79429

SiH2O4-2 + H+1 + Al+3 = Al+3_H+1_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:20.65(4SF)	1	ELC	

Reaction No. 79464

1.5<O2(g)> + Si(s) + 2<e-1> = SiO3-2

Note: The species SiO3-2 is an artefact; it should be SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-887.(3SF)kJ	0	CRV	7421

Reaction No. 79503

0.5<H2(g)> + 1.5<O2(g)> + Si(s) + e-1 = H+1_SiO3-2

Note: The species SiO3-2 is an artefact; it should be SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1015.2(5SF)kJ	0	EFE	28693

Reaction No. 79504							
0.5<H2(g)> + Na(s) + 1.5<O2(g)> + Si(s) = H+1_Na+1_SiO3-2 Note: The species SiO3-2 is an artefact; it should be SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1283.1(5SF)kJ	0	EFE	28693

Reaction No. 79505							
Al(s) + 1.5<H2(g)> + 2<O2(g)> + Si(s) - 2<e-1> = Al+3_H+1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-1782.5(5SF)kJ	3	EFE	28693

Reaction No. 79668							
1.99<Al(s)> + 0.657<Ca(s)> + 2.147<H2(g)> + 0.676<Na(s)> + 6.3235<O2(g)> + 3.01 <Si(s)> = Mesolite(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-5513.2(5.4SD)kJ	5	MCL	29317
2	25	0	Inf. Dilution	dH:-5947.1(5.4SD)kJ	5	CRV	29317

Reaction No. 79669							
CaAl2Si4O12.6H2O(s) + 8<H+1> = Ca+2 + 2<Al+3> + 4<H+1(2)_SiH2O4-2> + 2<H2O >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.51(4SF)	4	CRV	27383

Reaction No. 79670							
Ca2Al4Si4O16.9H2O(s) + 16<H+1> = 2<Ca+2> + 4<Al+3> + 4<H+1(2)_SiH2O4-2> + 9<H2O >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.96(4SF)	4	CRV	27383

Reaction No. 79674							
Fe+3 + H+1(2)_SiH2O4-2 = Fe+3_H+1_SiH2O4-2 + H+1							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4/m	lgK:-0.521(3SF)	4	MSP	29280
2	25	0.1	NaClO4/m	lgK:-0.527(3SF)	4	MSP	29280
3	25	0.3	NaClO4/m	lgK:-0.526(3SF)	4	MSP	29280
4	25	0.3	NaClO4/m	lgK:-0.532(3SF)	4	MSP	29280
5	25	0.7	NaClO4/m	lgK:-0.418(3SF)	4	MSP	29280
6	25	0.7	NaClO4/m	lgK:-0.564(3SF)	4	MSP	29280

Reaction No. 79675							
Eu+3 + 2<H+1_SiH2O4-2> = Eu+3_H+1(2)_SiH2O4-2(2)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4/m	lgK:11.73(4SF)	4	MSP	29280

Reaction No. 79676							
Eu+3 + H+1_SiH2O4-2 = Eu+3_H+1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.1	NaClO4/m	lgK:6.17(0.10SD)	4	MSP	29280
2	25	0.7	NaClO4	lgK:6.18(0.18SD)	5	MGL	29968
Note (2): Paper casts doubt on earlier results by various other authors							

Reaction No. 79849							
H+1_SiH2O4-2 + Am+3 = Am+3_H+1_SiH2O4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:8.13(0.18SD)	5	CRV	25812

Reaction No. 80285							
2CaO.SiO2(beta,s) + 2<CO2(g)> = 2<Ca+2_CO3-2_(Calc.,s)> + SiO2(am.,s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:22.7(3SF)	1	ELC	
2	25	0	Inf. Dilution	dG:-142.06(5SF)kJ	0	CRV	30199

Reaction No. 80306							
Si2O2(OH)5-1 = H+1 + Si2O3(OH)4-2							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-10.90(4SF)	4	EOD	30201
2	50	0	Inf. Dilution	lgK:-10.59(0.25SD)	4	EOD	30201
3	75	0	Inf. Dilution	lgK:-10.38(0.25SD)	4	EOD	30201
4	100	0	Inf. Dilution	lgK:-10.24(0.25SD)	4	EOD	30201

Reaction No. 80307							
Al+3_OH-1(4) + SiH2O4-2 = Al(OH)3.HSiO4-3 + H2O							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	20	0	Inf. Dilution	lgK:-0.42(0.10SD)	4	EOD	30201
2	25	0	Inf. Dilution	lgK:-0.42(0.15SD)	4	EOD	30201
3	50	0	Inf. Dilution	lgK:-0.38(0.25SD)	4	EOD	30201

4	75	0	Inf. Dilution	lgK:-0.35(0.25SD)	4	EOD	30201
5	100	0	Inf. Dilution	lgK:-0.33(0.25SD)	4	EOD	30201

Reaction No. 80308

OHSodalite(s) + 21<H2O> = 7<Na+1> + 6<Al+3_OH-1(4)> + OH-1 + 6<H+1(2)_SiH2O4-2 >

Note: Stoichiometric analog of actual reported solid

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-60.34(4SF)	4	EOD	30201
2	30	0	Inf. Dilution	lgK:-61.18(4SF)	4	EOD	30201
3	45	0	Inf. Dilution	lgK:-64.20(4SF)	4	EOD	30201
4	60	0	Inf. Dilution	lgK:-66.69(4SF)	4	EOD	30201
5	75	0	Inf. Dilution	lgK:-68.89(4SF)	3	EOD	30201
6	90	0	Inf. Dilution	lgK:-71.26(4SF)	4	EOD	30201
7	100	0	Inf. Dilution	lgK:-72.68(4SF)	4	EOD	30201

Reaction No. 80408

2<H+1(2)_SiH2O4-2> = H+1(4)_SiH2O4-2(2)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:1.2(2SF)	3	CRV	30909

Note (1): Corrected misprint in source

Reaction No. 80409

2<H+1(2)_SiH2O4-2> = H+1(3)_SiH2O4-2(2) + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0.6	NaCl	lgK:-7.75(3SF)	3	CRV	30909

Note (1): Corrected misprint in source

Reaction No. 80410

Al(s) + 2<H2(g)> + 0.5<N2(g)> + 4<O2(g)> + 3<Si(s)> = NH4AlSi3O8(disord,s)

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-3525.1(5SF)kJ	3	ETR	30953

Reaction No. 80588

Th+4 + 3<H+1(2)_SiH2O4-2> + 3<H2O> = Th+4_H+1(3)_OH-1(3)_SiH2O4-2(3) + 6<H+1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-27.8(0.35SD)	5	CRV	31301

Reaction No. 80595							
U+4 + H+1(2)_SiH2O4-2_(s) = UO2.SiO2(s) + 4<H+1>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.98(1.SD)	5	CRV	31301

Reaction No. 80660							
C(s) + 15<H2(g)> + 3<Ca(s)> + 12.5<O2(g)> + S(s) + Si(s) = Ca3Si(OH)6CO3SO4.12H2O(s)							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-7564.5(5SF)kJ	5	CRV	31308
2	25	0	Inf. Dilution	dH:-8700.(4SF)kJ	5	CRV	31308

Reaction No. 80662							
Ca3Si(OH)6CO3SO4.12H2O(s) = 3<Ca+2> + H+1_SiH2O4-2 + SO4-2 + CO3-2 + OH-1 + 13<H2O >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-24.75(4SF)	5	CRV	31308

Reaction No. 80666							
Na2Al2Si3O10.2H2O(Natr.,s) + 8<H2O> = 2<Na+1> + 2<Al+3_OH-1(4)> + 3<H+1(2)_SiH2O4-2 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-25.19(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-7.20(3SF)	0	CRV	31308
Note (2): For species SiO2(aq) !							

Reaction No. 80880							
Illite(s) + 8.8<H+1> + 1.2<H2O> = 2.7<Al+3> + 3.3<H+1(2)_SiH2O4-2> + 0.7<K+1 >							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.5(3SF)	5	MSL	31760
Note (1): Slightly impure solid							
2	40	0	Inf. Dilution	lgK:9.6(2SF)	5	MSL	31760
Note (2): Slightly impure solid							

Reaction No. 80886							
NaAlSiO4.2H2O(s) + 4<H+1> = Na+1 + Al+3 + H+1(2)_SiH2O4-2 + 2<H2O>							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:12.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:11.1(0.2SD)	0	MSL	31772

Reaction No. 80887

NaAlSiO4.2H2O(s) + 2<OH-1> = Na+1 + Al+3_OH-1(4) + SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-5.74(0.2SD)	5	MSL	31772

Reaction No. 80888

NaAlSiO4.2H2O(am.,s) + 2<OH-1> = Na+1 + Al+3_OH-1(4) + SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.15(3SF)	5	MSL	31772

Note (1): Table 4

Reaction No. 81040

Ni+2 + H+1(2)_SiH2O4-2 = NiSiO3(s) + 2<H+1> + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:2.09(3SF)	4	CRV	32222

Reaction No. 81041

2<Ni+2> + H+1(2)_SiH2O4-2 = 2NiO.SiO2(s) + 4<H+1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-19.24(1.32SD)	4	CRV	32222

Reaction No. 81104

UO2+2 + H+1(2)_SiH2O4-2 = UO2+2_H+1_SiH2O4-2 + H+1

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.84(0.1SD)	4	CRV	32222

Reaction No. 81105

UO2*2.SiO4.2H2O(s) + 4<H+1> = 2<UO2+2> + H+1(2)_SiH2O4-2 + 2<H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.74(0.21SD)	4	CRV	32222

Reaction No. 81346

3CaO.Fe2O3.3SiO2(s) + 12<H+1> + 2<e-1> = 3<Ca+2> + 2<Fe+2> + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:54.81(4SF)	4	CRV	32224

Reaction No. 81347

CaO.SiO2(Pseudo.,s) + 2<H+1> + H2O = Ca+2 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:13.87(4SF)	4	CRV	32224

Reaction No. 81348

CaO.SiO2.TiO2(s) + 6<H+1> + e-1 = Ca+2 + Ti+3 + H+1(2)_SiH2O4-2 + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.4(2SF)	1	ELC	32224
2	25	0	Inf. Dilution	lgK:-9.140(4SF)	0	CRV	32224

Reaction No. 81349

CaAl2Si4O12.4H2O(s) + 8<H+1> = Ca+2 + 2<Al+3> + 4<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:11.66(4SF)	6	CRV	27383
2	25	0	Inf. Dilution	lgK:14.25(4SF)	0	CRV	32224
3	25	0	CHEMVAL1	lgK:11.85(0.01SD)	0	ENS	5156
4	25	0	CHEMVAL2	lgK:11.85(0.01SD)	0	ENS	5156
5	25	0	CHEMVAL3	lgK:11.85(0.01SD)	0	ENS	5156
6	25	0	CHEMVAL4	lgK:11.80(0.01SD)	1	ENS	5156

Note (6): query mineral name "laumontite"

Reaction No. 81350

CaAl4Si2O10(OH)2(s) + 14<H+1> = Ca+2 + 4<Al+3> + 2<H+1(2)_SiH2O4-2> + 4<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:42.33(4SF)	4	CRV	32224

Reaction No. 81351

Ca2Al3Si3O12(OH)(clino,s) + 13<H+1> = 2<Ca+2> + 3<Al+3> + 3<H+1(2)_SiH2O4-2 > + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.16(4SF)	4	CRV	32224

Reaction No. 81352

Ca2FeAl2Si3O12(OH)(s) + 13<H+1> + e-1 = 2<Ca+2> + 2<Al+3> + Fe+2 + 3<H+1(2)_SiH2O4-2 > + H2O

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:44.29(4SF)	4	CRV	32224

Reaction No. 81358

Ca(UO2)2.(SiO4H)2(s) + 14<H+1> + 4<e-1> = Ca+2 + 2<U+4> + 2<H+1(2)_SiH2O4-2> + 4 <H2O>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:35.85(4SF)	4	CRV	32224

Reaction No. 81414

CuSiO3.2H2O(s) + 2<H+1> = Cu+2 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:4.92(3SF)	4	CRV	32224

Reaction No. 81444

FeSi(s) + 4<H2O> = Fe+2 + H+1(2)_SiH2O4-2 + 4<H+1> + 6<e-1>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:65.4(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:63.84(4SF)	0	CRV	32224

Reaction No. 81445

FeSi2(s) + 8<H2O> = Fe+2 + 2<H+1(2)_SiH2O4-2> + 8<H+1> + 10<e-1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:128.4(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:125.97(5SF)	0	CRV	32224

Reaction No. 81446

FeSi2.33(s) + 9.32<H2O> = Fe+2 + 2.33<H+1(2)_SiH2O4-2> + 9.32<H+1> + 11.32<e-1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:152.7(4SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:150.18(5SF)	0	CRV	32224

Reaction No. 81447

Fe3Si(s) + 4<H2O> = 3<Fe+2> + H+1(2)_SiH2O4-2 + 4<H+1> + 10<e-1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:94.1(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:87.82(4SF)	0	CRV	32224

Reaction No. 81452

KAlSi3O8(Feld.,s) + 4<H+1> + 4<H2O> = K+1 + Al+3 + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:0.05(1SF)	4	CRV	32224

Reaction No. 81453

KAlSi3O8(Mur.,s) + 4<H2O> + 4<H+1> = K+1 + Al+3 + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:1.25(3SF)	4	CRV	32224

Reaction No. 81454

KFe3AlSi3O10(OH)2(s) + 10<H+1> = K+1 + 3<Fe+2> + Al+3 + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:23.13(4SF)	4	CRV	32224

Reaction No. 81455

KMg3AlSi3O10(OH)2(s) + 10<H+1> = K+1 + 3<Mg+2> + Al+3 + 3<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:38.17(4SF)	4	CRV	32224

Reaction No. 81467

Mg2Si(s) + 4<H2O> = 2<Mg+2> + H+1(2)_SiH2O4-2 + 4<H+1> + 8<e-1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:209.12(5SF)	4	CRV	32224

Reaction No. 81473

2MgO.2Al2O3.5SiO2.H2O(s) + 16<H+1> + H2O = 2<Mg+2> + 4<Al+3> + 5<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:51.39(4SF)	4	CRV	32224

Reaction No. 81474

Mg48Si34O85(OH)62(s) + 96<H+1> = 48<Mg+2> + 34<H+1(2)_SiH2O4-2> + 11<H2O >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:483.58(5SF)	4	CRV	32224

Reaction No. 81475

Mg4Si6O15(OH)26H2O(s) + 8<H+1> + H2O = 4<Mg+2> + 6<H+1(2)_SiH2O4-2 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:30.94(4SF)	4	CRV	32224

Reaction No. 81485

Mo3Si(s) + 16<H2O> = 3<MoO4-2> + H+1(2)_SiH2O4-2 + 28<H+1> + 22<e-1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-13.08(4SF)	4	CRV	32224

Reaction No. 81504

H+1(2)_NH3(2)_SiF6-2_(hex,s) + 4<H2O> = 2<H+1_NH3> + H+1(2)_SiH2O4-2 + 6<F-1> + 4<H+1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.26(4SF)	4	CRV	32224

Reaction No. 81505

H+1(2)_NH3(2)_SiF6-2_(cubic,s) + 4<H2O> = 2<H+1_NH3> + H+1(2)_SiH2O4-2 + 6<F-1> + 4<H+1 >

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-28.27(4SF)	4	CRV	32224

Reaction No. 81506

NaCa2Mg4Al3Si6O22(OH)2(s) + 22<H+1> = Na+1 + 2<Ca+2> + 4<Mg+2> + 3<Al+3> + 6 <H+1(2)_SiH2O4-2>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:103.45(5SF)	4	CRV	32224

Reaction No. 81508

NaSi7O13(OH)3.4H2O(s) + H+1 + 8<H2O> = Na+1 + 7<H+1(2)_SiH2O4-2>

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-25.36(4SF)	2	CRV	32224

Note (1): H2O number changed (assumed corrected)

Reaction No. 81543

PbO.SiO2(s) + 2<H+1> + H2O = Pb+2 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.75(3SF)	4	CRV	32224

Reaction No. 81544

2PbO.SiO2(s) + 4<H+1> = 2<Pb+2> + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:18.19(4SF)	4	CRV	32224

Reaction No. 81599

SiO2(Tridy.,s) + 2<H2O> = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.58(3SF)	4	CRV	32224

Reaction No. 81600

SiO2(Coesite,s) + 2<H2O> = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.19(3SF)	4	CRV	32224

Reaction No. 81601

SiO2(Stish.,s) + 2<H2O> = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:5.36(3SF)	4	CRV	32224

Reaction No. 81602

SiO2(glass,s) + 2<H2O> = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.00(3SF)	4	CRV	32224

Reaction No. 81603

H+1(2)_SiO3-2_(s) + H2O = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-3.84(3SF)	4	CRV	32224

Reaction No. 81604

H+1(2)_SiO3-2 + H2O = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-1.56(3SF)	4	CRV	32224

Reaction No. 81605

H+1(2)_SiH2O4-2_(s) = H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-4.44(3SF)	4	CRV	32224

Reaction No. 81613

SrSiO4(s) + 4<H+1> + 2<e-1> = Sr+2 + H+1(2)_SiH2O4-2

No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	lgK:-56.78(4SF)	4	CRV	32224

Reaction No. 81709							
ZrSiO4(s) + 4<H+1> = Zr+4 + H+1(2)_SiH2O4-2							
No	**t**	**I-Str**	**Medium**	**Value (Dev)**	**W**	**Tech.**	**Ref. #**
1	25	0	Inf. Dilution	lgK:-14.8(3SF)	1	ELC	
2	25	0	Inf. Dilution	lgK:-5.64(3SF)	0	CRV	32224

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CMN	Several experimental values (crit.)
CRV	Critical review
EAE	Automated extrapolation
EDH	Debye-Hueckel corrections
EES	Personal estimate
EFE	Free Energy relations
ELC	Linear Combination of Reactions
ENS	Not specified
ENT	Nearest enthalpy & entropy changes
EOD	Calculated from data of others

EOR	Other Reaction Constants
ESC	Standard conditions anchor
ETR	Thermodynamic relationships
ETS	Ternary (statistical)
EVH	Van't Hoff Equation (dH = constant)
MAX	Anion exchange
MCL	Calorimetry
MCO	Colorimetry (often for measuring pH)
MGC	Gas chromatography
MGL	Glass electrode
MHY	Emf with hydrogen electrode
MIS	Ion selective electrode
MKN	Rate of reaction
MMR	Nuclear magnetic resonance
MPT	Potentiometry
MSE	Solvent Extraction
MSL	Solubility
MSP	Spectroscopy
MTD	Temperature difference

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